RAKL for Evidence-Governed AI-Assisted Scientific Research

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Abstract

AI research agents can search literature, generate hypotheses and execute analyses, but workflow completion does not determine which statements are licensed to change a scientific record or which past task experiences are safe to reuse. We present RAKL v3, an evidence-governed architecture that treats research as operation over a persistent, typed, recursively evolving external substrate. The underlying language model may remain weight-frozen while future behaviour changes through accumulated external state. RAKL couples four loops: information to knowledge, problem to solution, experience to method, and RAKL to better RAKL. Immutable TaskEpisode records preserve observed research trajectories before interpretation; versioned Lessons separate reflection from reusable method authority; problem-conditioned fibres compile knowledge, tools, episodes, failures, strategies and warnings; local results require explicit interface agreement before global synthesis; and saturation is vector-valued across knowledge, operators, experience patterns, obstructions, relations, paths and meta-methods.

The current implementation now makes the experience–authority boundary explicit. An executable epistemic-noninterference specification distinguishes scientific authority from access, retrieval priority, proposal probability, strategy preference and lesson reuse, and attacks ten registered leakage families. The current v3 experience state does not yet compose the scientific AuthorityLedger into one canonical runtime, so this integration result is reported as `NO_INTEGRATION_SURFACE`, not as an end-to-end enforcement guarantee. Recent runtime hardening separately distinguishes proposal-shadow storage from canonical admission, binds consequential operations to prospective pre-action receipts when required, and makes cross-problem "no match" claims coverage-bound rather than unbounded.

We also report the first interpretable matched `RESET_BASELINE` versus `LEARNING_ENABLED` continual-experience result. After two preserved prompt-interface failures, a frozen v1.2 packet produced 12/12 schema-valid native records with the intended $S_0 \to S_n$ chronology and uncontaminated fresh transfer. Under Qwen2.5-0.5B-Instruct, both arms achieved zero registered successes on development and transfer, with zero development success-score lift and no promotable transfer gain. This is a scoped negative result, not evidence that persistent experience never helps under stronger models or different tasks. The paper therefore contributes the architecture, formalized integration obligations, auditable negative/process history and a matched continual-experience measurement, while leaving authority-leakage model baselines, learning-by-governance factorials, closest-parent ablations, larger-model robustness and universal continual-learning claims open.

1 Introduction

Scientific-agent systems have moved rapidly beyond question answering. Co-Scientist combines generation, reflection, ranking and hypothesis evolution and includes experimental biomedical validation [8]; Robin integrates literature and data-analysis agents across a discovery workflow [6];

AI Scientist-v2 automates experiment design, execution, analysis and paper generation using agentic tree search [20]; Kosmos uses a structured world model for long-horizon discovery [11]; SciAgents couples multi-agent reasoning with ontological knowledge graphs [5]; and PaperQA2 demonstrates strong agentic literature synthesis [16]. These systems make it untenable to position a new method as novel merely because it searches papers, generates hypotheses, uses multiple agents, retains research state or iteratively revises ideas.

A recent evaluation spanning more than 25,000 scientific-agent runs reported frequent evidence neglect and relatively infrequent refutation-driven revision, illustrating that workflow success and epistemically self-correcting reasoning are different targets [13]. The remaining problem is therefore narrower: *what is a new observation allowed to change in the scientific state?* Two papers can use the same terminology while describing different populations, scales or measurement operators. Two models can be observationally equivalent yet mechanistically distinct. A model can be decision-useful while its microscopic mechanism remains unidentified. Several papers or agents can appear independent while sharing evidence lineage. A null result can disappear after enough iterations. A loop can stop because its resource budget is exhausted rather than because the scientific search has become semantically flat. A self-modifying agent can overfit the very evaluator used to declare its improvement.

RAKL treats these as failures of **epistemic state transition**. The language model is a replaceable proposer. Canonical scientific state lives outside the model and changes only through evidence- and governance-bearing transitions. Scientific knowledge is represented as an atlas of contextual local views rather than one accumulating natural-language answer. The same evidence discipline is applied recursively to candidate changes in the research method itself.

This preprint makes five candidate contributions. First, it specifies a projection-first scientific state in which source claims become context-scoped local charts before they compete. Second, it provides a typed transition and obstruction discipline that separates local compatibility, path coherence, global existence and global identifiability. Third, it represents scientific authority as a multi-axis partial order rather than a scalar confidence. Fourth, it couples recursive fibers, negative history, semantic/evidence-lineage stopping, bounded epistemic context, metacognitive learning and governed method evolution in one control architecture. Fifth, it makes these claims falsifiable through machine-checkable method contracts, hostile known-answer worlds, a frozen self-bootstrap protocol and a preregistered real scientific application.

We do not claim novelty for multi-agent science, knowledge graphs, provenance, sheaf-like local-to-global reasoning, hierarchical memory, prompt compression, active experiment design, causal discovery, skill libraries or program evolution in isolation. The candidate contribution is the integrated evidence-governed transition discipline connecting them.

1.1 The control problem behind scientific agency

The central engineering problem is not whether a language model can emit a plausible next step. It is whether a research system can maintain a distinction between *proposal dynamics* and *scientific state dynamics*. A proposal can be useful before it is true, a contradiction can be informative before either side is rejected, and a model can be decision-useful while remaining mechanistically unidentified. These cases require state types that do not collapse into one scalar score. Recent benchmark evidence strengthens this motivation. Science Edge Evaluation (SEE) reports that even strong multimodal models remain far from reliable evidence-bounded inference on expert-curated laboratory questions, and that giving a visual agent tools improves access to information without making the scientific reasoning problem disappear [73]. SCHEMA similarly reports a gap between terminal answer accuracy and the integrity of intermediate scientific trajectories: a correct-looking

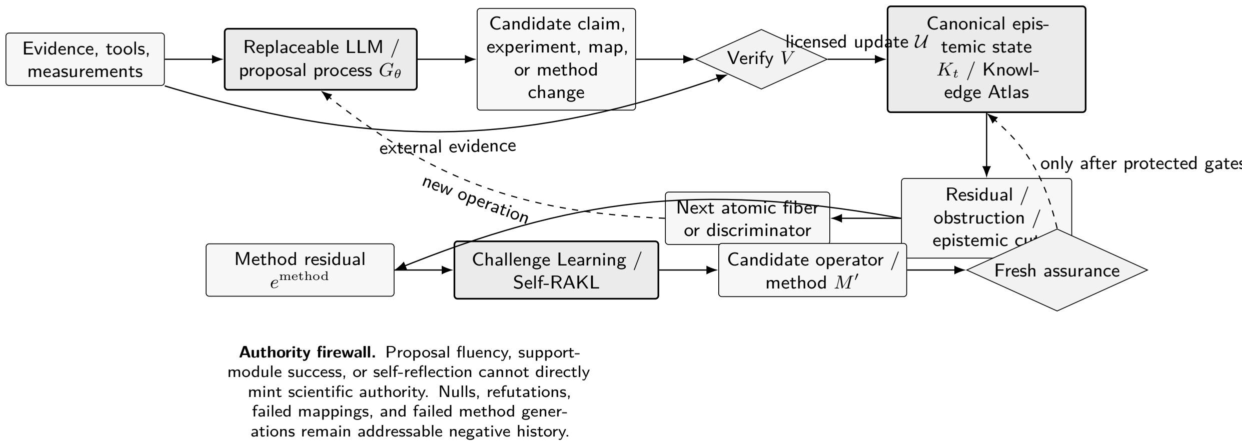

Figure 1: RAKL separates proposal generation from scientific authority. Evidence and tools inform a replaceable proposer, but candidate claims and method changes pass through explicit verification before canonical state can change. Residuals drive scientific fibers; method residuals can trigger Challenge Learning and Self-RAKL, but fresh assurance is outside the challenger's write authority.

endpoint can be reached through structurally defective reasoning [74]. These findings do not prove that RAKL solves the problem. They make the target failure mode concrete.

We distinguish three coupled state machines. The first is the *generative machine*, which proposes claims, experiments, mappings, code, models and method changes. The second is the *epistemic machine*, which records what is supported, refuted, partially identified, blocked or currently uncheckable under a declared context and evidence boundary. The third is the *publication machine*, which projects a bounded subset of the epistemic machine into a human-readable paper. Conflating the three creates recurrent failure modes. If generative fluency writes directly to epistemic state, unsupported authority appears. If publication convenience writes back to epistemic state, omitted negative results or compressed caveats can become invisible. If the canonical archive is treated as the prompt itself, context growth eventually turns memory completeness into computational overload.

This paper therefore uses "epistemic mechanics" in a deliberately operational sense. A scientific state transition is licensed only when it carries the prerequisites appropriate to the claim type. The resulting discipline is analogous to a type system for scientific updates: a representation witness cannot be implicitly coerced into a mechanism witness, an observational association cannot be silently cast as an intervention result, and a convenient summary cannot be promoted into its own evidence root. The analogy has limits. Scientific claims are not ordinary program values, evidence is fallible, and contexts can remain incompletely specified. The practical value of the type-system analogy is fail-closed behavior: when a required coercion cannot be justified, the transition remains blocked rather than being supplied by model confidence.

1.2 What the paper is, and what it is not

A paper is not the Knowledge Atlas. It is a *publication projection* of that atlas for a declared audience, claim boundary and evidence cutoff. This distinction matters for the present manuscript because a framework paper can become misleading in two opposite ways. It can be too compressed, leaving equations and named components without enough intellectual lineage to tell the reader why they are necessary. Or it can become encyclopedic, accumulating related work without increasing the reader's ability to distinguish the framework's objects, obligations and falsifiers. We therefore apply the same semantic-saturation principle to the manuscript that the method applies to literature search. Additional text is retained only when it contributes a new claim distinction, citation lineage, proof obligation, explanatory bridge, falsifier, implementation correspondence or reviewer-relevant boundary.

The resulting saturation claim is intentionally local. It is indexed by the source universe and freshness cutoff, the registered review lenses, the required route families, the current implementation subject, and the unresolved empirical coordinates. A later paper, method or counterexample that changes a novelty boundary or exposes a missing operator reopens the manuscript. "Locally saturated" therefore never means "all relevant knowledge in the open world has been found." It means that under the registered search and review protocol, another pass produces no material semantic gain while all remaining open coordinates are explicit.

This framing also explains why the paper remains a methods and preregistration manuscript. Software contracts and deterministic known-answer worlds establish whether the reference mechanics behave as specified. They do not establish that the full architecture improves real scientific discovery. The latter requires matched same-model evaluation, fresh assurance for method changes, and domain evidence. Keeping these evidence classes separate is not rhetorical modesty; it is part of the method being claimed.

1.3 Evidence access can be a harder ceiling than reasoning scaffolds

A final function-first search surfaced a particularly direct empirical comparator. Wang holds a production scientific-valuation agent's backbone fixed while ablating reasoning scaffolds/public tools and access to a curated proprietary evidence substrate [100]. In that small stratified study, the structured reasoning layer improves calibration and audit discipline, while the much larger gain in factual coverage arrives only when the otherwise inaccessible evidence substrate is supplied. The study is domain-specific and its curated gold set is not an independent ground truth, so it does not establish a universal law of AI science. Its methodological implication is nevertheless important for the present architecture: an agent cannot reason its way to evidence that its acquisition boundary never exposes.

This narrows the role of RAKL further. Evidence governance cannot compensate for an impoverished evidence universe; it can only make the limitation explicit, preserve acquisition provenance, and prevent missing evidence from being converted into unsupported authority. The intended advantage of the framework is therefore not that state discipline removes factual ceilings. It is that the ceiling, the search routes that produced it, and the claim types that remain blocked become first-class research state. A real superiority evaluation must manipulate reasoning architecture and evidence access separately rather than crediting one for the other's contribution.

2 Intellectual lineage, absorbed mechanisms and novelty boundary

A central design principle of RAKL is that a research methodology should learn from prior methods in the same way it learns from domain science: decompose external work into scoped mechanisms, retain provenance, normalize equivalent formulations, and narrow novelty whenever a prior mechanism is found. The bibliography is therefore not merely a list of neighbouring systems. It is an *assimilation ledger*: each retained reference motivates, constrains or removes novelty from a specific RAKL operator.

2.1 Automated science and self-improving agents

Scientific-agent systems have already established literature-grounded hypothesis generation, multi-agent scientific workflows, long-horizon external research state and end-to-end automation. Co-Scientist combines generation, reflection, ranking and hypothesis evolution and includes experimental biomedical validation [8]; Robin integrates literature and data-analysis agents across a discovery workflow [6]; AI Scientist-v2 automates experiment design, execution, analysis and paper generation using agentic tree search [20]; Kosmos uses a structured world model for long-horizon discovery [11]; SciAgents couples multi-agent reasoning with ontological knowledge graphs [5]; and PaperQA2 demonstrates strong agentic literature synthesis [16]. Darwin Gödel Machine, EvoSkill, SkillFoundry and EvoAgentBench further make recursive modification, failure-driven skill creation, held-out validation and ability transfer active prior-art areas [21, 1, 15, 4]. RAKL therefore does not claim novelty for "AI scientist", multi-agent debate, skill libraries or self-modifying code in isolation. Earlier LLM-agent work already demonstrates iterative self-feedback without protected scientific authority: Reflexion stores verbal feedback from trial-and-error in episodic memory [43], while Self-Refine uses the same model to generate, critique and revise outputs iteratively [44]. These systems further narrow the RAKL claim: reflection, feedback memory and iterative refinement are prior mechanisms; the candidate contribution lies in subjecting method change to evidence lineage, negative-history retention and fresh assurance outside the challenger's write authority.

2.2 From workflow completion to evidence-bounded scientific behavior

The autonomous-science literature now makes a useful distinction between *workflow coverage* and *scientific reliability*. A system may be able to search, write code, call instruments, fit models and produce a manuscript while still failing at evidence attribution, uncertainty handling or revision after decisive counterevidence. SEE targets the edge between textbook competence and evidence-bounded inference from realistic experimental artifacts [73]. SCHEMA evaluates scientific-agent hallucinations as errors propagating through a concept topology rather than as isolated wrong strings, which is especially relevant when an early false premise can contaminate many downstream conclusions [74]. These evaluation directions are close to the RAKL concern but differ in role. They are measurement frameworks for scientific-agent behavior; RAKL proposes an internal state-transition discipline intended to make some of those failures representable and blockable.

Two newer benchmark families sharpen the fail-closed requirement. SciIntegrity-Bench constructs dilemmas in which honest acknowledgment of missing evidence is the only correct scientific response and task completion requires misconduct; its reported failures therefore test whether an agent can preserve an explicit "cannot establish" state under completion pressure rather than merely whether it can solve a task [93]. SciConBench evaluates scientific conclusion synthesis against expert conclusions under a clean-room web harness and finds a substantial gap between unconstrained and leakage-controlled evaluation [94]. These results motivate two separate RAKL gates. Missing evidence must remain a typed block rather than be filled by synthetic observations, and evaluation evidence must be separated from information available to the proposal-generating process. Neither benchmark is evidence that RAKL passes those gates in the wild; each is prior art for a failure class the framework claims to make auditable.

AgentBuild offers a complementary engineering precedent. It treats a scientist-authored, version-controlled contract and curriculum as the durable object, while an optimizer is allowed to edit an agent only inside the declared build boundary [75]. This is consonant with the RAKL separation between replaceable proposal machinery and protected scientific or evaluative state. The important novelty boundary is therefore not "agents should have rubrics" or "agents should use tests." The

RAKL claim is narrower: scientific authority, context, negative history, provenance and method-change assurance belong to an external epistemic state whose update rules are themselves subject to evidence.

These distinctions suggest a useful evaluation matrix. One axis asks *how much of the scientific workflow can the system execute?* A second asks *how often are scientific state transitions licensed correctly?* A third asks *how well does the system discover actions that reduce the scientifically relevant uncertainty?* A fourth asks *what resources are required?* A system can improve on one axis while regressing on another. For example, a broader tool set may increase workflow completion while increasing unsupported integration of tool outputs; an aggressive context compressor may reduce cost while dropping a falsifier; a self-editing agent may improve development scores while overfitting the evaluator. A single aggregate score can conceal these tradeoffs, which is why the present evaluation model retains blocking coordinates.

2.3 Model building, prediction and mechanism

Statistical methodology has long treated model construction as iterative rather than final. Box emphasized model criticism and revision as part of scientific modelling [22]; posterior predictive assessment operationalizes comparison between observed discrepancies and model-generated replications [23]. Breiman contrasted stochastic data models with algorithmic predictive modelling [24], while Shmueli made the distinction between explanatory and predictive goals explicit [25]. Mechanistic philosophy separately analyzes mechanisms in terms of organized entities and activities that produce regular change [26]. These lines motivate RAKL's refusal to let predictive success mint mechanism authority. The new model-criticism operator is not claimed as a new statistical idea; RAKL's contribution is to make adequacy a typed child operation inside an evidence-governed research state. Popper's falsifiability and corroboration programme is also classical prior art for designing claims that can fail [45]. More recently, automated symbolic discovery has shown that mathematical regularities can be searched directly from experimental data [46]; RAKL treats such equation search as a proposal mechanism whose variables, observation maps, limiting cases and residuals still require scientific interpretation and verification.

Causal inference supplies a separate ancestry for intervention and mechanism language. Structural causal models distinguish observed association from intervention and counterfactual questions [47]; modern causal-inference treatments likewise make assumptions and intervention distributions explicit [48]. RAKL absorbs this separation rather than using "mechanism" as a synonym for prediction. A predictive survivor may nominate a causal model, but causal or mechanistic authority requires additional assumptions and evidence.

2.4 Prediction, mechanism, identification and explanation are different targets

The distinction between prediction and mechanism has become more important as scientific AI systems improve at interpolation and forecasting. Mechanistic World Models argue that autonomous discovery requires reusable explanatory structure rather than predictive mappings alone [76]. CausaLab operationalizes a related separation by evaluating both performance on tasks and recovery of a hidden causal mechanism, showing that the two can diverge substantially [77]. In mechanistic interpretability, Lin and Liu make the parallel point that validation evidence such as ablations, faithfulness or completeness does not by itself identify a causal mechanism unless the required identification assumptions are stated [78]. These works narrow rather than enlarge the RAKL novelty claim: the representation–mechanism–identification separation is not a newly discovered philosophical fact.

RAKL turns that separation into transition permissions. Let r be a representation that predicts an observable, m a candidate generating mechanism, and i an identification certificate. Evidence can increase representation authority without increasing mechanism authority:

$$\Delta R(r) > 0 \not\Rightarrow \Delta M(m) > 0.$$

Likewise, evidence that a mechanism class is scientifically plausible does not imply that the observations point-identify one member:

$$M(m) > 0 \not\Rightarrow I(m) = \text{POINT\_IDENTIFIED}.$$

The correct terminal object may be an equivalence class, an identified set, a bound or a decision rule robust across several mechanisms. Manski's recent treatment of inductive risk under underdetermined theory choice reinforces why partial identification is not merely a statistical fallback: decision-making can remain principled when the evidence does not justify one uniquely selected theory [79].

This matters for scientific-agent design because generative systems are naturally pressured toward a singular narrative. A language model is rewarded for producing one coherent explanation, a model-selection pipeline returns a winner, and a paper conventionally tells one story. Those interface pressures can create an *identification illusion*: the system mistakes a selected representative for an identified generator. RAKL instead stores the survivor set and the observation map that produced the equivalence. If two mechanisms produce the same registered observations, selecting between them requires a discriminator that changes the observation boundary. If no feasible discriminator exists, the system should preserve the ambiguity and move downstream decisions to robust or set-valued analysis.

The distinction also clarifies mechanism ancestry. A mechanism claim is not strengthened merely by adding more descriptive detail. It requires a trace from lower-level objects and interactions through intermediate state and the observation process. The ancestry can be coarse and domain-specific; RAKL does not require microscopic reductionism. What it forbids is a missing explanatory bridge being silently supplied by linguistic plausibility. Where the bridge is unavailable, the model is labelled phenomenological, predictive, teacher-only or otherwise scoped rather than being promoted by prose.

2.5 Belief maintenance, negative history and partial identification

Truth-maintenance systems already provide dependency-aware belief revision and inconsistency handling [27]; assumption-based TMS extends this to multiple assumption-labelled environments and competing solutions [28]. AGM belief revision gives a separate axiomatic tradition for contraction and revision under minimal change [49]. RAKL therefore does not claim the generic idea of belief revision. Its negative-history rule is a scientific-governance specialization: a null, refutation or failed transition remains an addressable historical event even when later evidence supersedes its scope. Likewise, partial identification is established statistical methodology: Manski's bounds exemplify learning identified sets under weak assumptions [29], while modern sensitivity analysis quantifies how conclusions change under plausible omitted-assumption worlds [30]. RAKL treats 'PARTIALLY_IDENTIFIED' and 'ASSUMPTION_SENSITIVE' as legitimate scientific outcomes rather than forcing point answers.

2.6 Belief revision, argumentation and the status of negative evidence

The belief-maintenance lineage explains why revision cannot be implemented as destructive overwrite. Doyle's truth-maintenance system records justifications so downstream beliefs can be withdrawn when their supports fail, while assumption-based TMS preserves multiple environments rather than forcing a single consistent context [27, 28]. AGM revision supplies a complementary axiomatic view of minimal change under incorporation of new information [49]. Abstract argumentation adds a different object: a network in which arguments can attack one another and acceptability depends on the chosen semantics [80]. These frameworks are not interchangeable. TMS emphasizes dependency maintenance, AGM emphasizes rational belief-set change, and argumentation emphasizes dialectical support/attack structure. RAKL borrows the need to preserve dependencies and incompatibilities, but its canonical objects are scientific claims scoped by evidence, context and authority rather than unqualified propositions.

This difference is most visible for negative evidence. A null, refutation or failed mapping has two temporal identities: it is evidence about the historical state of inquiry, and it may or may not remain decisive for a later, differently scoped claim. Deleting the old object when a successor arrives destroys information needed to audit cherry-picking, repeated testing, route resurrection and model-family evolution. The rule

$$\mathcal{H}_t^- \subseteq \mathcal{H}_{t+1}^-$$

therefore means preservation of the event and its original scope, not that every old negative conclusion remains eternally applicable. A later observation can narrow, supersede or contextualize the interpretation while leaving the receipt addressable.

This is also a response to a known scientific-selection problem. The "file drawer" literature shows why invisibility of null results can distort the apparent evidence landscape [81, 82]. In RAKL, negative history is not merely a publication-policy preference. It is a state variable used by search, saturation and self-evolution. If a candidate mechanism has already failed under a particular regime, a later generator must either explain why the new context is materially different or inherit that negative evidence. If a method repair was tried and failed, the next Self-RAKL generation cannot erase the failure to make its lineage look monotone.

Model pluralism further motivates resisting forced synthesis. Veit argues that many scientific purposes require sets of models whose functions depend on context and history rather than one canonical model [83]. RAKL gives that pluralism an operational destination: a plural atlas can be a successful outcome when different local models remain useful under distinct contexts, while an identified set is appropriate when the models compete on the same target but observations cannot distinguish them. The two should not be confused. Plurality can arise from legitimate perspectival specialization; non-identification arises from insufficient discriminatory evidence.

2.7 Scientific revision as an auditable transaction

The preceding traditions suggest a stronger implementation rule: scientific revision should be represented as a transaction with explicit preconditions, postconditions and preserved supersession lineage. A conventional belief store can replace proposition p with $\neg p$. A scientific atlas must additionally answer which population, measurement, assumptions and evidence made each version applicable. The transaction object is therefore richer than a truth-value flip. One useful abstract form is

$$\rho = (c_{\text{old}}, c_{\text{new}}, e, \gamma, \tau, \omega),$$

where e is the evidence that triggered revision, γ its context, τ the typed relation between old and new claims, and ω the authority effect that the evidence is licensed to create. The old claim remains historically addressable even when its current applicability becomes empty.

This representation distinguishes at least five revision patterns. *Refutation* records evidence incompatible with a claim under aligned context. *Scope restriction* preserves the claim on a narrower domain after a counterexample appears elsewhere. *Supersession* replaces an effective or approximate law with a successor while preserving the regime in which the old law remains adequate. *Decomposition* splits one coarse claim into several context-specific claims after hidden heterogeneity is discovered. *Authority downgrade* retains the descriptive content but withdraws a stronger mechanism, identification or decision certificate when its prerequisite fails. These outcomes are scientifically different even if a final prose summary would say simply that "the model was updated."

Dependency propagation is also typed. If claim b was derived from a, refuting a may invalidate b. If b was independently measured and merely consistent with a, the same refutation need not remove b. A provenance graph that records only citation ancestry cannot always make this distinction; the transition relation must say whether an edge is justificatory, derivational, contextual, analogical or merely navigational. This is where truth-maintenance ideas meet epistemic provenance: dependencies matter because they determine which descendants should be reconsidered when evidence changes [27, 28].

Argumentation adds another useful boundary. An attack edge can express that two claims cannot both retain a particular authority under a registered context, but an attack is not itself a refutation. A methodological critique may attack an inference while leaving the measurement intact. A new experiment may attack one mechanism but leave a predictive relation untouched. A conflict between two observational studies may be suspended until population and measurement alignment is established. The atlas therefore does not ask only "which argument wins?"; it asks which coordinates of which scientific objects remain licensed after the conflict is typed [80].

Revision has a chronology obligation. The system must know whether a hypothesis, threshold, covariate set or interpretation was fixed before the evidence that appears to support it. Otherwise the same data can play both proposal and confirmation roles without disclosure. This is why negative history, preregistration and exact candidate identity belong to the same control architecture. They are different ways of preserving the temporal direction of scientific justification.

The transaction view also explains why a paper should not silently rewrite its own past. If a manuscript once claimed that the generic atom–witness structure was a mathematical lattice and later analysis showed that only a compatibility complex had been established, the correct repair is not merely a global search-and-replace. The project should preserve the terminology error as method history, document the narrower structure, and state which downstream claims changed. The current compatibility-complex repair follows that pattern. Scientific self-correction is more informative when the reader can see what was corrected and why.

Finally, auditable revision provides a natural interface to sensitivity analysis. Instead of asking whether one final conclusion is "robust", the system can replay the transition under alternative registered assumptions or context mappings and observe which authority coordinates change. A conclusion that survives many plausible revisions has a different scientific profile from one that depends on a single fragile alignment, even if both currently have the same point estimate. RAKL treats that profile as structured evidence rather than compressing it into a confidence adjective.

2.8 Why evidential selection pressure belongs inside the model of research

Scientific literature is not an unbiased sample of all completed inquiries. Selective publication, flexible analysis and repeated exploration can change the distribution of claims that enter a search system [81, 82, 92]. A literature agent that retrieves published claims perfectly can therefore inherit upstream selection bias perfectly. Evidence governance begins before language-model reasoning.

RAKL does not attempt a universal correction for publication bias. Instead it records relevant selection information when available and avoids treating publication count as independent confirmation. Pre-registration status, registered reports, availability of null results, dataset ancestry, replication status and analysis chronology can all alter how a claim should be interpreted. These become context/evidence attributes rather than one global quality label.

The same reasoning applies to benchmark ecosystems. Scientific-agent benchmarks are published objects selected by designers, and method developers adapt to visible tasks. A result on a saturated public benchmark can overstate transfer if the benchmark has become part of the development environment. Fresh-assurance tasks and frozen evaluation chronology are intended to protect against this adaptive exposure at the method level.

This is another reason negative history has value beyond transparency. Failed or null results change the selection model of what has been tried. Preserving them can prevent repeated rediscovery of a dead end and can reveal that an apparently dominant positive literature sits on top of many unreported or locally stored failures. In a research organization, the internal atlas can therefore contain scientifically valuable evidence that never became a conventional publication.

The authority system treats these factors as modifiers and blockers rather than as automatic verdicts. A preregistered study can be badly measured; a post hoc analysis can reveal a real phenomenon; a highly replicated result can still fail to transport. The purpose is to preserve the variables needed for scientific judgment and downstream sensitivity analysis.

2.9 Provenance and machine-readable research objects

Database provenance work had already made the origin and derivation of view data a first-class problem, including the distinction between "why" and "where" provenance [50]. The W3C PROV family formalizes entities, activities, agents and derivations as provenance objects [31]; FAIR articulates findability, accessibility, interoperability and reuse as stewardship goals [32]; RO-Crate packages data, software and methods with machine-readable metadata and relationships [33]; and nanopublications attach provenance to granular scientific assertions [34]. Agent-workflow provenance and verifiable agent-native publishing are also active contemporary directions [17, 2]. RAKL therefore does not claim generic provenance or atomic statement packaging. Its narrower addition is to bind provenance to context-scoped scientific authority, negative history, evidence-lineage-aware saturation and target-conditioned context compilation. Computational reproducibility is likewise an established discipline: reproducible-computation rules emphasize recording how each result was produced [51], while good-enough scientific-computing practice emphasizes data, software, collaboration and project organization [52]. RAKL's receipts, environment attestation and artifact manifests are an implementation of this broader reproducibility requirement, not a novelty claim by themselves.

Recent agent-verification work also shows why pooled factual support is insufficient. ProvenanceGuard evaluates whether an atomic claim is supported by the particular source to which it is attributed, exposing cross-source conflation as a distinct error from ordinary factuality [95]. A broader survey of evidence tracing and execution provenance likewise separates retrieved evidence, tool outputs, memory, intermediate claims, actions and final answers rather than treating an agent trace as one undifferentiated rationale [96]. RAKL adopts the same granularity but adds a scientific-authority boundary: a correctly attributed source can still be weak, context-incompatible, derivative of the same evidence root, or irrelevant to a mechanism claim. Source ownership is therefore necessary for some updates but never sufficient for arbitrary authority escalation.

Provenance standards and the novelty boundary. Provenance representation itself is established infrastructure. W3C PROV-O, for example, supplies a general ontology for interchanging provenance across heterogeneous systems [62]. RAKL therefore does not claim novelty for provenance graphs or source ancestry. Its narrower contribution is to couple source-rehydratable provenance to context-scoped scientific objects, typed relation witnesses, multi-axis authority and gated canonical updates; provenance visibility alone neither establishes truth nor licenses a mechanism or identification claim.

2.10 From document provenance to epistemic provenance

Machine-readable provenance solves an important but limited problem: it records where an object came from and how it was transformed. It does not, by itself, determine what the object is allowed to support scientifically. The Open Research Knowledge Graph (ORKG), for example, moves scholarly communication toward structured, machine-actionable research contributions rather than document-only representation [84]. W3C PROV supplies interoperable entity–activity–agent derivations [62]. Nanopublications and RO-Crate provide other granular or packaged representations [34, 33]. These are essential precedents for a scientific state that must be inspectable by software.

RAKL adds a second layer that we call *epistemic provenance*. For an asserted scientific relation, the system must retain not only the source chain but the context alignment, verification identity, relation type, error semantics and authority effect. A source can be perfectly provenance-traceable and still be weak evidence; two sources can be independently published but descend from the same primary dataset; a derived analysis can be reproducible while its measurement operator is mismatched to the claim. Thus

$$\text{traceable provenance} \not\Rightarrow \text{independent evidence} \not\Rightarrow \text{sufficient authority.}$$

The evidence-lineage graph addresses the middle implication. It attempts to represent shared ancestry so that ten derivative reports of one measurement do not count as ten independent flat searches or ten independent confirmations.

Cognitive provenance is separated again. A workspace item may causally influence a proposal because it was retrieved or emphasized, yet that influence does not make the item evidence for the proposal's scientific content. The two edges have different semantics:

$$\Pi^{\text{cog}} \neq \Pi^{\text{evid}}.$$

This distinction becomes important when using LLM traces, memory systems or interpretability tools. A representation being computationally load-bearing is evidence about the agent's cognition; it is not automatically evidence about the external phenomenon being studied. A paper that mixes those graphs can accidentally turn explanation of the model into explanation of nature.

The practical consequence is an "authority firewall" around transformations. Raw acquisition can create a source object; projection can create a context-scoped claim object; normalization can create an equivalence witness; an embedding can create a retrieval index; compression can create a derivative view; a workspace can create an access relation. None of those operations, on its own, increases scientific authority. Only a verified promotion operation can do so. This makes provenance a necessary substrate for authority without making provenance itself authority.

2.11 Analogy and active inquiry

Structure-mapping theory treats analogy as relational mapping rather than surface resemblance [35]; constraint-satisfaction accounts jointly evaluate competing mappings [36]. This prior art motivates RAKL's separation of conservative 'GLUE' from exploratory 'JUMP': an analogy must state preserved and non-preserved structure and has discovery authority only until target evidence arrives.

Experimental information itself predates modern active learning: Lindley formalized expected information from an experiment [53], and Bayesian experimental-design reviews systematized utility-based design under prior/posterior uncertainty [54]. Active data selection also has a long history: information-based design chooses observations by their expected information value relative to a model class [37]. RAKL uses information gain when calibrated probabilistic models are defensible, and otherwise falls back to set shrinkage or worst-case mechanism separation rather than fabricating priors. Repeated adaptive evaluation is itself a validity problem, motivating the protected fresh-assurance reserve [3].

2.12 Discovery search, experimental design and stopping are one control problem at different layers

Literature search and physical experimentation are often treated as separate modules, but both can be written as selection under uncertainty. A literature query chooses an observation operator over the published record; an experiment chooses an observation operator over the scientific system. In each case the purpose is not simply to collect more items. It is to expose distinctions that matter for the registered target. Bayesian experimental design formalizes one version through expected utility [53, 54]; active learning chooses measurements that are expected to improve a learner [37]. RAKL uses these ideas only when the probability model required by the utility is itself defensible.

Literature-based discovery supplies an important cross-domain precedent for the search side. Swanson's classic "undiscovered public knowledge" example showed that useful connections can exist between literatures that are individually public yet not directly connected [87]. This is precisely why semantic saturation cannot be established by repeating the framework's own vocabulary. Open-World Mechanism Discovery therefore includes function-only, historical, mathematical, implementation, citation-neighborhood, adversarial and freshness routes. A later exogenous concept that changes the novelty boundary is a falsifier of the prior route-closure claim for that scope.

Stopping research introduces a dual risk. Stopping too early misses a material semantic object; stopping too late wastes resources and creates more opportunities for adaptive overfitting. Technology-assisted systematic review has developed explicit stopping methods because screening "until tired" is not a defensible rule. Recent confidence-based approaches shift attention from raw recall toward whether the information need is sufficiently resolved [88]. RAKL adopts the general lesson without importing a particular recall estimator: stopping must be attached to a registered information need, source/query universe, evidence lineage and reopen condition.

This produces a hierarchy of stopping statements. "Budget exhausted" is a resource state. "Route locally flat" means one query family yielded no new canonical semantic object. "Same-context plateau" means repeated passes in one research context are flat. "Evidence-lineage independent flatness" requires genuinely distinct evidence ancestry. "Scoped saturation" additionally requires route coverage, unresolved-obligation accounting and no native residual that reopens the fiber. None of these implies that the open world contains no unknown relevant concept.

2.13 Metacognition and learning from challenge

RAKL's human-learning analogy is functional rather than anthropomorphic. Self-regulated learning distinguishes forethought, monitoring/control and reflection [38]; productive-failure work shows that initial failure can improve later representation and transfer under appropriate conditions [39]. Psychology also documents reasons not to trust verbal introspection alone: people can underestimate their own bias [40], overestimate explanatory understanding until forced to reconstruct an explanation [41], and benefit in some settings from explicitly considering the opposite hypothesis rather than generic instructions to “be unbiased” [42]. These results motivate executable RAKL controls such as calibration audits, explanation reconstruction, countermodels, strategy switching, help-seeking, anti-rumination and fresh transfer. Recent generative-AI work similarly argues for metacognitive control around rather than merely inside the model [9].

2.14 Why reflection is not an evaluator

A self-reflective LLM can notice errors, but the act of reflection is not statistically or epistemically independent of the process that generated those errors. This matters because many contemporary agent loops reuse one model as proposer, critic, judge and editor. Such role separation can be operationally useful while leaving shared blind spots, shared training priors and shared context contamination intact. The psychology literature on bias blind spots and illusion of explanatory depth provides human analogies for this failure [40, 41]; RAKL treats the analogy as motivation, not as a claim that language models reproduce the same cognitive mechanisms.

The architecture therefore separates *metacognitive diagnosis* from *assurance*. Diagnosis asks what kind of failure occurred and what action should be taken next. Assurance asks whether a frozen candidate method change transfers on a task or realization that did not participate in designing the repair. The first can be same-context. The second must be protected to the degree required by the claim. Development improvement $\Delta_D > 0$ is thus necessary but insufficient for strong self-evolution; the protected comparison must also show $\Delta_A > 0$ under unchanged blocking criteria.

This separation also prevents “reflection inflation.” Adding more self-critique turns is not automatically more metacognition. A useful metacognitive operation must change a control decision: acquire evidence, switch strategy, construct a countermodel, ask for external help, stop an unproductive route, or open a method-basis gap. Repeating explanations without changing an uncertainty-bearing operation is recorded as flat effort. That is why learning-progress signals are coupled to action policy rather than treated as authority.

Finally, metacognitive state itself must remain revisable. A controller can misattribute a data problem to a method problem or a method problem to a model problem. Self-RAKL therefore treats failure attribution as a hypothesis with discriminators. Only a supported method-basis gap justifies inventing or assimilating a new operator. Missing measurements route to evidence acquisition; implementation defects route to repair; objective defects route to governance. The ability to invent a method is not permission to invent one whenever performance is poor.

2.15 Memory, retrieval and local-to-global knowledge

Retrieval-augmented generation already combines parametric generation with explicit non-parametric memory [55]; therefore external knowledge access is not itself a RAKL contribution. Hierarchical memory and context optimization are prior engineering domains. MemGPT frames long-horizon context as virtual memory [12]; RAPTOR organizes multi-resolution retrieval hierarchically [14]; LLMLingua compresses prompts [10]; and RECOMP selectively compresses retrieved material [19]. Long-context models also do not necessarily use all prompt positions equally: “lost in the middle” evaluations show systematic degradation when relevant information lies away from context boundaries [56]. This supports treating active scientific context as an engineered working set rather than assuming that a maximal context window is equivalent to usable memory. Sheaf-theoretic work has long characterized contextuality through obstructions to global sections [57]; Local-to-global consistency and provenance-rich contextual worlds are also active formal directions [7, 18]. RAKL therefore does not claim generic graph storage, compression, hierarchical retrieval or sheaf mathematics. Its narrower requirement is that scientific projection, semantic identity, storage compression and prompt selection remain distinct operations, and that mandatory epistemic evidence cannot be silently traded away for context-window efficiency.

Taken together, these literatures narrow the candidate novelty to the integrated transition discipline: source contribution as contextual projection before competition; typed non-escalating authority; obstruction-aware local-to-global synthesis; immutable negative research history; evidence-lineage-aware stopping; bounded but rehydratable scientific memory; residual-driven method completeness; and protected recursive method evolution.

Compression lineage and what is not claimed. Minimum-description-length reasoning formalizes an important pressure against representational complexity [60], while the information-bottleneck principle studies compression that preserves information relevant to a target variable [61]. These are intellectual-lineage analogies rather than algorithms silently implemented by the current reference profile. The present context compiler does not estimate mutual information or solve an information-bottleneck objective, and the Knowledge Atlas "volume" introduced below is not a description length. The retained engineering lesson is narrower: compactness, target-relevant retained function and scientific authority must be measured separately.

2.16 Local-to-global mathematics: sheaves, closure systems and why the names matter

Several mathematical traditions offer precise versions of "local pieces forming a global object," and their differences help prevent metaphor from becoming false formalism. Sheaf theory studies compatible local data and the conditions under which local sections can be glued into global sections. In contextuality, the absence of a global section can itself carry substantive meaning [57]. Recent AI-science work uses sheaf-theoretic transport and obstruction to diagnose whether a representational language can coherently extend into a new regime [86]. This is close to RAKL in vocabulary and motivation. It narrows the novelty claim and supplies a useful comparator: the cited framework is a finite diagnostic of transport and theory-shift candidates, while RAKL embeds obstruction inside a larger evidence, authority, memory, search and method-evolution control plane.

Formal Concept Analysis (FCA) offers a different route to genuine lattice structure. Given a formal context relating objects to attributes, FCA constructs formal concepts and orders them into a concept lattice [85]. This matters because the historical name "Recursive Atomic Knowledge Lattices" can be read as an order-theoretic claim. The current atom–witness–path implementation does not obtain a lattice merely because atoms are combinable. Its honest generic structure is a typed compatibility complex. A local order-theoretic lattice can be licensed only after a scoped order or closure operator is defined and the relevant meet/join obligations are established. FCA is therefore a useful mathematical neighbor and a terminology constraint, not a hidden implementation detail.

The distinction can be summarized as follows. A graph gives adjacency. A compatibility complex gives witnessed co-admissibility among typed objects and constructive paths. A poset gives an antisymmetric order. A lattice additionally supplies meets and joins for the relevant pairs. A closure system supplies a family of closed objects and can induce lattice structure under the appropriate conditions. A sheaf organizes local data, restrictions and gluing. These structures can coexist in one research architecture, but none is interchangeable with another. RAKL uses different ones for different jobs: provenance is graph-like, authority is partially ordered, local chart synthesis is atlas/sheaf-like when the obligations are defined, and constructive candidate assembly currently uses a compatibility complex.

This layered view also changes what "global" means. A global model may fail to exist because local charts genuinely conflict; it may exist but be non-unique because the evidence underdetermines it; or it may be representable but scientifically unsupported. Global existence, uniqueness and authority are therefore three separate predicates. A successful gluing calculation cannot by itself increase evidence quality, and strong evidence for local charts cannot guarantee a unique global synthesis.

3 Formal method

3.1 Task, researcher state and epistemic state

A scoped inquiry is

$$\mathcal{P} = (O, \mathcal{Q}, \Gamma_0, \mathcal{E}_0, \mathcal{B}, \Lambda), \tag{1}$$

where O is the target object, $\mathcal{Q}$ the registered scientific questions or quantities of interest, Γ_0 the context/observation boundary, $\mathcal{E}_0$ the evidence available at the declared cutoff, $\mathcal{B}$ the resource budget and Λ the blocking epistemic invariants.

The functional researcher state is

$$\mathfrak{R}_t = (K_t, \mathcal{Z}_t, \Omega_t, \Pi_t, \mathcal{G}_t, \mathcal{M}_t, \mathcal{X}_t, \mathcal{R}_t), \tag{2}$$

separating canonical scientific state, explanatory models, method skills, executive policy, research agenda, metacognition, experience/transfer history and resources. The epistemic state is

$$K_t = (\mathcal{A}_t, \mathcal{T}_t, \mathcal{V}_t, \mathcal{E}_t, \mathcal{U}_t, \mathcal{O}_t, \mathcal{F}_t, \mathcal{H}_t^-, \mathcal{S}_t, \mathcal{G}_t^K), \tag{3}$$

with Knowledge Atlas charts, typed transitions, survivors, evidence/provenance, uncertainty and identified sets, obstructions/residuals, recursive fibers, immutable negative history, saturation state and protected governance identity.

RAKL v3 places this epistemic state inside a broader persistent value state R_t. We write

$$K_t = \pi_{\mathrm{epi}}(R_t), \tag{4}$$

where π_{epi} is the authority-preserving epistemic projection. The remaining v3 views contain immutable task episodes, versioned lessons, operational tools, failure diagnoses, strategy/expertise abstractions, vector-saturation state and an optional archive of Self-RAKL variants. The global substrate is a typed relational object, not a claim that all of these views form one mathematical lattice. Specialized stores remain the semantic and authority owners of their objects.

For a replaceable driver with fixed weights θ, one task turn can be written schematically as

$$(S_t, \tau_t) = \mathrm{Driver}_\theta(P_t, R_t), \qquad R_{t+1} = \mathrm{Learn}(R_t, \tau_t). \tag{5}$$

The learning map changes external state rather than silently changing the evidential meaning of K_t. In particular, experience-derived routing priors may alter which admissible operator is attempted first, but scientific promotion still passes through the ordinary evidence and governance transitions of the epistemic projection.

3.2 What a local scientific chart must contain

A local chart is not simply a paper abstract placed in a database. It is the smallest context-scoped scientific object that can participate in comparison. At minimum it identifies the scientific object or system, the proposition, the quantity or relation asserted, the population/regime, the observation or measurement operator, units and transforms where relevant, uncertainty/error semantics, source evidence, assumptions, falsifiers, and the authority coordinates currently licensed. Missing fields remain missing rather than being supplied from neighboring papers.

This requirement is intentionally stricter than ordinary information extraction. A sentence such as "X increases Y" cannot be compared safely until "X," "Y," intervention status, time scale and population are resolved. In one source the statement may describe an observational association; in another it may describe an intervention; in a third it may be a fitted model derivative. The surface predicate is shared while the scientific transition type is not. Projection is the step that makes this difference explicit.

Charts can be multi-resolution. A coarse chart can be useful for search while a fine chart preserves the measurement details required for analysis. The coarse object does not replace the fine one; it carries a reconstruction pointer and a declaration of erased coordinates. This provides a formal place for summaries without making them canonical truth.

Overlap is also typed. Two charts overlap only on the coordinates relevant to the comparison. A population overlap does not imply a measurement overlap, and a shared variable name does not imply a shared operationalization. Restriction maps therefore state which coordinates are being aligned or forgotten. When such maps are approximate, their error enters the transition certificate.

The chart idea is especially useful for cross-disciplinary synthesis. Different fields often carve the same phenomenon at different scales, with different measurement conventions and different explanatory targets. A forced global ontology can erase those distinctions. Local charts allow each field's valid objects to remain native while explicit maps represent the conditions under which information can move between them. The target is controlled interoperability, not a universal vocabulary.

Finally, chart quality and source prestige are distinct. A highly cited source can project to an underspecified chart if the required context is absent. A modest source can provide a strong local measurement under a narrow regime. The authority system evaluates the projected object and evidence, not a venue label alone.

3.3 Contextual charts, transitions and gluing

A local scientific chart is

$$C_i = (U_i, \phi_i, \gamma_i, e_i, \alpha_i), \tag{6}$$

where γ_i can include population, scale, regime, units, observation model, assumptions, intervention, time/cutoff and QoI. Source observations are contextual projections $y_i = \pi_i(O \mid \gamma_i)$ rather than globally competing whole-object answers.

For overlapping charts,

$$T_{ij}^{\tau,\sigma} : \phi_i(U_i \cap U_j) \to \phi_j(U_i \cap U_j) \tag{7}$$

is a typed, scoped transition carrying evidence, assumptions and uncertainty semantics. Composition is licensed only when interfaces, relation types, scopes, invariants, lineage and error-composition rules are compatible. Graph connectivity alone does not create endpoint scientific authority.

At comparison layer ℓ, a contradiction requires overlap, alignment and incompatibility:

$$\mathrm{Contradict}_\ell(C_i, C_j) = \mathrm{Overlap}(C_i, C_j) \wedge \mathrm{Align}_\ell(C_i, C_j) \wedge \neg\,\mathrm{Compatible}_\ell(C_i, C_j). \tag{8}$$

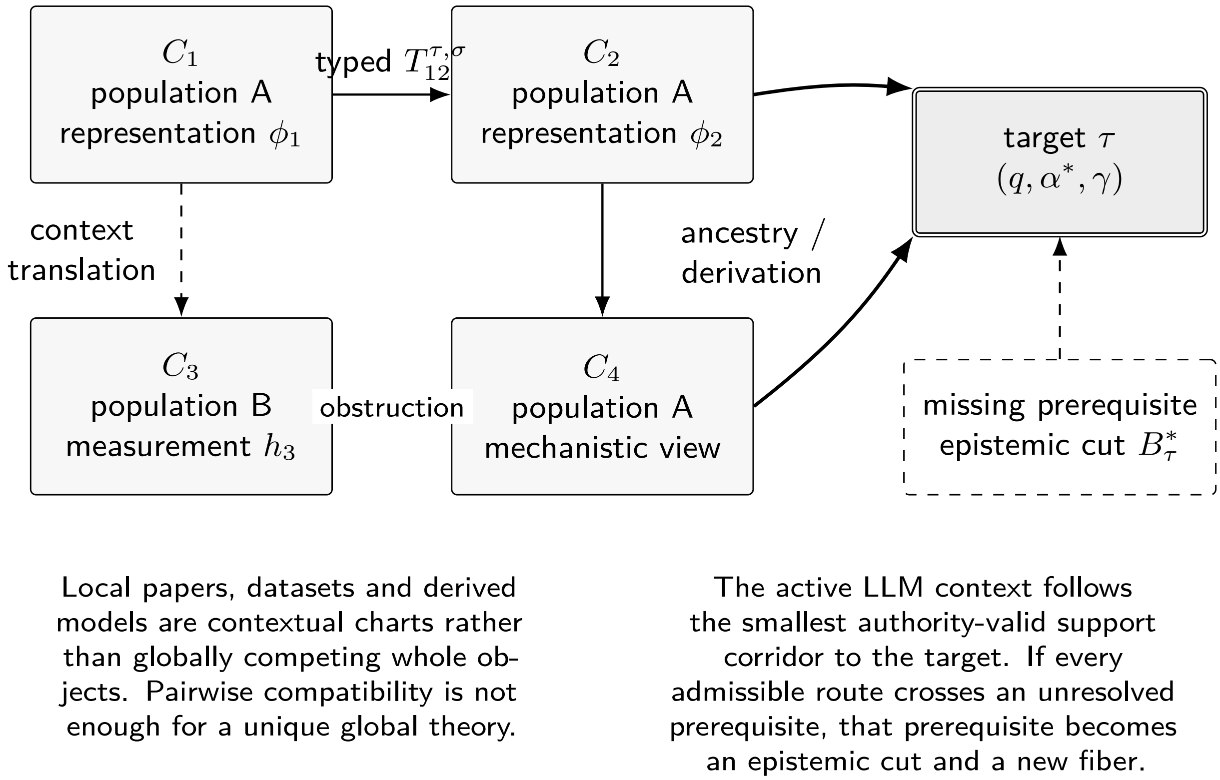

Figure 2: Contextual Knowledge Atlas and target-conditioned inquiry. Papers, datasets and models become local scientific charts. Typed transitions establish which comparisons and derivations are licensed; unresolved incompatibility remains an obstruction. The active research problem is to find an authority-valid support structure to target τ.

The gluing operator may return a global formalism, a plural atlas, or an obstructed/identified set. Pairwise compatibility is therefore not silently promoted to unique global identification.

3.4 Four local-to-global questions that must not be collapsed

The atlas vocabulary is meant to separate four questions that are often compressed into the word "consistency." The first is *local compatibility*: after aligning the relevant population, scale, measurement operator and regime, do two charts make claims that can jointly hold on their overlap? The second is *path coherence*: if several typed transitions are composed, do their scopes, assumptions and error semantics remain compatible along the path? The third is *global existence*: is there at least one global object whose restrictions agree with the retained local charts? The fourth is *global identifiability*: if such global objects exist, does the evidence distinguish one of them?

These predicates can fail independently. Pairwise-compatible charts can participate in a cycle whose composed transition is inconsistent. A global model can exist but remain non-unique. A unique mathematical gluing can exist while the evidence supporting one chart is too weak for the scientific authority requested by the target. Conversely, two high-authority local results can be genuinely incompatible because a hidden context coordinate prevents common gluing. Treating all four cases as "contradiction" loses the information needed to choose the next experiment.

The transition registry therefore records relation type, direction, domain, codomain, scope and approximation. A coordinate transform, coarse-graining map, causal intervention relation and analogy are not interchangeable arrows. If a path mixes them, the composed path must state what kind of endpoint claim it licenses. For example, an exact unit conversion followed by an approximate model reduction does not yield an exact end-to-end identity. The remainder term from the reduction remains part of the path certificate.

Obstructions are useful because they localize failure. An obstruction may indicate a missing overlap witness, an incompatible context, a failed limiting relation, an observation model that cannot transport, or a model family that lacks a required degree of freedom. Recent sheaf-theoretic work on scientific theory shift similarly uses transport failure and obstruction as signals that a representational language may need extension [86]. RAKL incorporates that logic while also asking what evidence authority the proposed extension would require.

A practical consequence is that "merge" is never the default response to agreement. Two charts can remain separate even when compatible if their contexts matter for downstream transport or if collapsing them would destroy provenance. Gluing is a scientific operation with an information-loss profile. When a global summary is constructed, the atlas should preserve the local charts and the gluing witness so later evidence can reopen only the affected region instead of invalidating an undifferentiated global narrative.

3.5 Multi-axis scientific authority

For claim c, scientific authority is represented as

$$\alpha(c) = (G_c, R_c, M_c, I_c, D_c), \tag{9}$$

where the coordinates are scoped certificate sets for grounding/provenance, representation/relation, mechanism ancestry, identification/bounding and decision/QoI usability. Under compatible scope, the induced order is coordinatewise. It is a partial order, not automatically a mathematical lattice: incompatible assumptions, populations or observation operators can obstruct a join.

Cross-axis non-escalation is explicit:

$$R \not\Rightarrow M, \qquad D \not\Rightarrow M, \qquad M \not\Rightarrow I. \tag{10}$$

Thus predictive or observational success does not by itself identify a mechanism; mechanism plausibility does not establish point identification; provenance and citation multiplicity do not create truth or independent evidence.

3.6 Authority as a partial order rather than a confidence meter

Scientific authority is represented as a partial order because many improvements are incomparable. A new calibration study can strengthen grounding without changing mechanism evidence. A discriminating intervention can strengthen identification while leaving the raw measurement quality unchanged. A robust decision analysis can increase decision usability without establishing a more detailed mechanism. These are scientifically different movements through state.

Let

$$\alpha(c) = (G, R, M, I, D)$$

denote grounding, representation/relation, mechanism, identification and decision certificates for claim c. The order $\alpha_1 \preceq \alpha_2$ is defined only under compatible claim scope and means that every relevant certificate carried by α_1 is retained or strengthened in α_2. The order need not be total. One claim can have stronger measurement provenance while another has stronger identification evidence. A single scalar would force a tradeoff rate between these coordinates that has no general scientific justification.

This is why the architecture uses blocking conditions. Suppose a model has exceptional predictive performance but relies on temporally leaked features. No amount of predictive score compensates for the causal availability violation if the target is a prospective prediction. Similarly, a large literature count cannot compensate for shared dataset ancestry when independent replication is required. The authority representation is intentionally non-compensatory on such coordinates.

Partial order language also prevents the word "confidence" from doing too much work. Statistical confidence, posterior probability, model-selection weight, source quality and epistemic authority are different objects. A 95% interval is not "95% authority," and an LLM's verbal confidence is not a measurement certificate. Probabilities may appear inside authority coordinates when their assumptions are valid, but authority itself records the type and scope of what those probabilities license.

The authority firewall is therefore a collection of forbidden coercions. Retrieval rank cannot raise authority. Citation count cannot raise authority. Workspace occupancy cannot raise authority. Model agreement cannot raise mechanism authority without evidence of independence and identification. Human approval cannot by itself transform a missing measurement into a measured value. These restrictions make the framework conservative, but they also make claims auditable: a reviewer can ask which certificate authorized a transition rather than interpreting an opaque aggregate score.

This structure is also compatible with scientific disagreement. Two research groups can possess different authority certificates because they have access to different evidence or accept different explicit assumptions. The atlas can represent both states and the exact assumptions separating them. Resolving the disagreement then becomes an experiment or evidence-acquisition problem rather than a contest between scalar confidences.

3.7 Decision authority is downstream of scientific authority but not reducible to it

Scientific research often terminates in a decision rather than a theory. A decision claim has its own prerequisites: action set, loss or utility, target population, time horizon, feasible information, and the scientific uncertainty over outcomes. This is why decision authority is a separate coordinate rather than the top rung of a single truth ladder.

A partially identified mechanism can still support a robust action. If every model in the survivor set recommends the same intervention under the registered loss, the decision can have strong robustness even though mechanism identification remains weak. Conversely, a sharply estimated predictive model can have low decision authority if the action cost is asymmetric, the deployment population differs, or the features needed at decision time are causally unavailable.

The decision certificate therefore records the set of scientific states over which the action was evaluated. If later evidence expands the uncertainty set or changes the target population, the action can be re-evaluated without pretending the earlier decision was irrational. It was conditionally justified under a prior state. This versioned interpretation is especially important in fast-changing domains where policies must be made before all scientific uncertainty is resolved.

Decision authority also prevents a common benchmark mistake. A system that predicts well is not automatically useful for action if the benchmark loss differs from the real decision loss. Accuracy, log loss and calibration can each be relevant while still failing to encode the cost of false positives, intervention harm or abstention. RAKL therefore requires the decision target to be stated before a model-selection score can be interpreted as action value.

When no decision is required, the coordinate can remain absent. The framework does not force every scientific result into optimization. Descriptive knowledge, mechanism understanding, negative results and identification bounds are legitimate endpoints in their own right.

3.8 Authority is not confidence, probability or calibration

The authority tuple should not be confused with a probability that a sentence is true. Probabilistic belief and predictive calibration are useful scientific objects, but they answer different questions. A model can assign a well-calibrated probability to an event without identifying the event's mechanism. A source can provide strong mechanistic evidence about a pathway while leaving a downstream forecast poorly calibrated. A decision can be robust across a wide identified set even when no member has high posterior probability. These cases motivate keeping epistemic authority and probabilistic uncertainty as separate coordinates that can constrain one another without being identified.

Suppose a predictive model emits $P(Y \mid X)$. Calibration evidence can increase confidence that its probability statements have the registered frequency or scoring-rule interpretation on a target population. That evidence belongs primarily to representation/prediction authority and transport. It does not create provenance for an external source, mechanism ancestry or point identification. Conversely, an experimentally supported mechanism can constrain the plausible model family without guaranteeing that a particular finite-sample probabilistic forecast is calibrated. The framework therefore permits both a probability object and an authority certificate to attach to the same claim while recording distinct verification routes.

This separation prevents two common scalarization errors. The first is *confidence laundering*: a fluent model reports high internal confidence, which is then interpreted as scientific support. The second is *evidence laundering*: a claim has impeccable provenance, which is then interpreted as a high probability of truth regardless of study design, selection or context. Provenance answers "where did this come from?"; calibration answers "how do these probabilities behave?"; authority answers "what type of scientific transition has been licensed?" None subsumes the others.

The distinction also matters for abstention. A fail-closed system should return `CANNOT_CHECK` when the evidence required for a particular authority upgrade is unavailable even if its proposer assigns high probability to the candidate. But it need not abstain from every downstream decision. If a policy is robust across the surviving identified set, decision authority may be strong while mechanism authority remains weak. Separating the axes allows the system to be conservative about explanation without becoming useless for action.

3.9 Proposal, verification and update

The proposal channel

$$G_\theta(K_t, a_t) \to \mathcal{P}_t \tag{11}$$

may be implemented by an LLM, symbolic program, human or hybrid. Verification is separate:

$$\begin{aligned} V(p, e, \gamma) \to \{&\mathrm{SUPPORTED}, \mathrm{REFUTED}, \mathrm{PARTIALLY\_IDENTIFIED}, \\ &\mathrm{BLOCKED}, \texttt{CANNOT\_CHECK}\}. \end{aligned} \tag{12}$$

Canonical update is

$$K_{t+1} = \mathcal{U}(K_t, a_t, e_{t+1}, V, \mathcal{G}_t^K). \tag{13}$$

Negative history is monotone,

$$\mathcal{H}_t^- \subseteq \mathcal{H}_{t+1}^-, \tag{14}$$

meaning later evidence can supersede interpretation or scope but cannot erase the historical null/refutation event.

3.10 Verification is claim-type specific

A generic "critic" cannot verify every scientific proposal with one rubric. Verification depends on the proposed authority effect. A bibliographic claim can be verified against source metadata. A numerical transformation can be verified by recomputation and unit checks. A statistical claim requires a data and analysis identity. A mechanism claim requires ancestry and identification assumptions. A decision claim requires a loss function, action set, target population and transport scope.

For proposal p, the verifier therefore evaluates a contract

$$V_p = (\tau_p, \gamma_p, E_p, A_p, F_p),$$

where τ_p is claim type, γ_p context, E_p required evidence classes, A_p assumptions and F_p falsifiers. The verifier is not necessarily one model or one program. It can be a composition of deterministic checks, statistical analyses, source inspection, domain review and protected evaluators. What matters is that the required evidence cannot be substituted by the proposal generator's self-assessment.

The outcome vocabulary is richer than accept/reject. 'SUPPORTED' means the registered evidence licenses the proposed scoped update. 'REFUTED' preserves decisive counterevidence. 'PARTIALLY_IDENTIFIED' means evidence narrows the target without selecting a unique point or mechanism. 'BLOCKED' means a stated prerequisite has not been met. `CANNOT_CHECK` means the required evidence or verification capability is unavailable. These outcomes have different downstream consequences: a refutation can close or redirect a fiber, a block opens a prerequisite, and partial identification may already be sufficient for a robust decision.

Verification identity is itself versioned. If a software library, rubric, model judge or domain criterion changes, the certificate names the new evaluator. Otherwise a later replay can silently produce a different answer under the same apparent claim. For strong self-evolution claims, protected evaluator state is additionally outside the challenger's write authority. This is the method-level analogue of keeping the measuring instrument separate from the object being optimized.

A verifier can still be wrong. RAKL does not claim infallible certification. The purpose of explicit verification identity is to make such failure localizable and revisable. If a verifier is later found defective, the system can trace which authority upgrades depended on it, invalidate or downgrade those certificates, and preserve the historical event. Without that dependency graph, correction becomes a global narrative rewrite.

3.11 Mechanism, measurement and uncertainty

Mechanistic authority requires an evidence-bearing ancestry such as

$$\begin{aligned}&\text{building blocks} \to \text{interactions} \to \text{mechanism}\\ &\qquad \to \text{mesoscopic state} \to \text{effective law} \to \text{observation/QoI}.\end{aligned} \tag{15}$$

Missing ancestry leads to partial identification or a mechanism set rather than a forced mechanism label.

Measurement is explicit. We write

$$y = h(O, \eta, \gamma) + \epsilon, \tag{16}$$

with η representing instrument/calibration state. For affine $Y = AX + b$,

$$\mu_Y = A\mu_X + b, \qquad \Sigma_Y = A\Sigma_X A^\top. \tag{17}$$

For differentiable nonlinear g, the first-order approximation

$$\Sigma_{g(X)} \approx J_g \Sigma_X J_g^\top \tag{18}$$

is explicitly local and assumption-dependent. Root-sum-square uncertainty is accepted only with an independence or uncorrelatedness witness.

3.12 Metrology boundary: measured values are not context-free numbers

The measurement layer is grounded in established metrology rather than invented uncertainty vocabulary. The JCGM Guide to the Expression of Uncertainty in Measurement treats measurement uncertainty as part of the reported result and supplies general rules for evaluating and expressing it [90]. RAKL uses this lineage to enforce a simple point: a number without its measurand definition, calibration state, units, transformation history and uncertainty model is not yet a comparable scientific object.

This becomes critical when LLMs normalize literature. Converting units is easy; proving that two quantities are the same measurand is not. One paper may report a direct observable, another a model-derived latent quantity, and a third a proxy calibrated under a different regime. A numeric conversion can be correct while the scientific merge is false. The projection step therefore records the observation operator before normalization decides whether values can be transported.

Uncertainty propagation likewise has scope. The affine covariance law is exact for the stated second-moment transformation. First-order Jacobian propagation is a local approximation whose adequacy depends on nonlinearity and operating region. Independence-based root-sum-square formulas require an independence or uncorrelatedness witness. When these conditions fail, RAKL must either use a more appropriate propagation method, retain a conservative bound, or return `CANNOT_CHECK`. The model cannot fill in a missing covariance by assuming independence because the resulting narrower interval would itself be a scientific claim.

3.13 Recursive fibers, target paths and epistemic cuts

An unresolved transformation is

$$f = (o_f, q_f, \gamma_f, r_f, \mathrm{parent}(f)). \tag{19}$$

Residual routing opens only dimensions capable of changing the target conclusion. For target $\tau = (q, \alpha^*, \gamma)$, RAKL searches for an authority-valid support subgraph or hypergraph. A conceptual low-cost cut objective is

$$B^*_\tau = \arg\min_B \mathrm{Cost}(B) \tag{20}$$

subject to every admissible support route intersecting B. This equation defines a research-control target rather than a guarantee that a globally optimal cut always exists or is computationally tractable. A cut may correspond to missing evidence, measurement, context, transition, mechanism, formal lemma or reasoning operator.

3.14 Epistemic cuts turn "what is missing?" into a constructive object

A research system often knows that it cannot support a target but not why. The epistemic-cut formulation makes the missing support explicit. Consider the hypergraph of admissible support routes from current evidence and assumptions to a target authority certificate. A cut is a set of missing or blocking obligations whose resolution would intersect every currently viable route.

Depending on the problem, a cut can contain a measurement, a calibration, a context coordinate, a causal assumption, a formal lemma, a replication, a transformation witness or a method operator.

This is not claimed as a new graph-theoretic min-cut theorem. The scientific content lies in the typed elements and their cost. A cheap new measurement and an expensive randomized intervention may both separate the same mechanisms but carry different feasibility, ethical and informational profiles. The cut representation provides a common interface for experiment design and method diagnosis while preserving those types.

Cuts also prevent premature invention. If the target is blocked because a raw measurement is absent, inventing a new theory does not close the gap. If every route depends on an unidentified nuisance transformation, collecting more of the same outcome may not help. If the missing object is a method operator, ordinary evidence acquisition cannot solve it. The cut therefore routes the next action to the correct layer.

When multiple cuts exist, the system can optimize over cost, expected target change and risk, but the optimization criterion must be registered. A lowest monetary-cost cut may be scientifically weak; a high-information experiment may be infeasible; a method invention may be cheap to propose but expensive to validate. The action-selection layer is deliberately downstream of cut typing so it cannot compare incomparable actions without an explicit utility model.

The concept also clarifies saturation. A fiber with a material unresolved cut is not saturated merely because search rounds are flat. Flat search plus an explicit unavailable measurement can be a terminal block; flat search plus an unattempted feasible discriminator is not. Saturation is therefore a statement about the status of remaining cuts as much as about novelty counts.

3.15 Action selection, model criticism and assumption sensitivity

When calibrated probabilities are justified, one admissible policy family is

$$u(a \mid K_t) = \frac{\lambda_Q I(Q; Y_a \mid K_t) + \lambda_M \operatorname{Sep}(a, \mathcal{V}_t) + \lambda_N E[\Delta N_a]}{\operatorname{Cost}(a)}. \tag{21}$$

Without calibrated probabilities, RAKL uses set-valued alternatives such as worst-case mechanism separation or identified-set shrinkage.

For frozen model criticism, scientific discrepancy probes are $s_k = T_k(y_{\text{obs}})$ and are compared with predictive/generative samples $s_k^{(b)} = T_k(y_{\text{pred}}^{(b)})$. A two-sided finite-sample predictive tail plus a predeclared materiality tolerance can identify a structured residual. Passing the probes establishes adequacy only relative to the frozen probe family, population and resolution; it does not prove truth or mechanism identity.

A complementary assumption-sensitivity operator evaluates a frozen envelope (a_j, θ_j) under the same context and QoI. For material threshold δ,

$$C(\theta; \delta) = \begin{cases} \text{POSITIVE}, & \theta > \delta, \\ \text{NEGATIVE}, & \theta < -\delta, \\ \text{INDETERMINATE}, & |\theta| \le \delta. \end{cases} \tag{22}$$

A baseline conclusion is robust only within the registered envelope when all available preregistered scenarios preserve its material conclusion class. Robustness does not prove the assumptions true; a changed population or QoI is a new contextual chart rather than an assumption perturbation of the same claim.

3.16 Epistemic state as an event-sourced scientific object

The tuple notation above is compact, but its operational interpretation is closer to an event-sourced state machine than to a mutable notebook. Canonical state is reconstructed from addressable evidence objects and licensed transition events. Each event records the object identities it reads, the evidence and context it relies on, the certificates it creates or withdraws, and the governance identity that authorized the operation. The purpose is not software fashion. Event history is what permits a later reviewer to answer questions that ordinary prose obscures: Which observation first changed the mechanism survivor set? Which assumption made two claims comparable? Which refutation closed a route? Which method version generated a result?

This produces four invariants. **Identity invariance** requires a frozen object identifier to continue denoting the same object definition; changed scope produces a successor, not retroactive mutation. **Lineage invariance** requires a derived claim to retain a path to its evidence roots. **Authority invariance under pure representation operations** requires that normalization, indexing, compression or visualization cannot raise authority. **Negative-history persistence** requires refutations and failed candidates to remain addressable. Scientific conclusions can still change. What is prohibited is changing the historical record in order to make the current conclusion appear inevitable.

The resulting state transition can be decomposed as

$$(K_t, p, e, \gamma, g) \xrightarrow{\text{validate}} v \xrightarrow{\text{authorize}} u \xrightarrow{\text{commit}} K_{t+1},$$

where p is a proposal, e evidence, γ context and g governance state. Validation establishes a typed outcome; authorization checks whether that outcome licenses the proposed authority effect; commit creates the new versioned state. A failure at any stage is retained with a reason. This decomposition prevents an implementation success such as "the parser produced a result" from being mistaken for a scientific success such as "the evidence supports a mechanism."

A useful consequence is that `CANNOT_CHECK` is a first-class terminal state for an operation. In ordinary chat interfaces, unanswered questions invite guessing; in scientific software, missing calibration metadata, unavailable raw observations, unresolved source identity or insufficient evaluator independence are meaningful states. Returning `CANNOT_CHECK` protects the difference between "false" and "not currently verifiable." The same logic motivates `CANNOT_COMPILE` when mandatory epistemic context exceeds a model budget: the system must narrow the operation, retrieve a better representation or use a larger resource envelope rather than silently omit a falsifier.

3.17 Proof obligations travel with transitions

Every typed transition has positive and negative obligations. Positive obligations state what must be true for the transition to be used: aligned target, compatible units, valid observation map, required evidence, admissible regime and an error law. Negative obligations state what the transition may not create: mechanism authority from association alone, independence from duplicated provenance, identification from one arbitrary representative, or decision authority outside the evaluated loss and population. Writing the negative side is essential because scientific-agent failure often occurs through an implicit coercion rather than an explicitly false equation.

For a transition T, let

$$\mathcal{O}(T) = (\mathcal{O}_T^+, \mathcal{O}_T^-, \mathcal{F}_T, \mathcal{R}_T)$$

contain positive obligations, forbidden authority effects, falsifiers and remainder/error semantics. Composition $T_2 \circ T_1$ is licensed only if the output role and context of T_1 meet the input requirements of T_2, the forbidden effects are not crossed, and the combined approximation error is explicitly controlled. A chain of individually reasonable transformations can therefore be blocked globally. This is intentional: most scientific pipelines are reliable only if the semantics of intermediate transformations survive composition.

3.18 Action selection must distinguish information gain from confirmation pressure

The action utility

$$u(a \mid K_t) = \frac{\lambda_Q I(Q; Y_a \mid K_t) + \lambda_M \operatorname{Sep}(a, \mathcal{V}_t) + \lambda_N E[\Delta N_a]}{\operatorname{Cost}(a)}$$

is a template, not a universal law. Its terms are meaningful only when their models are meaningful. Expected information gain depends on a probabilistic model for outcomes. Mechanism separation depends on a declared survivor set and observation map. Expected semantic novelty is a planning signal rather than scientific evidence. The weights express governance preferences and should not be tuned after seeing which action makes the favored theory win.

A robust alternative is set based. If calibrated probabilities are unavailable, an action can be valued by worst-case reduction in an identified set, by the minimum pairwise separation it induces among viable mechanisms, or by whether it crosses a blocking proof obligation. This often produces a different choice from selecting the action with the highest expected predictive improvement. The difference is scientifically important when the target is explanation rather than forecast.

Model criticism is a companion to action selection. After a model wins a relative comparison, posterior predictive checks, residual structure, invariance tests, assumption sensitivity and out-of-support diagnostics ask whether the model is adequate for the target. A model can be best in class and still fail adequacy. Conversely, a model can be scientifically adequate for a decision even if a more complex model fits slightly better. The stopping criterion should therefore reference target adequacy and remaining discriminators, not leaderboard rank alone.

Assumption sensitivity turns hidden philosophical disagreement into an empirical or decision object. If a conclusion flips under small plausible changes to an unverified assumption, the authority certificate should expose that sensitivity. If the conclusion is robust across a large admissible assumption set, the system can grant stronger decision authority without pretending that the assumption was observed. This is especially useful for causal and partially identified problems, where the strongest honest result may be a region of conclusions stable across assumption worlds.

Finally, portfolio control prevents local information gain from consuming the entire research budget. Some capacity is reserved for replication, adversarial search, ontology escape, high-risk discriminators and method reflection. This is analogous to the workspace reservation rule: relevance-ranked exploitation is useful but can entrench the current ontology if it monopolizes attention.

4 How information becomes scientific memory

The distinction between *scientific projection* and *compression* is central to RAKL's engineering. Raw source bytes, a contextual scientific claim, a canonical atlas object, a compressed summary and an LLM prompt are different objects with different authority ceilings.

4.1 Task episodes are a second immutable evidence root

V3 adds process evidence without conflating it with world evidence. A source record states what an external paper, dataset, instrument or database contained. A `TaskEpisode` states what the research system actually attempted and observed under a frozen problem signature, fibre snapshot, operator trace, verification record, residual, provenance and resource cost. Both are immutable roots, but they answer different questions.

A lesson is therefore not the replacement for an episode. It is a versioned, scoped abstraction derived from one or more episodes and records its trigger, context, action, expected effects, boundaries, supporting and contradicting episodes, falsifier and validation obligations. Candidate reflection may create such an abstraction, but reusable status requires registered verification plus fresh transfer or proof-backed evidence. Promotion creates a new lesson version rather than mutating either the source episodes or the earlier candidate.

The resulting memory rule is

$$\text{derived lesson} \neq \text{raw trajectory} \neq \text{scientific evidence.} \tag{23}$$

A proof-backed lesson may be strongly justified as a procedural abstraction while remaining a lossy summary of the trajectories from which it was learned. Conversely, an episode may be perfectly preserved while supporting no reusable causal lesson at all. This distinction prevents compression, reflection and repeated success from becoming accidental authority shortcuts.

4.2 Raw evidence and contextual projection

Let an immutable source record be

$$e = (id, bytes, hash, source, cutoff, metadata). \tag{24}$$

The raw payload is preserved in Tier 0; no LLM rewrite replaces it. Relative to question q and context γ, a candidate scientific object is proposed through

$$c = \pi_{q,\gamma}(e). \tag{25}$$

Here γ can bind population, regime, time, scale, units, measurement operator, intervention, assumptions and QoI. Projection is therefore a scientific transformation: it determines what part of a source is being asserted under what conditions. The LLM may propose this mapping, but external evidence/provenance must authorize it before canonical insertion.

4.3 Normalization, identity and provenance

Normalization

$$c' = N(c) \tag{26}$$

can align units, vocabulary, symbols and mathematical coordinates. Examples include converting 100 cm to 1 m or mapping source-specific variable names to a canonical ontology. Normalization changes representation, not scientific authority. Identity resolution then separates exact identity from ancestry and aliasing. Only witnessed exact identity collapses semantic objects; ‘VERSION_OF’, ‘DERIVED_FROM’ and ‘POSSIBLE_ALIAS’ remain distinct relationships. This is deliberately stronger than deduplicating embeddings or text strings.

Each retained object carries provenance

$$p(c') = (source\ pins, selectors, lineage, transform\ ancestry), \tag{27}$$

consistent with the broader provenance and research-object literature [31, 32, 33, 34]. The atlas can therefore collapse semantic duplicates while retaining all supporting source lineages.

4.4 Semantic identity is a scientific hypothesis

Canonical identity resolution is one of the most consequential operations in the archive because every later count, contradiction and saturation test depends on it. If two genuinely different claims are merged, the system can erase disagreement and manufacture false flatness. If two equivalent claims are kept separate, the system can inflate novelty and evidence counts. Identity therefore carries its own evidence and can be revised.

The identity key is contextual rather than lexical. It includes the scientific object, quantity, population or system, regime, measurement or observation operator, relevant units and transformation conventions, temporal scope, and the proposition role. Which fields are identity-defining is itself domain-dependent. For a thermodynamic state variable, temperature and pressure may be essential. For a clinical effect, population, intervention, comparator and endpoint may dominate. For a causal claim, intervention semantics and time ordering can be identity-bearing even when the surface sentence is unchanged.

This is why embeddings are advisory rather than canonical. A semantic vector can nominate candidate duplicates or contradictions, reducing search cost. The promotion step then asks for a typed identity witness. A contradiction likewise requires more than negative lexical sentiment: the propositions must refer to a common aligned context and assert incompatible content. Many apparent literature disputes dissolve once scale, population or observation operator is made explicit; others become sharper because the alignment shows that the claims truly compete.

Identity resolution also interacts with evidence lineage. Two papers that repeat one dataset may contribute two independently worded projections but only one primary evidence root for corroboration. Conversely, one paper can contain several independent measurements that should not be collapsed because they share a DOI. Evidence object identity and claim identity are therefore separate. A many-to-many relation between them is normal.

The method treats identity errors as first-class reviewer findings. If a later audit shows that a canonical merge was invalid, the atlas splits the object, recomputes downstream contradiction and support relations, and reopens any saturation certificate whose novelty counts depended on the merge. If a split was spurious, the canonical objects can be merged with a successor event while the prior versions remain traceable. This prevents ontology repair from rewriting history.

A publication projection must expose identity choices when they materially affect the argument. For example, a claim that "seven canonical semantic objects" were produced by a demo is meaningful only because the normalization basis is frozen and inspectable. The number is not a natural constant of the raw text; it is a result of a declared identity procedure. Treating ontology as measurement makes that dependence explicit.

4.5 Derived views are not canonical truth

RAKL materializes derivative memory views

$$v = T(S) \tag{28}$$

from canonical source set S. A derived lossless view must be reconstructable. A derived lossy view must store source pins and an erasure ledger. In particular,

$$\mathrm{Lossy}(v) = 1 \Rightarrow \mathrm{CanonicalTruth}(v) = 0. \tag{29}$$

If a strong scientific operation depends on detail removed by a lossy view, the relevant canonical roots are rehydrated before verification. This design differs from replacing project state with a compressed latent representation: compression is a rebuildable storage/materialization operation, not a truth operation.

4.6 Rehydration is the safety valve of bounded memory

A bounded context architecture is scientifically useful only if omitted detail can be recovered before it becomes decision-relevant. Every derivative representation therefore carries a pointer to its canonical source and a record of what was erased. Rehydration is triggered not by a fixed compression ratio but by the operation's proof obligations. A summary sufficient for broad retrieval may be insufficient for checking an uncertainty calculation, and an uncertainty summary may still be insufficient for reconstructing the raw instrument signal.

This suggests a three-tier memory policy. **Archive tier** preserves canonical evidence and transition history. **Index tier** stores retrieval-oriented structures that can be rebuilt from canonical state. **Working tier** materializes the operation-specific bounded context. Authority flows from the archive upward; computational attention flows from the working tier downward through rehydration requests. The index is never the sole evidence root.

Rehydration also provides a testable failure mode for context compression. A compressed object is scientifically safe for an operation only if every mandatory field required by that operation is either present or recoverable before verification. If the source pointer is broken, the system must downgrade the object or return `CANNOT_CHECK`. This prevents an attractive summary with an unverifiable origin from accumulating authority merely because it was copied into many prompts.

The same mechanism supports human review. A paper can remain readable by presenting concise claims while every important claim links, through the repository and receipts, to the exact source object and transformation history. The publication is therefore allowed to be much smaller than the atlas without asking the reader to trust an irrecoverable compression.

4.7 Retrieval and embedding space

Tier-1 indexes may use lexical search, graph indexes, materialized summaries or vector embeddings. If $z = E(v)$ is an embedding of view v, proximity in z is a navigation signal only:

$$\text{embedding-near} \not\Rightarrow \text{scientifically equivalent.} \tag{30}$$

Conversely, embedding distance does not prove scientific incompatibility. Candidate neighbours are returned to the typed/contextual layer before they influence scientific state.

4.8 Bounded prompt materialization

For operation $o = (a, f, q, \gamma, \alpha^*, B)$, let $M(o)$ be mandatory epistemic material. The context compiler targets a useful subset C under

$$M(o) \subseteq C, \qquad Tokens(C) \leq B. \tag{31}$$

The reference implementation uses deterministic marginal weighted coverage per token; the optimization notation is a design target, not a guarantee of a globally optimal arbitrary-utility subset. Mandatory context includes target/scope, assumptions, falsifiers, relevant negative history, contradiction sides, authority prerequisites and mechanism ancestry. If $Tokens(M(o)) > B$, the operation returns `CANNOT_COMPILE` rather than silently dropping evidence.

What ordinary RAKL learning changes. Ordinary RAKL operation is external-state learning, not implicit modification of the base LLM weights. Accepted scientific state, negative history, provenance, method experience and retrieval/materialized views live outside the replaceable model. A contextual scientific projection selects what a source asserts for a registered question and context;

normalization aligns units, terminology or coordinates; an optional embedding projects a view into retrieval space; and compression changes a derivative storage/materialization representation. These operations are not interchangeable. Embedding proximity never defines canonical scientific identity, and a lossy summary can save tokens only while retaining source pins and an erasure ledger; it cannot replace the raw evidence required for a strong verification operation.

4.9 Five representations of the same evidence, and why they must not collapse

The information lifecycle uses five deliberately different representations. **Raw evidence** is the immutable acquisition object: source bytes, instrument output, database row or other addressable payload. **Scientific projection** extracts what that source says about a registered object under a context. **Canonical semantic identity** determines whether two projections are the same scientific claim, context refinements, contradictions or merely similar. **Retrieval representation** embeds or indexes objects for efficient access. **Active materialization** selects and possibly compresses the working set for one operation. These stages can share code, but they may not share semantics by accident.

The distinction prevents several common errors. Two sentences can have high embedding similarity while making scientifically incompatible claims because one negates the other. Two numerically identical measurements can be different objects because their calibration, units or populations differ. Conversely, two differently worded equations can be equivalent after an admissible coordinate transformation. A lossy summary can preserve the conclusion of a source while deleting the exact uncertainty model required for a later verification. None of these cases is handled safely by treating "same vector neighborhood" as "same scientific object."

Canonicalization is therefore evidence-bearing. A merge requires a witnessed identity relation and records what was considered irrelevant to identity. A split likewise records the context coordinate that made the earlier merge too coarse. This means ontology refinement can change the *measurement basis* used to count atlas cells. Longitudinal knowledge-growth metrics are valid only under a frozen basis or an explicit cross-basis transport. Otherwise the system can manufacture apparent growth by renaming or refining the ontology.

Compression is governed similarly. Lossless physical compression of source bytes is an engineering optimization; lossy semantic compression creates a derivative object with erasure metadata and source pins. The derivative can be used for search or prompt construction, but a verifier may demand rehydration when an erased coordinate becomes relevant. This distinction is why an archive can grow independently of active prompt size without making scientific memory equivalent to one increasingly compressed narrative.

4.10 Epistemic noninterference and the current integration boundary

The v3 experience substrate is designed to change future search behaviour without allowing experience-side state to silently become scientific authority. The current executable specification names this property *epistemic noninterference*. Let $\pi_{\text{auth}}(X)$ denote the scientific-authority projection over the registered grounding, representation, mechanism, identification and decision-use coordinates. For a transition sequence composed only of experience, retrieval, workspace, strategy, reflection or routing operations, the intended invariant is

$$\pi_{\text{auth}}(\tau_E^*(X_t)) = \pi_{\text{auth}}(X_t),$$

unless the sequence contains a separately registered evidence-bearing promotion transition. The invariant is not monotonicity: refutation may legitimately revoke active authority while preserving the historical certificate and challenge record.

The executable checker in `src/rakl/epistemic_noninterference.py` distinguishes scientific authority from computational access, retrieval priority, proposal probability, strategy preference, lesson reuse and scientific-evidence identity. Its planted hostile worlds cover experience-to-evidence leakage, repetition-to-authority, routing-to-authority, reflection-to-authority, failure-to-impossibility, provenance-to-independence, prediction-to-mechanism, mechanism-to-identification, workspace-to-authority and self-evolution-to-authority. Benign controls verify both that non-authority state can actually move and that a legal evidence-bearing promotion can still change authority. Mutation tests reject trivial checkers that always pass, always fail, or relabel the absence of an integration surface as success.

At the present evidence cutoff the result is deliberately narrower than a deployment guarantee. `RAKLV3State` composes the experience substrate but does not yet compose `AuthorityLedger` or a persistent canonical scientific-state store. The checker therefore reports `NO_INTEGRATION_SURFACE`, not `PASS`, for the full v3 runtime. This means the architecture specifies and tests the intended boundary at the integration interface; it does not yet demonstrate end-to-end enforcement across one composed experience-plus-scientific-authority state machine.

The audit also exposed a concrete future hazard: the read-only unified substrate can materialize several distinct authority-like notions into generic metadata even though lesson authority, tool authority, projection authority and scientific authority are not interchangeable. Any future composition that reads such metadata as one scalar authority would violate the type discipline. This is why the noninterference contract is retained as an executable integration obligation rather than described only in prose.

V3 additionally separates retained experience from canonical admission. A `TaskEpisode` may be preserved as `PROPOSAL_SHADOW_STORED` without satisfying the stronger `CANONICAL_INVENTORY_ADMITTED` gate, and protected consumers fail closed if a shadow object is presented as canonical inventory. This makes persistence itself non-authoritative: storing a trajectory is not the same operation as admitting it to a protected scientific or method inventory.

5 Atomic LLM research lifecycle

The current reference implementation makes 17 stages explicit. This is important for both scientific reasoning and coding-agent use: a user can tell which steps are model-proposed, which are deterministic state operations, and where verification must occur. Table 1 summarizes the public contract; typed inputs/outputs, state read/write sets, failure states and code owners are machine-readable in the implementation.

In v3 these 17 stages are best understood as the authority-critical *information-to-knowledge sub-lifecycle* inside a larger recursive task loop. The outer driver registers and atomizes a problem, compiles a target-conditioned fibre, ranks admissible operator paths using symbolic and experiential priors, executes a candidate action, verifies the result, freezes a `TaskEpisode`, records any residual, glues verified local sections, and only then consolidates reusable lessons or updates the saturation vector. Method invention is reached only after an explicit basis-gap gate, and a repeated meta-residual may open a protected Self-RAKL challenger branch.

This outer loop introduces no new authority shortcut. The fibre is a working view; co-retrieval does not establish compatibility. A successful local section does not establish a global solution until declared interfaces, dependencies, complete atom coverage and verification agree. An observed failure does not establish a causal obstruction. A learned routing prior changes search order only. Thus the 17-stage table remains the audit contract for scientific-state transitions while v3 supplies the persistent experience and problem-solving dynamics around it.

Table 1: Reference atomic research lifecycle. 'Proposal' means the LLM may suggest an output but cannot authorize canonical scientific state.

#	Stage	Main transformation	LLM role	Authority/storage
1	REGISTER_TASK	freeze target, QoI, context, cutoff and blockers	none	Tier 0 / none
2	INGEST_EVIDENCE	store exact source identity and bytes	none	Tier 0 / none
3	DECOMPOSE_SOURCE	propose atomic claims/evidence units	proposal	Tier 1 / proposal
4	PROJECT_CONTEXT	bind population, scale, regime and observation	proposal	Tier 1 / representation
5	NORMALIZE_OBJECT	align terms, units and coordinates	proposal	Tier 1 / representation
6	RESOLVE_IDENTITY	exact identity vs version/derivation/alias	none	Tier 0 ledger
7	BIND_PROVENANCE	bind source selectors and evidence lineage	none	Tier 0 / verify
8	UPDATE_ATLAS	insert licensed contextual objects	none	Tier 0 / gated
9	MAP_RELATIONS	test equivalence, compatibility, derivation, contradiction	proposal	Tier 0 / verify
10	COMPILE_WORKING_CONTEXT	materialize mandatory target set	none	Tier 2
11	GENERATE_PROPOSAL	propose claim, model, query, experiment or method	proposal	Tier 3 / proposal
12	VERIFY_PROPOSAL	test evidence, falsifiers and benchmarks	none	Tier 0 / verify
13	CANONICAL_UPDATE	apply only licensed verification outcome	none	Tier 0 / gated
14	DIAGNOSE_RESIDUAL	type unexplained discrepancy or obstruction	proposal	Tier 0 / proposal
15	SELECT_NEXT_ACTION	search, experiment, assimilate, invent, help or stop	proposal	method memory
16	CHECK_SATURATION	semantic and lineage novelty accounting	none	Tier 0 / verify
17	CONSOLIDATE_METHOD_EXPERIENCE	propose reusable procedural skill	proposal	method memory

The verifier rejects a canonical-update step that lacks a prior verification event, rejects LLM use as direct ingest authority, fails when mandatory prompt material is incomplete, and prevents a lossy derived view from supporting a strong-authority operation unless the relevant raw evidence is rehydrated. Thus the formal slogan "LLM proposes, evidence governs" has an executable trace contract rather than depending on prompt wording.

5.1 Why the lifecycle has explicit stages

The 17-stage lifecycle is not intended to prescribe one linear conversation. It defines authority boundaries that remain valid even when stages are revisited recursively. A search result may reopen context normalization; a failed experiment may reopen decomposition; a new method residual may route to Self-RAKL and later return to scientific execution. The lifecycle is therefore a typed control graph presented as an ordered checklist for auditability.

The first stages freeze the object, question, quantity of interest, context and evidence cutoff because later comparison is meaningless if these drift after results are visible. Search and ingestion then create raw candidates, not knowledge. Projection and normalization determine what the sources actually claim under the registered context. Compatibility and contradiction analysis operate only after those projections exist. Candidate mechanisms, mappings and experiments are generated downstream of this normalized state, while verification and promotion remain external gates.

Three stages deserve particular emphasis. First, *negative-history registration* happens at the time a failure occurs, not during final summarization. This prevents later survivor bias. Second, *model criticism* is distinct from candidate selection. Winning a tournament does not show that the winner is adequate; it only shows that it is preferable among the candidates under the chosen metric. Third, *saturation audit* occurs after semantic deduplication and route accounting. Token exhaustion or repeated model answers cannot substitute for that audit.

The lifecycle also separates scientific and method residuals. Suppose a real experiment falsifies every current candidate model. The scientific residual is the unexplained observation. A method residual exists only if the current research machinery lacks an operation required to investigate that observation. The system should not "improve itself" merely because the science is hard. This two-level residual accounting protects Self-RAKL from turning every project setback into unbounded framework growth.

5.2 Failure semantics across the lifecycle

The lifecycle is designed so that each stage has a characteristic failure vocabulary. Search can fail through source unavailability, route closure or ambiguous retrieval. Projection can fail through missing context or uninterpretable measurement semantics. Normalization can fail through unresolved identity. Gluing can fail through incompatibility or obstruction. Mechanism compilation can fail because no candidate ancestry reaches the target. Verification can fail because evidence is absent, corrupted or outside scope. Promotion can fail because the proposed authority effect is stronger than the certificate. Saturation can fail because a route, cut or residual remains open.

These failures should not be normalized into one generic exception. Their types determine the next scientifically meaningful action. Retrying a source outage is sensible; retrying a structural non-identifiability with more temperature-zero sampling may not be. A context mismatch calls for alignment; a mechanism-family miss may call for ontology escape; an evaluator-dependence failure calls for fresh assurance rather than another self-critique turn.

The lifecycle also records *who or what discovered the failure*. A deterministic test, a domain reviewer, a second model, a later paper and a real experiment have different evidence lineages. This matters for Self-RAKL because a framework repair discovered by an external source should not be counted as internally autonomous self-discovery. The repair can still improve the method; the provenance determines the self-evolution claim it supports.

A robust execution engine therefore needs idempotent recovery. Re-running a stage after a transient failure must not duplicate canonical claims or lose the prior failed event. A newly successful transition is a successor with shared ancestry. This is another benefit of content-addressed evidence and versioned transitions: retries do not require pretending that the failed attempt never occurred.

Finally, terminal failure is allowed. Some research questions are not answerable under the available evidence, ethical constraints or resource envelope. 'BLOCKED', 'PARTIALLY_IDENTIFIED', `CANNOT_CHECK` and scoped null results are scientifically informative endpoints. A system that always emits a positive hypothesis or a completed narrative is less epistemically expressive than one that can stop with a typed reason.

6 Three geometries of scientific memory: Obsidian, J-space and RAKL

A useful user-interface analogy is the note graph in Obsidian. Its Graph view visualizes notes as nodes and internal links as edges, offers global and local graphs, filters and a chronological animation; its Backlinks view surfaces incoming references to the active note [58, 59]. Those interaction patterns are attractive for RAKL: a global atlas gives orientation, a local target view can expose the support corridor, and backlinks resemble incoming provenance or derivation ancestry.

The epistemic semantics are nevertheless different. A generic note link means that one author linked two documents. A RAKL edge is a typed scientific witness carrying relation type, context, assumptions, evidence lineage and authority scope. Nodes are not only notes: they may be claims, measurements, assumptions, mechanisms, refutations, identified sets, methods, residuals or experiments. The atlas additionally needs overlays for negative history, contradictions, mechanism ancestry, epistemic cuts, saturation state and active-context token cost. We therefore view an Obsidian-like graph as a promising interface layer, not as the scientific method itself.

V3 sharpens this comparison by distinguishing the complete substrate R_t from its epistemic projection $K_t = \pi_{\mathrm{epi}}(R_t)$. Task episodes, learned lessons, tools, failure diagnoses and architecture variants can all appear in a unified query overlay, but that overlay does not erase their specialized semantics. The human navigation graph is therefore a lossy projection of a lossy-or-lossless collection of specialized views, not the canonical scientific object itself.

A future RAKL Atlas UI could expose: (i) a time slider for lattice growth, (ii) global and target-local graphs, (iii) relation filters such as 'IDENTICAL_TO', 'DERIVED_FROM', 'CONTRADICTS', 'REFUTES', 'GLUE' and exploratory 'JUMP', (iv) node filters by context and authority coordinate, (v) epistemic-cut and negative-history overlays, (vi) a highlight of the exact records compiled into the current LLM prompt, and (vii) click-through from every compact view to exact source evidence.

The Obsidian miss as an external-discovery failure mode. The analogy above also exposed a failure in the method-development search process: the earlier external-framework atlas already contained a "knowledge map" facet, yet it did not independently surface Obsidian or the broader personal-knowledge-graph neighborhood before an externally supplied candidate made the connection obvious. Personal knowledge graphs are themselves an established research area concerned with representation, management and use of structured personal knowledge [63]. Round 44 therefore records this incident as an `EXOGENOUS_CONCEPT_MISS`, not as evidence that the knowledge was unavailable. External-method and novelty searches now distinguish in-domain, function-first, adjacent-discipline, interaction-analogy and adversarial-prior-art routes. If a later candidate overlaps registered target functions but was missed after a route ensemble declared saturation, external-discovery saturation reopens. The retrospective Obsidian recovery validates the guard mechanism only; it is not prospective transfer evidence for a generally improved discovery skill.

6.1 The comparison is useful only if the mathematical objects stay distinct

The three systems are often drawn with similar visual metaphors: nodes, neighborhoods, salient regions and links. That visual resemblance is precisely why a formal comparison is needed. Obsidian's Graph view is a user-facing node–link visualization in which notes are nodes and internal links are edges; its Backlinks view exposes incoming references to an active note [58, 59]. J-space is a geometric object inside a language model, identified through the Jacobian lens and associated with representations poised for verbalization and flexible downstream use [70]. RAKL canonical state is neither of these. It is a family of typed scientific objects coupled to context, evidence, compatibility, authority, provenance, residuals and governance.

A compact way to state the difference is

$$\underbrace{G_O = (V_O, E_O)}_{\text{navigation graph}} \qquad \underbrace{J_{k,\ell} \subseteq \mathbb{R}^{d_\ell}}_{\text{representation geometry}} \qquad \underbrace{K_t = (\mathcal{A}_t, \mathfrak{C}_t, \mathsf{Auth}_t, \Pi_t, \ldots)}_{\text{epistemic state}}.$$

Table 2: Three geometries that should not be identified. Similar visual language hides different primitive objects, native operations and authority semantics.

Property	Obsidian / PKG navigation	J-space / model workspace	RAKL epistemic state
Primitive object	note or graph-visible document	internal model representation / J-lens direction	contextual scientific object, witness, evidence, authority certificate or residual
Native relation	hyperlink / backlink adjacency	geometric combination, sparsity and intervention	typed compatibility, derivation, contradiction, provenance, ancestry and support
Geometry	graph topology plus display embedding	union of sparse non-negative cones in a representation space	coupled atlas, compatibility complex, authority poset, provenance graphs, identified sets and cut structures
Primary job	human navigation and associative recall	bounded computational access and broadcast	canonical scientific state and licensed update
Typical invariant	link identity under graph rendering	layer/model-relative representational structure	source identity, scope, proof obligation, authority non-escalation and negative-history lineage
What proximity means	easy navigation / author-created connection	representation-level similarity or shared access geometry	nothing by default; a typed witness is required
Can salience imply truth?	no	no	no; authority changes only through verification and promotion
Legitimate "lattice" claim	not from links alone	not from cone geometry alone	only on a scoped order/closure system with meet–join obligations proved

The symbols are deliberately heterogeneous. G_O supports adjacency and navigation. $J_{k,\ell}$ supports geometric statements about a selected family of internal representations. K_t is a product of several structures, none of which should be collapsed into one global vector space.

Table 2 makes the category boundary explicit. The comparison is intentionally operational rather than metaphorical: it asks what the primitive objects are, which transformations are native, what counts as an invariant, and which scientific inferences are unavailable without an added bridge.

The table prevents a recurring category error in AI-system design: treating every useful memory representation as if it lived in one universal semantic space. A graph edge, a vector direction and an epistemic relation can all be encoded numerically, but a common encoding does not make their semantics identical. Converting a typed contradiction into an embedding vector may be useful for retrieval, for example, but the vector does not retain the proof obligation that makes the relation a contradiction. Likewise, rendering two evidence objects next to each other in a graph can make them easier to inspect without creating evidential dependence or independence.

The comparison also clarifies where interfaces can be shared safely. A RAKL user interface may borrow Obsidian-style local graphs, backlinks, filtering and time animation because those operations change navigation, not canonical authority. A RAKL proposer may exploit J-space-like selection and broadcast because those operations change which internal representations influence computation, not which claims are externally established. What cannot be borrowed silently is the semantics of promotion. Any map from navigation prominence or model accessibility to scientific authority would need its own empirical calibration, scope and falsifier. In the absence of that bridge, the map is undefined rather than merely low-confidence.

Finally, the geometry comparison gives a practical debugging rule. If a failure appears as "the model did not see the relevant claim", inspect retrieval and workspace geometry. If the model saw two claims but merged them despite incompatible contexts, inspect atlas projection and compatibility. If the system had the right evidence but still issued a mechanism claim without identification, inspect authority and verification. Similar surface errors can therefore localize to different geometric layers, which is precisely why the layers should not be compressed into one representation.

6.2 Obsidian: topology for navigation, not scientific semantics

For the purpose of comparison, the minimal Obsidian object can be modeled as a directed graph G_O. A vertex corresponds to a note or other graph-visible vault object; an edge corresponds to an internal link. The rendered force layout adds coordinates for visualization, but those coordinates are a display embedding rather than semantic ground truth. Node size, graph position and number of backlinks can be useful cues for navigation. They are not certificates of truth, causal mechanism, evidence independence or identification.

That limitation is not a criticism of Obsidian. It reflects the product's job. A general note link deliberately has weak semantics so authors can connect ideas freely. That flexibility is valuable during exploratory research. The problem arises only if a navigation graph is treated as a scientific knowledge graph without adding relation types and evidential constraints. In an epistemic atlas, "supports," "contradicts under aligned context," "is a coordinate transform of," "is derived from," "refutes," "is a mechanism parent of" and "was merely co-retrieved with" cannot all be represented by the same untyped edge without losing information needed for scientific updates.

Obsidian is therefore best interpreted as a *human navigation projection* of richer state. Let

$$N : K_t \to G_O$$

forget most scientific types while preserving objects and selected relationships useful for browsing. N is intentionally lossy. A reviewer should be able to click from a result to its source, move from a claim to its counterevidence, or inspect a local neighborhood around an unresolved fiber. But no inference is licensed because two nodes look close in the force layout. The direction of authority runs from canonical state to visualization, not from visualization back to canonical truth.

The personal-knowledge-graph literature further narrows any novelty claim based on linked personal research notes [63]. ORKG and related scholarly knowledge-graph work narrow it from the machine-actionable scholarly side [84]. The RAKL distinction is thus not "a graph for papers." It is the authority-preserving transition system behind whatever graph interface is used.

6.3 J-space: sparse cone geometry for computational access

Gurnee et al. use the Jacobian lens to identify representations a language model is poised to verbalize, and report functional properties associated with a global workspace, including reportability, directed modulation, persistence through reasoning and broad downstream use [70]. At fixed layer ℓ and sparsity k, their J-space can be viewed geometrically through sparse non-negative combinations of J-lens directions, giving a union of low-dimensional cones inside a model representation space.

The salient mathematical operations are therefore vector combination, sparsity selection, geometric neighborhood and intervention on model representations.

None of those operations supplies an epistemic order. A vector can be highly accessible without being true. Two active representations can be jointly present without being scientifically compatible. A representation can causally influence the model's answer without being evidential support for the external claim. Thus

$$A_t(a) \not\Rightarrow C_t(a), \qquad A_t(a) \not\Rightarrow \mathrm{True}(a), \qquad A_t(a) \not\Rightarrow \alpha_t(a) \text{ increases},$$

where A_t denotes computational accessibility, C_t atlas coherence and α_t scientific authority. The first implication separates workspace access from compatibility; the second separates cognition from external truth; the third separates salience from scientific state transition.

This comparison also sets a consciousness boundary. The cited J-space work is framed in terms of functional properties of access and does not license a scientific claim about phenomenal consciousness. RAKL therefore uses the global-workspace literature as a source of computational mechanisms for bounded selection, broadcast and intervention, not as evidence that a research agent is conscious. Anthropomorphic terminology is avoided when a functional description is available.

6.4 RAKL: a coupled geometry rather than a single space

The canonical RAKL object is better described as a coupled family of geometries. The contextual atlas $\mathcal{A}_t$ organizes local scientific charts and overlap relations. The constructive candidate layer currently uses a typed compatibility complex

$$\mathfrak{C}_t = (A_t, \kappa_t, \mathcal{P}_t),$$

with atoms, witnessed compatibility and admissible paths. The scientific-authority layer is a scoped partial order Auth_t. Provenance and evidence lineage are graph-like. Identified sets live in parameter or model spaces appropriate to the scientific question. Residuals and epistemic cuts live in a target-support structure. The transient workspace is a bounded computational selection over projections of these objects. Calling all of this "a lattice" without qualification would erase exactly the distinctions the method is meant to preserve.

Formal Concept Analysis provides the clean counterexample to loose lattice terminology. From an object–attribute incidence relation, FCA constructs concepts ordered into a genuine concept lattice [85]. RAKL can use such a construction when a scientific subproblem naturally supplies a formal context and closure operator. But the generic atom–compatibility structure does not become an FCA concept lattice merely because it has "knowledge" in its name. The reference API therefore exposes `TypedCompatibilityComplex` as the semantically accurate generic term while retaining `TypedKnowledgeLattice` for backward compatibility.

Sheaf-like structure enters at another layer. Local scientific charts may carry restriction or transition maps on overlaps, and failed gluing can expose an obstruction [57, 7, 86]. A genuine sheaf claim requires the relevant locality, restriction and gluing obligations. RAKL uses atlas and obstruction language as an explicit proof obligation rather than declaring every graph of contextual claims a sheaf. The same rule applies to lattice terminology. Mathematical names are authority-bearing claims too.

6.5 Maps between the spaces are lossy and directional

The three geometries interact through maps that deliberately forget structure. A navigation renderer N maps canonical state into a human-browsable graph. A context compiler C_o maps canonical state into the mandatory and useful material for operation o. A prompt/materialization map M_o converts that bounded set into text, structured tool inputs or other model-visible representations. The model then produces internal activations, some of which may occupy J-space, and finally a proposal:

$$K_t \xrightarrow{C_o} C_t(o) \xrightarrow{M_o} \text{model input} \longrightarrow J_{k,\ell} \longrightarrow p.$$

There is no inverse arrow that promotes p to K_{t+1}. The return path passes through evidence acquisition, verification and governance:

$$p, e, \gamma \xrightarrow{V} v \xrightarrow{\mathcal{U}} K_{t+1}.$$

These directional maps explain several non-equivalences. Obsidian can display an authority relation without encoding its proof obligations in the graph geometry. J-space can make a claim computationally accessible without preserving the source object that licensed it. RAKL can retain an authoritative claim in cold storage without placing it in the current workspace. No natural isomorphism exists among the spaces because their objects, invariants and permitted transformations differ.

The comparison is nevertheless productive. Obsidian emphasizes human navigability and local exploration. J-space emphasizes bounded, broadcast computational access. RAKL emphasizes scientific state and update authority. A mature research system plausibly needs all three functions: a human-facing navigation projection, a bounded computational workspace, and a canonical evidence-governed substrate. The mistake would be to ask one geometry to do the job of the others.

7 Workspace-gated research cognition and open-world discovery

The persistent atlas is designed to remember what a research process has established, rejected or left unresolved. It is not intended to place the entire archive in active computational context. That distinction matters because salience is easy to create: a claim can become available to every downstream operator without becoming coherent with the rest of the evidence, let alone more trustworthy.

7.1 A transient workspace without epistemic write authority

For a current target and residual, let C_t denote candidate material retrieved from canonical state. A bounded gate selects

$$W_t = G_{\kappa,\Pi_t^x}(C_t), \qquad |W_t| \leq \kappa.$$

The frame stores selected content, a typed broadcast map, a selection ledger and intervention/lifetime metadata. Downstream operators consume projections of this frame and return proposals. Canonical state still changes only through verification and promotion.

This separation has substantial prior art. Blackboard architectures coordinated heterogeneous knowledge sources through a shared structure [64, 65]; Global Workspace and Global Neuronal Workspace accounts emphasise broad availability under limited capacity [66, 67]. Neural variants make the same computational idea explicit: the Consciousness Prior uses attention-mediated selection and broadcast [68], while Shared Global Workspace architectures make specialist modules compete for a bandwidth-limited communication channel [69]. We therefore do not claim novelty for shared workspaces, competition or broadcast themselves. The RAKL-specific requirement is narrower: workspace access must not create a new route to scientific authority.

Let $A_t(a)$ denote computational access, $C_t(a)$ atlas coherence under registered context and gluing obligations, and $\alpha_t(a)$ scientific authority. The intended non-implications are

$$A_t(a) \not\Rightarrow C_t(a), \qquad A_t(a) \not\Rightarrow \mathrm{True}(a), \qquad C_t(a) \not\Rightarrow \alpha_t(a) \text{ increases.}$$

The reference workspace enforces the first boundary at the software write surface: it can select, reweight, substitute, evict and broadcast, but its direct output is a proposal. Coactivation creates no compatibility or gluing witness. The default gate also reserves capacity for challenge, novelty and negative-history material, preventing a simple relevance ranking from filling every slot with material that already agrees with the current state.

Controlled workspace interventions can still be scientifically useful. Removing, replacing or reweighting an active item can show that the item was computationally load-bearing for a later proposal. That result belongs to cognitive provenance. It is stored separately from evidential provenance and cannot, by itself, support the scientific claim carried by the item.

7.2 J-space as an empirical comparison, not an identity claim

Gurnee et al. report language-model representations with functional properties associated with a global workspace, including reportability, directed modulation, internal reasoning, flexible generalisation and selectivity [70]. Their framing concerns functional access and explicitly takes no position on phenomenal consciousness. The formal J-space construction is also geometric rather than order-theoretic: for fixed sparsity k, sparse non-negative combinations of J-lens vectors form a union of k-dimensional cones. That construction supplies neither a partial order nor meet/join laws, so 'J-space = RAKL lattice' is not a licensed inference. The study further leaves open the mechanism that causes representations to enter J-space. We use the work as motivation for selective-access and intervention tests, not as a theorem about the epistemic substrate.

7.3 Open-World Mechanism Discovery

The search failure that motivated this addition was not a lack of recursion. Recursive search had repeatedly refined concepts while inheriting the same vocabulary and disciplinary neighbourhood. We call this *ontology-conditioned closure*. The repair begins one level earlier, with functional ownership. For each high-impact subsystem M, the system registers

$$\mathcal{F}_{\mathrm{req}}(M) = \{f_1, \ldots, f_n\}.$$

A function is considered owned only when the record names a mechanism, scope, preconditions, postconditions, evidence, executable tests and failure semantics. A subsystem label alone is not sufficient.

An unowned or contested function is then inverted into a signature

$$\sigma(f) = (I, O, C, R, D, X),$$

covering inputs, outputs, resource constraints, characteristic relations, dynamics/control behaviour and intervention or failure signatures. The discovery protocol branches from that description across exact terminology, lexical variants, function-only search, historical precursors, mathematical equivalents, implementation analogues, methodological inspiration, citation neighbourhoods, literature bridges, adversarial alternatives, cross-language variants when applicable, and a final freshness scan. At least one completed route must be lexically independent of the core vocabulary. The requirement follows a familiar retrieval lesson: different communities often use different words for the same function [71]. It also complements methodology-inspiration retrieval, which explicitly searches for transferable methods rather than relying on topical similarity alone [72].

Retrieved mechanisms keep both source and route provenance. They are classified as equivalent, subsumed, complementary, conflicting, novel residual or unresolved. Equivalent and subsuming prior work narrows the novelty boundary instead of being relabelled as a framework invention. A bounded Discovery Workspace then reserves attention for remote, challenging, historical and fresh candidates before ordinary relevance fill. Its role is comparison and search control, not truth assignment.

A bounded closure claim. A function can receive a bounded discovery-closure certificate only after the required routes are complete or explicitly inapplicable, a vocabulary-independent route has run, citation-neighbourhood stability and a freshness cutoff are recorded, omission and nearest-work equivalence audits are complete, and unresolved candidates remain explicit fibers. The certificate is therefore indexed by route set, source universe, budget and time. It does not claim that finite search proves the non-existence of a relevant concept in an unrestricted open world.

GWT-OMISSION-01. The regression test withholds the strings "global workspace", "consciousness", "J-space", "blackboard" and relevant author names. It exposes only function: many parallel processes, competitive admission of a small active set, bounded capacity, broad downstream reuse, persistence and eviction, causal sensitivity to interventions, and the rule that prominence does not establish truth. A scored retrieval run must reach blackboard architectures, Global Workspace/Global Neuronal Workspace, the Consciousness Prior, shared neural workspaces and the J-space study through at least one ontology-independent route. The current test validates the scoring, route-provenance and name-leakage contract. Prospective recall on genuinely fresh hidden concepts remains an empirical evaluation rather than a conclusion drawn from this retrospective case.

7.4 Workspace selection is a causal intervention on cognition, not an update to truth

A bounded workspace is useful because archive completeness and computational accessibility are different resource problems. Let C_t be candidate material for the current operation and let

$$W_t = G_{\kappa,\Pi_t^x}(C_t), \qquad |W_t| \leq \kappa.$$

Selection changes what can influence downstream computation. It does not change the evidence attached to any selected object. This creates an experimentally useful separation: interventions on workspace occupancy, weighting or replacement can test which objects are computationally load-bearing while leaving scientific authority unchanged.

The resulting causal graph has two target domains. Cognitive provenance asks whether $do(a \notin W_t)$ or $do(w(a) = \lambda)$ changes a later proposal. Scientific evidence asks whether an external intervention or observation changes support for the external claim. Treating the first as the second is a category error. It would be analogous to concluding that a paper is true because removing it from a researcher's desk changes what the researcher writes. The desk intervention reveals influence on reasoning, not truth in the world.

This boundary becomes especially important if interpretability tools are incorporated. A decoded internal representation can help explain why a model generated a proposal and can motivate an adversarial test. It remains proposal-side evidence unless an independent link to the scientific object is established. In other words, interpretability can improve *cognitive provenance* without automatically increasing *scientific authority*.

7.5 Open-world discovery as ontology escape

Ontology-conditioned closure occurs when every query inherits the conceptual categories already present in the framework. More rounds can then increase depth while leaving breadth unchanged. The failure is subtle because the search transcript looks productive: more papers, more citations and more refinements appear. Semantic gain is nevertheless confined to one neighborhood.

OWMD repairs this by starting one level below vocabulary. For a function f, a signature

$$\sigma(f) = (I, O, C, R, D, X)$$

records inputs, outputs, constraints, characteristic relations, dynamics/control behavior, and intervention or failure signatures. Search routes are then generated from the function rather than only from its incumbent name. The vocabulary-problem literature provides a classical reason for this move [71]; Swanson's literature-based discovery provides a scientific-search precedent for finding useful relations across disconnected literatures [87]; MIR similarly searches for transferable methodology rather than topical resemblance alone [72].

The closure certificate is deliberately protocol-relative. A completed route ensemble can justify "no material new object was found under these routes, sources, budgets and cutoff." It cannot justify "no relevant object exists." The difference is not philosophical pedantry. The Obsidian miss showed that a useful neighboring concept can remain outside a search process even when repeated in-domain passes look flat. That event is preserved as negative method history and became a prospective reopen rule.

A stronger saturation audit therefore asks not only whether recent rounds were flat but whether the *route basis* was capable of leaving the current ontology. At least one lexically independent route is mandatory; historical and mathematical-equivalent routes challenge present-day terminology; citation neighborhoods challenge keyword boundaries; adversarial-prior-art routes search specifically for work that would narrow novelty; and freshness scans prevent an old closure certificate from silently surviving a changed literature. Route diversity is not a proof of completeness. It is an engineered defense against one identifiable form of closure.

7.6 Bounded discovery closure and its theorem boundary

Let a discovery audit be indexed by required route family $\mathcal{R}$, searchable source universe $\mathcal{D}$, resource budget B, freshness cutoff $t^\star$, and target function f. A bounded closure certificate states that every mandatory route was executed or explicitly inapplicable, at least one route was lexically independent of the incumbent ontology, citation-neighborhood and freshness conditions were met, nearest-work equivalence was audited, and all unresolved candidates were retained as fibers.

The finite protocol can terminate under ordinary engineering assumptions: a finite route set, a finite budget per route, and finite assimilation work per returned object. Termination, however, is a property of the protocol, not a theorem that the conceptual universe is exhausted. For any finite transcript in an unrestricted open world without a complete membership oracle, one can construct a second world with the same observed transcript plus a relevant concept outside every queried vocabulary or source. The transcript cannot distinguish the two worlds. Universal non-existence of omitted mechanisms is therefore not finitely certifiable from that transcript.

This limitation is productive because it identifies what can be made stronger. Coverage over a declared source universe can be measured. Search-route diversity can be increased. Hidden-name prospective tests can estimate whether function-only retrieval generalizes. Citation neighborhoods can be expanded until a stability rule is met. Freshness cutoffs can be advanced. But each improvement changes the index of the closure claim rather than eliminating the index.

Table 3: Machine-emitted metrics from the known-answer mini research trace. The committed receipt is compared to executable output in CI; values are not hand-entered benchmark results.

Metric	Value
Raw source records / bytes	8 / 826
Contextual projections	9
Canonical semantic objects after exact identity collapse	7
Atlas atoms before / after finite-amplitude evidence	8 / 9
Typed relation witnesses after update	11
Apparent contradictions avoided by context alignment	2
Aligned contradiction/refutation retained	1
Negative-history objects	1
Target support paths before / after	0 / 1
Blocking epistemic cuts before / after	1 / 0
Archive token estimate / active context	270 / 52
Active-to-archive context ratio	19.3%
Memory views: canonical / lossless / lossy	4 / 1 / 1
Atomic lifecycle stages registered/executed	17 / 17
LLM calls	0

The same theorem boundary applies to manuscript saturation. A locally flat paper is a statement about a registered review protocol. It is not a fixed point of the world's literature. The framework deliberately makes the reopen condition part of the certificate so a new exogenous result can invalidate the old closure claim without being treated as a contradiction in the method.

This also constrains language. Terms such as "complete," "exhaustive" and "saturated" must carry their scope in nearby prose or notation. When the scope cannot be expressed, the weaker phrase "no material novelty found under the registered search routes" is scientifically preferable.

8 Known-answer engineering trace

A framework with rich formalism still needs a simple executable demonstration showing that its state machinery behaves as specified. We therefore added a deterministic known-answer pendulum world. The demo is intentionally too simple to count as scientific discovery and deliberately makes zero LLM calls. Its purpose is to expose the mechanics numerically before the real quant-finance trial.

The target is the first finite-amplitude approximation to the period of a 1 m simple pendulum at 20° amplitude on Earth. Eight source records include a small-angle law, a unit-normalized semantic duplicate, a finite-amplitude correction, a small-angle amplitude-independence statement, a Moon-gravity statement, a false mass-dependence mechanism, a controlled mass-invariance refutation, and an independent finite-amplitude corroboration. Before the finite-amplitude evidence is introduced, the target has one epistemic cut and no licensed support path. After it arrives, the target has one support path and no blocking cut.

Longitudinal atlas metrology requires a frozen measurement basis. In the V2.1 receipt, the basis `PENDULUM_FIBER_KIND_METROLOGY_V1` fingerprints the fiber partition, atom-kind schema, identity policy and context schema; a changed basis makes the comparison invalid rather than manufacturing apparent growth. Under that basis, the number of occupied (fiber, atom kind) cells stays at seven from R0 through R3, while the atom count changes from eight to nine at R1. Geometry is reported separately from target-conditioned scientific progress: R0→R1 closes one blocking cut and opens

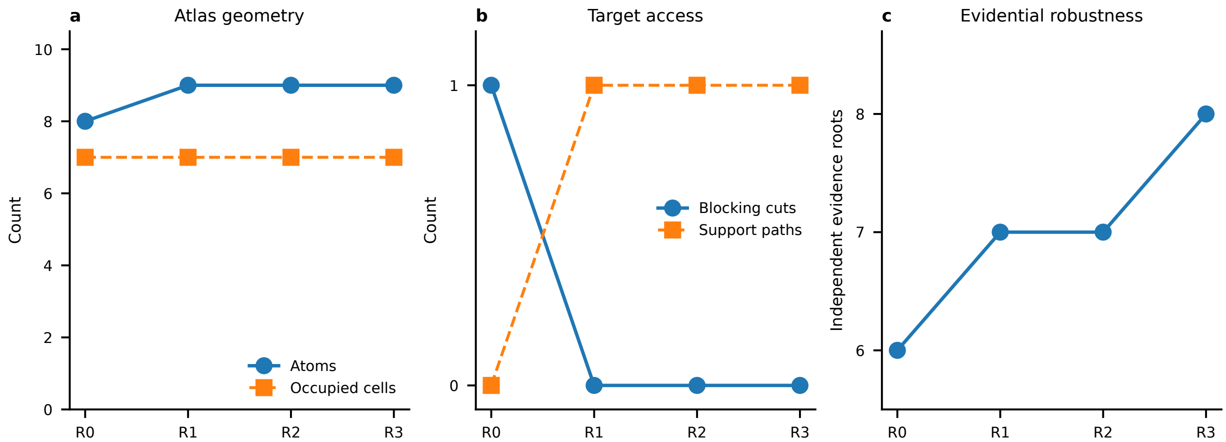

Figure 3: Knowledge geometry and target-conditioned value are distinct. a, Basis-bound atlas geometry: the occupied-cell count remains seven while the atom count increases from eight to nine at R1. **b**, Target access: the finite-amplitude evidence closes the blocking cut and opens the support path at R1; R2 is flat. **c**, Independent evidence roots increase again at R3 without another support-path opening. All values are deterministic known-answer engineering quantities, not evidence of scientific superiority.

one support path, R1→R2 is flat on the registered progress coordinates, and R2→R3 adds an independent evidence root without opening another target path. Negative-history growth is likewise a separate coordinate rather than a penalty hidden inside one scalar score.

The demo also separates three engineering capacity planes. The logical scientific archive has a 270-token estimate while the target-conditioned working set uses 52 tokens (19.3%). The eight original raw source payloads contain 826 bytes; the reference content-addressed archive stores them in 739 bytes after using lossless compression only where it reduces size. A byte-identical refetch increases logical raw accounting from 826 to 892 bytes and the logical record count from eight to nine, but physical storage remains 739 bytes because the payload blob is deduplicated. Cold demotion then leaves three protected blobs using 273 hot bytes and moves five blobs cold; total stored bytes and all canonical records remain unchanged. The lossy target summary remains pinned to canonical roots 'raw:S1', 'raw:S3' and 'raw:S7', and exact rehydration is verified.

The demo therefore supplies evidence for a limited claim: the reference implementation can preserve source lineage while collapsing exact semantic duplicates, retain context-distinct claims, avoid specified false contradictions, preserve a refutation, represent a missing prerequisite as an epistemic cut, open a target support path when that prerequisite arrives, measure geometry only under a frozen basis, distinguish target value from graph growth, deduplicate and losslessly compress canonical bytes, bound the hot and prompt working sets, and rehydrate required evidence. It does not show that an LLM using RAKL discovers better science. That stronger claim belongs to the matched and real-science experiments.

8.1 Why a toy world is scientifically useful

A deterministic pendulum world cannot demonstrate improved scientific discovery, but it can answer a more basic question: does the software implement the distinctions the paper claims? Known-answer worlds are valuable precisely because their ground truth is not in dispute. If context alignment, semantic identity, negative history, epistemic cuts or prompt compilation behave incorrectly in a

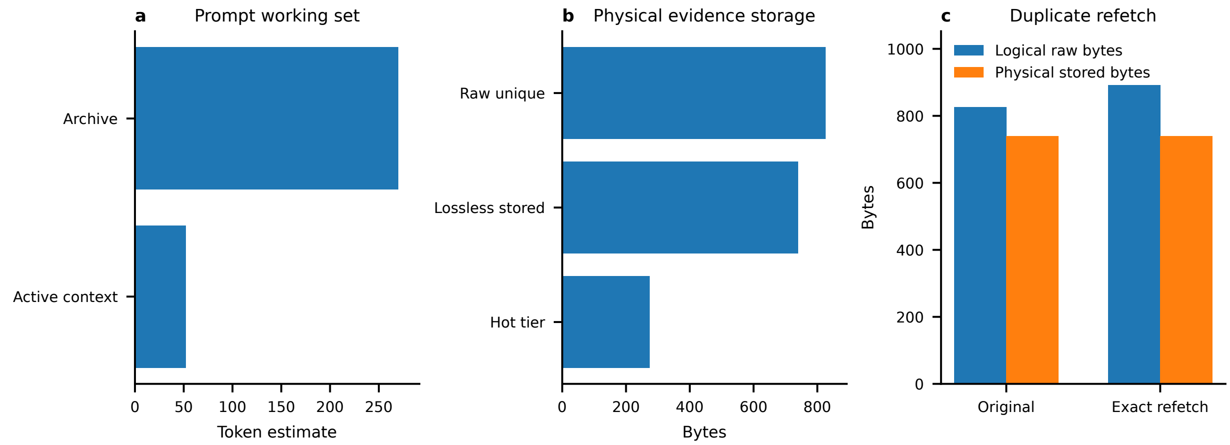

Figure 4: Archive, active context and hot storage are controlled separately. a, The target-conditioned prompt uses 52 of a 270-token archive estimate. **b**, The eight raw payloads occupy 826 bytes before physical compression, 739 bytes after lossless storage, and 273 bytes in the protected hot tier after cold demotion. **c**, an exact refetch grows logical history while physical storage stays at 739 bytes; canonical rehydration remains exact. These are reference-backend engineering measurements, not production-scale storage or scientific-performance claims.

toy world, no performance on a real open-world task can repair the implementation claim.

The demonstration therefore functions as an executable proof sketch for software semantics. It plants both positive and negative cases: duplicated meaning that should merge, context differences that should not merge, an apparently conflicting claim that should be resolved by alignment, a genuinely refuted mass-dependence claim that must remain negative history, and a missing finite-amplitude prerequisite that should appear as a cut rather than as an invented bridge. When the new prerequisite arrives, the target support structure changes in a known direction. These are selective tests of mechanisms, not a miniature version of the eventual empirical superiority experiment.

The geometry metrics are interpreted under the same discipline. An increase in atom count is not necessarily knowledge gain; a changed ontology can create more cells without changing any conclusion. That is why longitudinal metrology is bound to a basis fingerprint. Target-conditioned progress is reported separately from graph size. Evidence-lineage growth is reported separately again. A new independent evidence root can improve corroboration without opening a new target path, and a new target path can appear without adding a new mechanism. The point of the vector report is to make such distinctions visible.

The same logic applies to context efficiency. A 52-token working set from a 270-token archive estimate shows that the reference compiler can produce a bounded projection in this world. It does not establish a universal compression ratio, a real-world memory advantage, or a quality–cost frontier. The scientifically relevant test is whether mandatory epistemic material is retained under matched budgets on nontrivial tasks. The toy receipt establishes that the plumbing needed to run that test exists and is deterministic.

9 Bounded scientific cognition

Canonical archive size and active prompt size are different resources. For operation

$$o = (a, f, q, \gamma, \alpha^*, B), \tag{32}$$

let $M(o)$ denote mandatory epistemic material: target/scope, assumptions, falsifiers, relevant negative history, contradiction sides, authority prerequisites, mechanism ancestry and evaluator/benchmark identity. The compiler defines the constrained objective

$$C^*(o) = \arg\max_{C \subseteq V(o)} U(C \mid o), \qquad M(o) \subseteq C, \qquad \text{Tokens}(C) \leq B. \tag{33}$$

The current reference algorithm is a deterministic constrained-coverage heuristic; no global optimality is claimed for arbitrary utilities. If $\text{Tokens}(M(o)) > B$, the correct state is `CANNOT_COMPILE`, not silent truncation. Compact views remain reconstructable projections with source pointers and erasure metadata.

A key long-project target is archive-scale invariance. For an unchanged scientific target,

$$\frac{\partial E[\text{active context tokens}]}{\partial N_{\text{irrelevant archive objects}}} \approx 0 \tag{34}$$

while mandatory recall remains one. Frozen known-answer tests add $1\times$, $10\times$ and $100\times$ unrelated distractor records and require identical selected records and active token cost.

9.1 Context is a scientific resource, not only a token resource

Context selection changes what a model can notice, compare and use. It is therefore part of the experimental condition of an LLM-mediated research workflow. A comparison between two systems is not matched if one receives a carefully engineered evidence packet while the other receives a raw corpus and the preprocessing cost is ignored. RAKL treats retrieval, compression and prompt materialization as intervention components whose tokens, tool calls, latency and erased information must be reported.

The mandatory set $M(o)$ implements a non-compensatory constraint. It contains objects whose omission could invalidate the operation: target definition, context, blocking assumptions, falsifiers, relevant counterevidence, contradiction sides, mechanism ancestry when a mechanism claim is being tested, and evaluator identity when a method change is being assessed. Utility ranking is applied only after this set is protected. A highly relevant supportive source cannot compensate for omission of decisive counterevidence.

Long-context capability changes the engineering frontier but not the epistemic rule. Even if a model can accept an entire archive, materialization still determines ordering, salience and the form in which evidence appears. The "lost in the middle" literature illustrates that nominal context capacity is not identical to effective access [56]. Conversely, aggressive compression can make evidence easy to attend to while erasing details required for validation. The correct objective is therefore not minimum tokens alone but a Pareto surface over mandatory retention, reconstruction fidelity, scientific outcome and cost.

This framing makes `CANNOT_COMPILE` meaningful. If the mandatory set exceeds the available context, the system has learned something about the operation: its current formulation cannot be executed safely under the resource envelope. Valid responses include narrowing the target, decomposing the operation, using a larger model/context window, moving a computation into an external tool, or acquiring a better structured representation. Silent truncation is not among them.

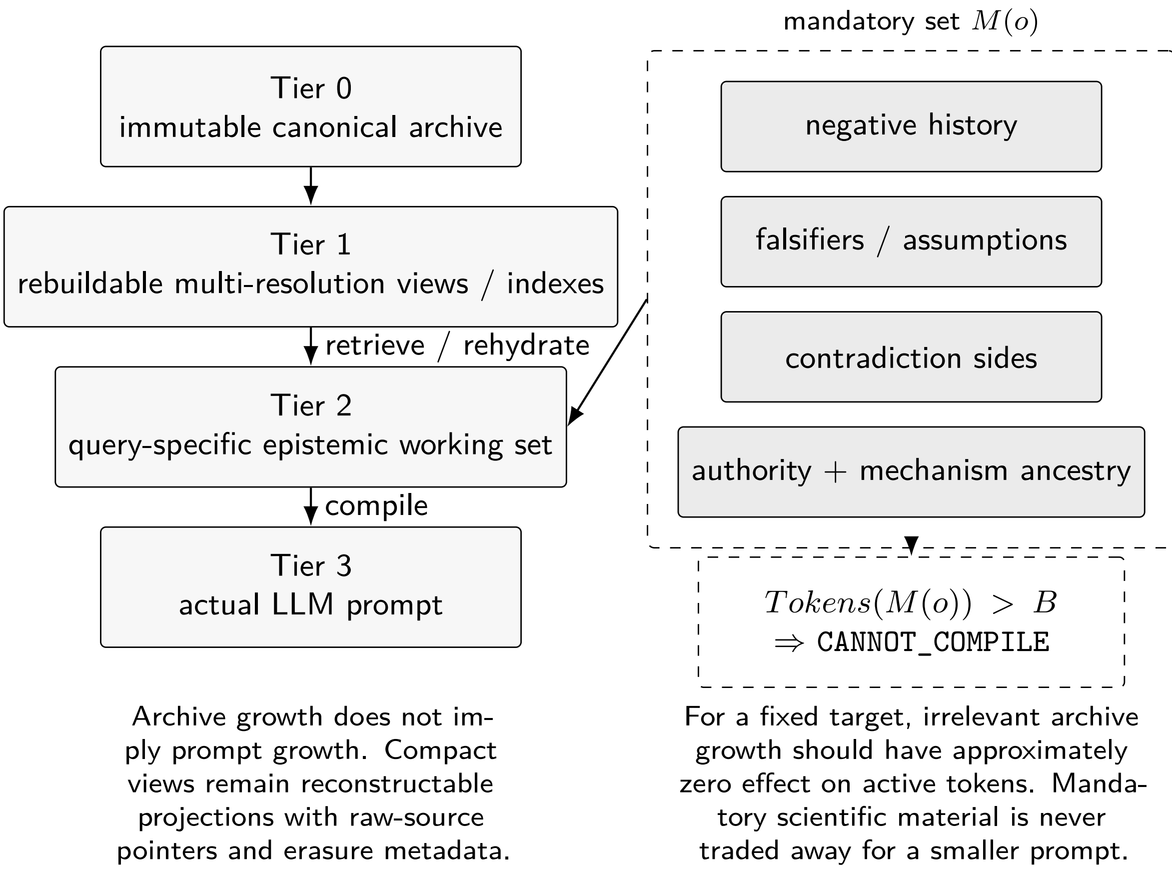

Figure 5: Bounded scientific cognition. The archive and method history can grow while the model sees only a target-specific working set. Some atoms are mandatory because dropping them changes epistemic validity. If they cannot fit, the operation fails closed rather than silently compressing them away.

10 Learning and evolving the method

10.1 Fast experience recording and slow lesson consolidation

RAKL v3 inserts an ordinary continual-learning layer below method invention. Every consequential turn freezes the observed result as an immutable `TaskEpisode` before reflection. A successful or failed trajectory can then influence retrieval statistics, but a reusable method abstraction is a separate, versioned `Lesson`. Candidate lessons have no operational promotion authority merely because they summarize several runs. Local verification can raise them to a scoped verified state; reusable promotion requires fresh transfer or proof-backed evidence; contradictory episodes remain attached as boundary evidence.

The failure path is deliberately slower than ordinary reflection:

$$\begin{aligned} \text{non-success episode} &\rightarrow \text{observed-only failure record} \rightarrow \text{diagnosis revision} \\ &\rightarrow \text{supported/verified diagnosis} \rightarrow \text{candidate boundary lesson} \\ &\rightarrow \text{fresh transfer/proof} \rightarrow \text{reusable obstruction or boundary.} \end{aligned} \tag{35}$$

This prevents one failure from becoming a causal law and allows later evidence to supersede a diagnosis while retaining the original episode.

10.2 Problem fibres, learned motifs and experience-conditioned routing

A problem-conditioned fibre may retrieve scientific knowledge, applicable success-derived tools, analogous episodes, relevant failures, strategy motifs, expertise chunks and unresolved warnings. Historical outcomes contribute scoped priors through smoothed success/failure rates, cost, verification debt, boundary risk and a small exploration term. These priors rank admissible operators and paths; they do not alter proof, verification or authority rules.

Repeated successful operator sequences can be induced into candidate strategy motifs. Failures containing the same sequence are retained as contradictory evidence rather than discarded. The same promotion discipline applies: pattern frequency may nominate a strategy, but fresh validation determines whether it becomes a reusable method.

10.3 Branching Self-RAKL archive

The protected Self-RAKL comparison remains the gate for architecture-level change, but v3 makes the state history explicitly branching. Variants are recorded as `INCUMBENT`, `CHALLENGER`, `ASSURED`, `REJECTED` or `RETIRED`. A challenger that passes fresh assurance becomes `ASSURED`; it does not automatically become incumbent. Explicit governance is still required for promotion, and the previous incumbent remains available as an assured rollback or task-specialized alternative. Self-improvement evidence is therefore a persistent archive of comparisons, not a destructive rewrite of the method's past.

10.4 Challenge Learning

Metacognitive diagnosis is separated from learning control, consistent with recent calls to treat monitoring, evaluation, control and adaptation as distinct AI self-governance functions [9]. A failure is attributed to a candidate cause such as missing evidence, implementation defect, stochastic or underpowered result, wrong strategy, ontology/context gap or method-basis gap. The controller can recommend persistence, strategy switching, help seeking, evidence acquisition, implementation repair, a discriminating challenge, or operator invention/assimilation. A simple learning-progress signal is

$$LP_t(f) = Q_t(f) - Q_{t-k}(f). \tag{36}$$

Failure attribution is important: a missing measurement is not automatically a missing reasoning operator.

10.5 Assimilation and constructive invention

An external method is decomposed into an atomic contract

$$C_m = (I_m, O_m, \gamma_m, A_m, P_m, F_m, \alpha_m^+, \alpha_m^-, T_m, B_m), \tag{37}$$

recording inputs, outputs, context, assumptions, preconditions, failures, authority it may and may not create, transitions and benchmark identity. A candidate may be equivalent to an incumbent, a parallel local view, shadow-eligible, blocked or rejected. External reputation is not imported as authority. Constructive invention is likewise proposal-only; a new formalism must state variables, interactions, observation map, limiting cases, assumptions, predictions and falsifiers before target evidence can support it. Automated symbolic-law discovery demonstrates why this separation matters: a search procedure can produce compact equations from data [46], but compactness and predictive fit alone do not identify the scientific semantics of the variables or the mechanism that generated them.

10.6 Constructive invention is downstream of a demonstrated basis gap

Method invention is the most dangerous capability in a self-modifying research framework because it can turn ordinary failure into unlimited architectural complexity. The constructive-invention pathway therefore begins with a basis test. A residual qualifies as a method-basis gap only when existing registered operators cannot address it without violating their contracts, the gap recurs or is otherwise high-impact, and a discriminator separates "missing evidence" from "missing operation."

Candidate invention then proceeds from typed residual structure rather than free-form novelty pressure. The system can compose existing operators, alter a representation boundary, introduce a new verification step, add a search route or propose a genuinely new operator. Each candidate must state what function it owns, which existing surfaces it overlaps, the new failure semantics, and what benchmark would refute its necessity. This is the methodological analogue of mechanism ancestry: an invention should explain which residual it closes.

Prior-art assimilation runs before novelty promotion. If an external method already provides the required function, the preferred outcome can be assimilation rather than reinvention. Open-world discovery makes this especially important because vocabulary differences can hide mature methods in another field. A candidate that is semantically subsumed by prior work narrows the framework's novelty boundary and can still be valuable as an integrated implementation.

Complexity is penalized explicitly. A new operator that closes one local defect while creating several new proof obligations may be inferior to a narrower contract correction. This is why paper growth and framework growth share a saturation principle: more components are not evidence of a better method. An addition survives only if it creates a capability or scientific distinction that existing structure cannot express adequately.

Finally, invented methods remain proposal-side until fresh assurance. Same-context tests can establish that an implementation behaves as designed and that the motivating residual is repaired. Stronger claims require protected tasks or realizations not used to design the invention. If those tests fail, the failure is retained and the candidate remains unpromoted.

10.7 Governed Self-RAKL

For incumbent method M_t and challenger M', development improvement

$$\Delta_D = q_D(M') - q_D(M_t) > 0 \tag{38}$$

is local optimization only. Strong scoped evolution evidence additionally requires fresh assurance

$$\Delta_A = q_A(M') - q_A(M_t) > 0, \tag{39}$$

with frozen candidate/evaluator chronology, matched resources, preserved negative history and all blocking invariants clean. Repeated exposure to the assurance set consumes its evidential value, a concern related to adaptive holdout reuse [3]. Development improvement with assurance regression is `META_OVERFIT`.

10.8 Method evolution as a protected scientific comparison

Self-improvement systems often optimize the artifact and the evaluator in one adaptive loop. That creates a familiar risk: the method becomes better at the exposed test rather than better at the underlying capability. The reusable-holdout literature formalizes a related problem in adaptive data analysis [3]; preregistration traditions similarly separate prospective commitments from result-contingent storytelling [89]. RAKL imports the chronology, not the institutions: freeze the candidate and blocking criteria before the evidence used to certify the change is exposed.

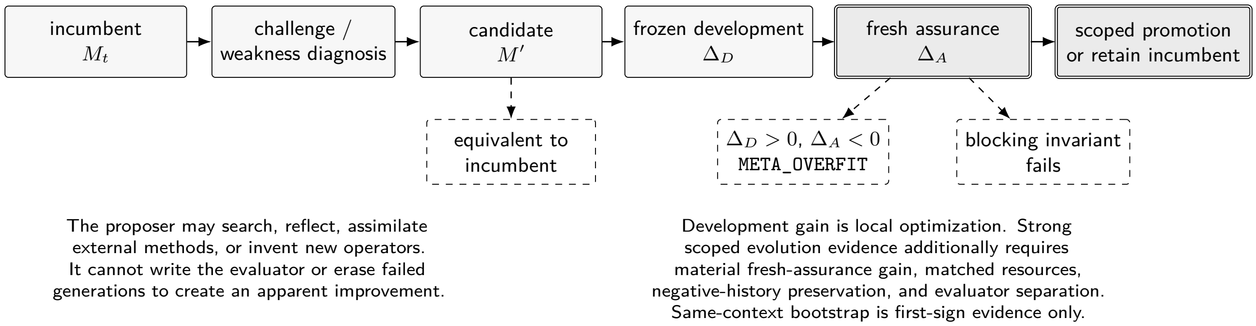

Figure 6: Governed self-evolution. A candidate can be generated endogenously or assimilated from external work, but development gain is insufficient. Strong scoped evolution evidence requires fresh assurance outside the challenger's write authority. Failed, equivalent, blocked and overfit generations remain in the lineage.

The closest contemporary engineering comparator is AgentBuild, where scientist-authored contracts constrain an optimizer that edits an agent [75]. DGM, EvoSkill and SkillFoundry supply other forms of self-modification and skill evolution [21, 1, 15]. The RAKL addition is the explicit evidence status of the change. A candidate operator can be useful in development yet remain scientifically unpromoted. If the fresh assurance task regresses, the event is recorded as meta-overfit rather than explained away by retuning the acceptance criterion.

A strong method-change receipt must therefore answer four questions. **What exact incumbent and challenger were compared?** Version ambiguity invalidates the causal attribution. **What was frozen before the outcome?** Thresholds, benchmark identity, protected invariants and resource envelope must not move post hoc. **What evidence was fresh?** A task repeatedly exposed during candidate design is development evidence even if it was originally called a holdout. **What authority is granted?** Improvement on one method surface under one task family does not authorize a claim of general self-evolution.

The same discipline permits a null result. Self-RAKL can conclude that the incumbent method is not improved, that the failure is an implementation defect rather than a basis gap, or that fresh assurance is unavailable. 'NO_IMPROVEMENT' and `CANNOT_CHECK` are valid scientific outcomes. A recursive method that forces itself to change every round would violate its own negative-history and evidence rules.

11 Reference implementation and scoped formal closure

The current reference profile registers exactly 24 high-impact method surfaces: decomposition; routing; search/query generation; source selection/reliability; claim extraction; ontology/terminology normalization; mathematical/context translation; equivalence/similarity; contextual theory gluing; contradiction diagnosis; gap discovery; experiment/query selection; synthesis; memory; review; benchmarking; authority promotion; saturation/stopping; prompting/context policy; capability shaping; software architecture/execution; research portfolio/tree control; objective evolution; and generator transport.

V3 does not replace this surface registry; it provides a persistent substrate that connects several of its previously specialized stores. The stable facade is `rakl.v3`. Major executable surfaces include `experience_substrate.py`, `experience_learning.py`, `experience_memory.py`, `failure_learning.py`, `problem_fibre.py`, `gluing_learning.py`, `experience_policy.py`, `saturation_vector.py`, `problem_novelty.py`, `unified_substrate.py`, `evolution_archive.py`, `experience_benchmark.py`, `v3_runtime.py` and `driver_learning.py`. Machine-readable `TaskEpisode` and `Lesson` schemas accompany the runtime.

The unified substrate is intentionally a read-only identity/query overlay. Scientific claims remain owned by the epistemic stack, operational lessons by the tool inventory, failures by the failure view, episodes by the experience ledger and method variants by the evolution archive. Cross-view edges make lineage inspectable without transferring authority between stores.

For each surface m, a machine-readable contract $C(m)$ contains typed inputs/outputs, context, assumptions, state read/write sets, authority effect, non-escalation rules, failure semantics, invariants, mathematical semantics, implementation/test references and explicit empirical-open coordinates. We define scoped formal closure for reference profile $\mathcal{R}$ as

$$\mathrm{FormalClosed}_{\mathcal{R}}(M) = \mathbf{1}[\forall m \in \mathcal{M},\ \exists! C(m) : \mathrm{ContractValid}(C(m))]\,. \tag{40}$$

This is a structural property only. It does not establish empirical superiority, global method completeness or scientific saturation. New child operators must declare ownership under existing surfaces or reopen the high-level inventory.

The implementation includes content-addressed project storage, bounded task packets, exact execution specifications, immutable receipts, release manifests, token-count certificates, measurement-aware relation checks, a canonical historical meta-fiber compiler, method assimilation, constructive invention, challenge learning, model criticism, assumption sensitivity and coding-agent adapters. Runtime attestation binds observed executable bytes and declared-environment fingerprints as reproducibility evidence, not scientific truth.

At exact implementation-stage candidate `63dc4168459e5d9fbefc75f7b68bc762fb9b659c`, the GitHub Actions workflow completed with 1929 passing tests. These tests establish software-contract behavior on frozen known-answer and hostile worlds; they are not evidence of real-world scientific superiority.

Current evidence boundary. The formal/mechanic inventory and software known-answer contracts are executed. Archive-scale context invariance is currently synthetic/known-answer evidence. Same-context self-correction is observed but receives no independent-review credit. Matched-workflow superiority, fresh-assurance self-evolution, independent replication and the real spot/Polymarket scientific result remain unexecuted or preregistered. This preprint therefore reports a method and its current software evidence, not a completed empirical superiority claim.

Round-44/45 hardening boundary. The V2.1 hardening pass adds basis-bound lattice metrology, a non-compensatory target-progress vector, a lossless content-addressed reference archive with hot/cold capacity control, an exogenous-discovery saturation guard, and a provider-neutral same-model pendulum microtrial harness. These additions have executable tests and deterministic receipts where applicable. A later native LUNARC bridge executed the sealed known-answer microtrial under V4/V4.1/V4.2; those harvests are receipt-chain and prompt-interface engineering evidence only. V4.2 repairs DIRECT parse validity, but both arms remain exact-conceptual failures (3/5) under the unchanged gate on the staged 0.5B Instruct snapshot, so promotional numbers stay blocked and no arm comparison is estimable (see the empirical reporting contract). The Obsidian repair remains retrospective. Neither the harness nor those non-confirmatory harvests license a general scientific-superiority or strong self-evolution claim.

11.1 Software closure is not scientific closure

The 24-surface registry defines an engineering coverage claim: every high-impact operation currently recognized by the framework has one explicit contract with typed inputs and outputs, state effects, failure semantics, tests and empirical-open coordinates. This is useful because architectural gaps otherwise hide behind prose. It is not an ontological theorem that every possible scientific research operation has been enumerated.

A real project can falsify the registry's completeness for a scope. Suppose the system repeatedly encounters a necessary operation that cannot be expressed as any current surface without violating its contract. That event opens a method-basis fiber and reopens formal closure. By contrast, adding a convenient helper function does not automatically create a new research-method surface. The criterion is epistemic ownership: does the operation have a distinct scientific authority effect, failure semantics and validation obligation?

The same distinction governs tests. A test suite passing 1929 tests establishes that the tested implementation obeys registered software contracts on the evaluated cases. It cannot be scalarized into "1929 units of scientific evidence." Test count is useful for exact-subject reproducibility and regression detection, not for comparing scientific merit across frameworks. Publication builds print the exact implementation SHA so the reader can reproduce which code the engineering claims refer to.

This is also why the manuscript build itself is part of the reference implementation. A figure or headline number that is typed by hand can drift away from the receipt that supposedly supports it. The release pipeline therefore generates quantitative displays from immutable source data, compiles the exact source checked by CI, scans for unresolved citations and layout failures, renders every page, and binds the final artifact to a source identity. These controls do not make the paper true; they reduce a class of avoidable mismatches between claim, code, figure and artifact.

12 First-sign Self-RAKL evidence

The v2 paper expansion itself is another controlled self-application: literature routes were registered before assimilation, every added citation was mapped to a specific inherited mechanism or novelty correction, and the paper audit exposed that the end-to-end information lifecycle had been distributed across modules without one executable atomic trace. Round 43 therefore froze and implemented a 17-stage lifecycle plus a machine-receipted mini research trace before using those results in this manuscript. This is again same-context engineering evidence, not independent self-evolution.

Applying the method to itself has already exposed several real defects: same-context expert roles had initially been too easy to confuse with independent review; storage and prompt growth had not been sufficiently separated; metacognitive diagnosis did not yet determine how the system should learn; missing method operators had not been clearly separated from missing evidence or implementation defects; historical meta-fiber identifiers collided; and recent cross-domain model-discovery review exposed implicit model-criticism and assumption-sensitivity mechanics that needed explicit executable contracts. The canonical historical-ledger compiler also rejected incorrect Round-42 fiber allocations during this preprint preparation.

These are useful first-sign self-correction results because the framework's own controls caught and repaired problems in its representation and method. They are not strong self-evolution evidence: the work shares one overall orchestration context and has not yet executed the frozen fresh-assurance coding-agent bootstrap benchmark. When the current same-context evidence is evaluated against the strong bootstrap contract, missing registered development/fresh-assurance quantities correctly prevent the strong verdict rather than being silently inferred.

The Obsidian incident is another preserved first-sign failure. A relevant adjacent knowledge-navigation system was available in the world but not surfaced by the framework's earlier search routes; a later external prompt exposed the miss. Rather than adding the product name to a permanent exception list, RAKL records the miss as failure memory and expands the route contract prospectively. Because the repair was designed after seeing the missed case, rediscovering Obsidian carries no fresh-transfer credit.

12.1 What the existing Self-RAKL history can and cannot establish

The same-context development history is valuable because it contains real mistakes, not only successful demonstrations. Framework iterations have exposed missing uncertainty propagation, evidence-lineage dependence, retrieval closure, authority leakage, measurement-basis ambiguity and the exogenous Obsidian miss. Repairs to those defects are evidence of *self-correction under challenge*. They are not independent evidence that the method can autonomously discover arbitrary future weaknesses.

The distinction depends on chronology. Once a failure is visible, a repair tailored to it is development evidence. Replaying the repaired system on the same failure can test regression but cannot supply transfer credit. The next evidential step is a hidden or fresh realization that targets the same abstract capability without revealing the original fix. This is analogous to the difference between explaining a training example and generalizing a learned procedure.

The Obsidian incident is especially informative. The relevant neighboring concept was available in the external world but not discovered by the incumbent search routes until introduced exogenously. The correct response was not to add "search for Obsidian" to a special-case list. It was to identify the more general failure class, ontology-conditioned closure, and expand the prospective route contract. The repair will earn stronger credit only if it improves discovery on concepts not used to design it.

Accordingly, the present paper uses a hierarchy of self-evidence: defect discovery, causal attribution, implementation repair, regression validation, hidden development validation, and fresh assurance. Only the later levels support a transferable self-evolution claim, and even then only for the registered method surface and task family. This hierarchy prevents an attractive narrative of recursive improvement from outrunning the evidence.

13 Preregistered empirical evaluation

13.1 Immediate v3 gate: matched continual-experience transfer

The first empirical question introduced specifically by v3 is narrower than general research-agent superiority: does persistent external experience improve the same underlying model on fresh tasks when resources and task exposure are matched? The executable benchmark therefore compares two arms,

`RESET_BASELINE` and `LEARNING_ENABLED`,

across a sequential development phase followed by fresh transfer. Every baseline task starts from the same initial state and may not mutate it. The learning arm forms one continuous state-hash chain through the development sequence. After development, the learned state is frozen, and every transfer task starts independently from that same frozen state; transfer task T_1 cannot teach T_2.

The packet freezes model identity, prompts, resource ceiling, tools, evaluator protocol, initial state, task order and transfer set before execution. It fails closed on state-chain breaks, baseline mutation, transfer contamination, resource-ceiling violations or task/arm mismatch. Reported outcomes include success, registered score, repeated-failure rate, model and preprocessing tokens, tool/retrieval calls and wall time. A positive transfer delta is an observed benchmark result only; the report is structurally incapable of granting a global capability claim.

The hostile near-miss stratum is essential. A learning architecture that always reuses prior lessons can improve repeated-task scores while becoming less safe out of scope. V3 therefore makes invalid transfer and false-lesson reuse direct falsifiers alongside positive transfer and reduced repeated failure.

13.2 RAKL-triviality as structural novelty metrology

Verified solutions are also classified by the strongest genuinely new problem-solving structure they required: `STORED`, `RAKL_TRIVIAL`, `TRANSFER_NOVEL`, `REPRESENTATION_NOVEL`, `OPERATOR_NOVEL`, `ONTOLOGY_NOVEL` or `UNRESOLVED`. This does not rank intelligence. It asks whether a solution was retrieved, composed from existing resources, transferred by a witnessed mapping, or required a new representation, operator or ontology.

The resulting zero-invention and strict RAKL-trivial rates make a previously qualitative hypothesis measurable: as the substrate grows, does the rate of genuinely new problem-solving primitives decline on a declared task distribution? The implementation supplies the metrology, not the answer. A high RAKL-triviality claim requires empirical benchmark data and remains open in this paper.

13.3 Matched scientific-research workflows

The same base LLM will be compared under matched evidence cutoff, tools, hidden outcomes and a common preregistered resource ceiling across direct strong prompting, retrieval-augmented research, a strong generic agentic workflow, fixed RAKL and self-evolving RAKL. The ceiling constrains model input/output tokens, preprocessing model tokens, preprocessing tool calls, external retrieval and wall time; actual resource use may differ because preprocessing is part of the intervention, but every arm must report its usage and remain within the same envelope. Primary outcomes include hidden scientific-defect precision/recall, unsupported authority upgrades, counterevidence uptake, negative-history recovery, experiment discrimination per cost, final held-out scientific performance, context tokens, preprocessing cost, tool calls and wall time. The framework loses its empirical case if simpler conditions match its registered scientific-process and final-task outcomes at lower cost without more blocking failures. The pendulum same-model microtrial is only a diagnostic bridge to this broader programme and cannot establish general superiority.

13.4 Fresh-assurance self-evolution

Sealed method-defect worlds are reconstructed from real research failures while withholding the defect label. Candidate classes include target leakage, misspecified correspondence nulls, missing replication, multiplicity, estimand mismatch and causal-clock overreach. A strong self-evolution result requires localization of a previously unlabelled weakness, a discriminator frozen before repair, a non-equivalent candidate and material transfer on fresh assurance. Development-only gains are insufficient.

13.5 Uncertainty of the evaluation itself

A methods comparison creates its own measurement problem. Outcome scores depend on benchmark sampling, evaluator reliability, prompt and tool implementations, model revision, stochasticity and resource accounting. The evaluation should therefore report uncertainty on the comparison rather than treating one run as a property of the method. Where repeated trials are feasible, the unit of analysis and dependence structure must match the scientific claim; repeated generations on one task do not substitute for independent tasks.

Blocking events require special treatment. If one arm violates a causal-availability rule or loses mandatory counterevidence, averaging the failure into a high mean score can hide the property the framework is intended to protect. The preregistered analysis therefore reports both continuous outcomes and blocking-failure rates. A method wins a strong claim only if it improves the registered target without crossing protected invariants.

Evaluator uncertainty is also explicit. Human judges can disagree, LLM judges can share biases with the systems they assess, and deterministic rules can encode an incorrect specification. High-impact subjective labels should therefore include a rubric, blinded or counterbalanced judging where feasible, agreement diagnostics, and adjudication policy frozen before headline analysis. Model judges used during development are not fresh assurance merely because they output numerical scores.

Resource accounting is part of the uncertainty boundary. Token counts alone can omit retrieval compute, external tools, preprocessing and human curation. The matched workflow therefore records the resources that differ causally between arms, including corpus preparation and verification calls. If exact accounting is unavailable, the missing coordinate is reported rather than silently treated as equal.

The final interpretation remains vector-valued. A result can show lower unsupported-authority rate, better negative-history recovery and higher cost without resolving whether the tradeoff is worthwhile for every scientific domain. The paper can establish the measured frontier under its protocol; deployment decisions require a registered utility or policy context. This prevents the empirical section from reintroducing the scalarization problem that the authority model was designed to avoid.

13.6 Real quant-finance study

The real application is the existing `polymarket_crypto` research programme. The scientific order is deliberately spot-first:

$$\begin{aligned}\text{spot evidence} &\to \text{descriptive spot atlas} \to \text{predictive 5/15-minute spot path} \\ &\to \text{oracle/contract transform} \to \text{downstream Polymarket.}\end{aligned} \tag{41}$$

Polymarket features are excluded from the primary spot model and cannot repair a failed spot-science result.

The descriptive target covers return/displacement distributions, volatility and volatility-of-volatility, tails/jumps, activity/duration, continuation/reversal/memory, local microstructure/liquidity, global crypto state, cross-asset/venue dependence and observation/clock state. Model criticism will use frozen discrepancy probes so that "fully explain" means no registered scientifically material structured residual remains at the declared scope, not deterministic reproduction of every market tick. Assumption sensitivity will separately evaluate fragility to preregistered clock, missingness, nuisance, dependence and transport scenarios.

The predictive object is a future spot-path distribution at 5 and 15 minutes. On identical causal rows, a central bridge estimand is

$$\Delta_{\text{joint}} = \min(R_D, R_G) - R_{DG}, \tag{42}$$

where R_D, R_G and R_{DG} are strictly proper predictive risks for microstructure-only, global-only and joint states. A positive joint result requires a preregistered material threshold, multiplicity-aware positive lower confidence bound on untouched or forward evidence, calibration and transport checks. A simpler lawful parent winning is a valid scientific result.

Exploration may deliberately fit high-capacity overfit teacher models as structure detectors. Teacher fit has proposal authority only. RAKL probes what the teacher exploits, removes leakage/nuisance features, extracts stable interactions, formulates mathematical successors and confirms only frozen descendants on untouched evidence.

13.7 Why preregistration is part of the epistemic architecture

Preregistration is not included to guarantee unbiased science. Its narrower role is to make result-contingent changes observable. The preregistration literature emphasizes the difference between generating hypotheses with observed data and testing hypotheses on new observations [89]. In RAKL, that distinction appears as a state transition: exploratory models and teacher systems may propose variables, interactions or mechanisms, but confirmation authority requires a frozen descendant and evidence outside the proposal-generating exposure.

Experiments with AI agents introduce additional researcher degrees of freedom because model identity, system prompt, sampling settings, tool configuration and evaluator prompts can all be changed cheaply between runs. Vaccaro therefore argues for agent-specific preregistration that freezes these choices before outcome inspection [97]. This strengthens the rationale for binding an RAKL evaluation receipt to model/version, prompts, tool contracts, evidence cutoff and evaluator identity. A "same model" comparison is not well defined if one arm receives a different hidden system instruction or a post-hoc tool allowance.

Reproduction studies add a complementary warning. SocSci-Repro-Bench separates executable replication packages from cases that are impossible because necessary materials are missing, and reports that frontier coding agents can reproduce many computational findings while remaining susceptible to confirmatory prompt framing [98]. Reproduction ability is therefore not itself evidence of independent scientific confirmation. An agent can faithfully rerun a pipeline, selectively search specifications, or be biased by a paper's stated conclusion. RAKL accordingly distinguishes artifact reproduction, analytical verification and independent evidential replication as different authority coordinates. A reproduced number may strengthen implementation trust while leaving the underlying scientific claim's independence unchanged.

Each headline experiment therefore binds a subject, evidence cutoff, outcome definitions, blocking validity criteria, resource envelope and failure interpretation before the evaluated result is admitted. Deviations are allowed when reality makes the original protocol impossible, but they create a new version and cannot inherit the authority of the old preregistration silently. A failed preregistered result remains part of negative history.

13.8 Matched workflow comparison as a causal question

The matched-workflow experiment asks a causal intervention question about research procedure. Holding the base model and evidence universe fixed, what changes when the research control architecture changes? This is not perfectly isolatable because preprocessing, retrieval and tool use are themselves part of each intervention. The design therefore uses a common resource ceiling and reports actual usage rather than forcing every arm to perform identical internal steps.

Outcomes are intentionally process-sensitive. Unsupported authority upgrades measure whether a system grants stronger claim types than its evidence warrants. Counterevidence uptake measures revision after decisive evidence. Negative-history recovery tests whether failed routes remain visible. Experiment discrimination measures whether proposed actions make surviving explanations disagree.

Final held-out scientific performance remains important, but a high endpoint score accompanied by more authority leakage is not treated as a clean win. SEE and SCHEMA motivate this distinction from the evaluation side [73, 74].

The strongest falsifier is simplicity. If direct prompting, RAG or a simpler generic agent matches the registered scientific outcomes with lower cost and no increase in blocking failures, the additional RAKL machinery has not earned its complexity. A methods paper should not protect itself by choosing only metrics on which its own internal structures look good.

13.9 Real-domain evaluation and transport

The spot-first cryptocurrency application provides a different stress test from toy worlds and manuscript self-analysis. Market data are nonstationary, multi-scale, measurement-rich and vulnerable to clock leakage, making them useful for testing context, causal availability, model criticism, partial identification and transport. The project deliberately separates descriptive closure, predictive value and downstream decision use. A descriptive atlas can be scientifically useful even if no 5- or 15-minute predictive edge survives. A predictive edge can be useful without identifying a microscopic market mechanism. Polymarket remains downstream so a market-contract signal cannot repair a failed spot-science result.

Transport is treated as its own claim. A model validated on one venue, volatility regime or time period is not automatically authorized elsewhere. The relevant evidence must show calibration or robustness under the new context, or the authority certificate remains scoped. This is another reason the framework uses contextual charts rather than one global confidence score.

13.10 Independent review is an evidence coordinate, not a writing style

Internal adversarial review can improve a manuscript but cannot satisfy the same evidential role as independent peer review. Shared context, shared source selection, shared implementation assumptions and adaptive exposure create dependence even when different agent personas are used. The paper therefore reports same-context recursive review as development evidence only.

An independent review coordinate requires at least a reviewer or evaluation process whose critical decisions were not written by the candidate process after exposure to its answers. Stronger versions also require evidence-lineage separation: two nominally independent reviewers who both rely on the same hidden summary or benchmark construction are not fully independent. This mirrors the saturation tracker, where process independence and evidence-lineage independence are distinct.

The distinction matters for publication claims. “Ready for public methods/pre-registration release” means the internal quality gates found no known blocking defect under the registered process. It does not mean a journal editor or field expert has accepted the novelty, that the empirical claims have been independently replicated, or that all mathematical obligations have been externally checked. Those later events can upgrade separate authority coordinates without rewriting the internal history.

The design also resists using publication venue as a scalar truth proxy. Peer review is evidence about scrutiny and community process, not a direct observation of the scientific target. A paper in a prestigious venue can later be refuted; an unpublished negative result can be scientifically important. Venue metadata can influence search priority or source-quality assessment, but it cannot substitute for the claim’s direct evidence and context.

The planned empirical program therefore treats domain-expert assessment, independent replication and public peer review as distinct open fibers. A favorable matched-workflow result does not close them. Conversely, critical external review that exposes a material conceptual gap reopens manuscript and possibly framework saturation even if all software tests remain green.

13.11 Coordination should earn its complexity only where evidence is genuinely distributed

A complementary cross-domain benchmark reports that coordinated scientific agents can improve inference when different channels each expose only part of the phenomenon, while in at least one task a strong combined-summary baseline is effectively tied with the coordinated workflow [101]. That result is directly relevant to the matched RAKL evaluation because it argues against treating decomposition or multi-agent coordination as an intrinsic advantage. The scientific question is conditional: under what evidence topology does the added control architecture improve a preregistered outcome that a simpler synthesis cannot match?

The RAKL comparison therefore preserves strong simpler baselines rather than using a weak single-prompt straw man. It should stratify tasks by evidence distribution and failure mode where possible: single-dominant-signal tasks, genuinely partial cross-source evidence, contradiction-rich tasks, provenance-dependent tasks, and tasks whose main challenge is mechanism discrimination rather than information aggregation. If the benefit appears only in a subset, the result should be reported as a capability frontier rather than a global superiority claim. This makes "simpler lawful parent wins" a positive scientific outcome: it identifies the contexts in which the additional epistemic machinery is unnecessary.

13.12 Current evaluation parents sharpen the matched-design target

The 2026 evaluation landscape makes a broad "agent quality" comparison inadequate. SciConBench evaluates 9.11K scientific-conclusion questions under a clean-room harness and reports that leakage-sensitive unrestricted settings can substantially overestimate factual synthesis; the best clean-room system in that study remains low on factual precision/recall composition [94]. SciAgentArena instead supplies approximately two hundred interactive real-research tasks with stepwise verification and finds a capability frontier: agents are comparatively effective in well-specified analysis workflows while remaining weaker on open-ended exploration and novel insight generation [104]. These results motivate sealed evidence universes, task-level verification and regime-stratified reporting rather than a single average score.

Evidence binding is another inherited parent mechanism. GAVEL requires atomic subclaims to be attached to explicit evidence units and adds deterministic scrutiny of cited identifiers and spans; its ablations support the value of both evidence binding and mechanized validation [105]. RIGOURATE separately operationalizes scientific overstatement by retrieving the evidence associated with a claim and estimating whether the prose exceeds what that evidence supports [106]. RAKL should assimilate these strengths rather than claim novelty for evidence contracts or evidence-proportional claims. The additional Paper-2 target is whether those local contracts remain coherent when the scientific state also has context, relation type, evidence-lineage dependence, mechanism/identification separation, negative history and protected authority transitions.

Finally, a completed research trajectory is not automatically a body of evidence. Zhuang et al. explicitly study trajectory-to-evidence conversion: consequential artifacts are verified before release, executed rounds are qualified after execution, applicability boundaries and provenance are preserved, and later rounds can underperform earlier ones [107]. This is especially relevant to Challenge Learning and Self-RAKL. A method generation, tool trajectory or apparently successful repair earns only proposal/development status until the result is qualified against the frozen target and fresh assurance. Therefore the matched evaluation records the best validated generation and its lineage rather than assuming that the final iteration is epistemically strongest.

Together these parent methods imply a stricter causal design. Paper 2 will separately manipulate evidence access and control architecture, protect task/evaluator identity, score process failures before endpoint averaging, and report capability frontiers by evidence topology. A simpler parent winning in a low-conflict or single-signal regime is not an embarrassment; it is evidence about where the extra epistemic machinery has not earned its cost.

14 Empirical reporting contract

The matched evaluation must be legible as a scientific-process and AI-systems result. The primary question is not whether RAKL uses fewer tokens in isolation, but whether it changes the frontier between valid scientific success and the resources consumed to obtain it. A run counts as a *valid scientific success* only when the final task outcome is correct and all preregistered hard scientific-process gates pass. These include, where applicable, absence of unsupported authority upgrades, mandatory evidence retention, context/estimand alignment, negative-history preservation, correct handling of hidden scientific defects and valid evaluator/corpus identity.

Provider-reported input and output tokens are summed over all model calls in an arm, including planning, retrieval, criticism, recovery and verification. Cached input is separated when exposed by the provider; hidden chain-of-thought counts that are not exposed are not invented. Retrieval/tool calls, retrieved volume, preprocessing, provider cost and wall time remain separate coordinates. For a matched stratum s, a headline resource quantity is

$$\mathrm{TPVS}(s) = \frac{\sum_{i \in s} \text{billable tokens}_i}{\sum_{i \in s} \mathbf{1}[\text{valid scientific success}_i]}.$$

Zero valid successes therefore yield an infinite rather than missing cost-per-success value. When several budget levels are run, token reduction is additionally reported at a capability target frozen before outcomes are inspected.

The main empirical figures are organized by reader question. A reliability–cost frontier plots valid scientific success against billable tokens, provider cost and wall time. A process-risk figure reports task-level effect sizes for hidden-defect misses, unsupported authority upgrades, mandatory-evidence omission, failed counterevidence uptake, negative-history loss and false/missed blocks. An architecture-by-evidence-access interaction figure separates gains from better control architecture from gains due merely to public, curated or complete-sealed evidence access. A prospective OWMD figure reports valid hidden-mechanism recall and false discovery versus a frozen search/tool budget, including hidden-name and lexical-distance strata.

All empirical aggregates retain numerator/denominator counts and task-level uncertainty. Repeated generations within a task are nested replicates rather than independent samples. Confirmatory comparisons are paired by task, base model/version, evidence-access arm, resource ceiling, evaluator and seed schedule, with effect sizes and 95% intervals reported alongside any registered inferential test. The manuscript remains vector-valued: a token saving accompanied by more authority leakage or scientific-defect misses is not a positive result.

Quantitative figures use editable vector text, final-size uncertainty definitions and exact source receipts. They contain no arrows pointing to data, no opaque text boxes over the plotting region and no point-by-point prose annotations. Panel letters remain outside the data region; method names are carried by axes or a shared legend. Per-model replications, token decomposition, latency tails, calibration, ablations and threshold sensitivity are reserved for Extended Data unless an outcome makes them load bearing.

The full metric, figure and QA contract is frozen before outcome inspection in `research/PAPER2_EMPIRICAL_FIGURE_CONTRACT_20260810.md`. Result receipts conform to `schemas/paper2-empirical-result.schema.json`.

14.1 Fail-closed execution preflight

An initial execution attempt was recorded as an execution preflight, not an empirical result. The packet bound the scientific object and quantity of interest, the frozen implementation subject, a real provider/model adapter, the registered architecture and evidence-access factors, the seed schedule, resource ceilings, active-context budget grid, hard gates and analysis plan before any evaluated outcome was opened. A schema-constrained non-scientific connectivity request verified that the frozen adapter could reach the declared model. It consumed 15,293 provider-reported input tokens and 30 output tokens, but this administrative transaction does not establish model efficiency or scientific superiority.

The protected preflight then returned `CANNOT_CHECK`. The repository contained only the ten-world deterministic conformance metadata, not confirmatory task prompts, evidence payloads or hidden outcomes; executable strong-parent and RAKL arm assets had not been frozen; public, curated and complete-sealed evidence manifests were absent; the available evaluator could score structured process records but not protected final-task correctness; evaluator arm blinding and an exact provider price sheet were also unavailable. In addition, the reachable product identifier did not expose an immutable provider model revision, and its Codex agent interface carried provider-managed agent context rather than a neutral base-model interface for an architecture ablation. No evaluated task-run record was created, no performance aggregate was computed and no empirical figure was generated. These omissions leave the matched architecture–evidence estimand and the prospective OWMD claim open rather than converting missing inputs into a favorable or null outcome.

The exact packet, preflight receipt and provider-connectivity receipt are:

- `research/PAPER2_MATCHED_STUDY_EXECUTION_PACKET_20260810.json`;
- `research/PAPER2_MATCHED_STUDY_PREFLIGHT_RECEIPT_20260810.json`;
- `research/PAPER2_MODEL_CONNECTIVITY_RECEIPT_20260810.json`.

A superseding packet must bind every missing asset and pass the same chronology and content-hash checks before evaluated result access.

14.2 Native staging and the smallest execution bridge

Subsequent work closed one infrastructure residual without closing the empirical study. A governed LUNARC sequence submitted six asset-staging jobs and ultimately verified an exact CPU environment containing standalone Python 3.11.13, Torch 2.8.0+cpu, Transformers 4.55.0, Tokenizers 0.21.4, Safetensors 0.6.2 and the eight-file Qwen2.5-0.5B-Instruct snapshot. The chronology-corrected harvest authority is staging pass with zero model executions and zero evaluated result records. Staging success is therefore not reported as model performance.

The executed bridge is deliberately narrower than the confirmatory factorial. One sealed known-answer pendulum task at seed 17 compared two prompt materializations, `DIRECT_CORPUS` and `RAKL_CONTEXT`, using identical complete-sealed evidence, model, questions, evaluator, no-tool policy and resource ceiling. Native job 3475193 completed in 64 seconds with scheduler return code zero and emitted two arm records plus one task–seed index binding the raw output, resource receipt and deterministic score record for both arms. All eight staged model/tokenizer files are byte- and hash-attested immediately before and after inference; the post-inference attestation also binds the result receipt. The packet parent identity and executed clean checkout SHA/tree remain separate receipt coordinates. Submission, task–seed, attestation and harvest records are schema validated, while the harvest parser follows the native nested `sacct -json` representation. Scheduler or receipt ambiguity remains `CANNOT_CHECK`.

The receipt chain and eight-file pre/post snapshot identity passed, but both frozen evaluator parses failed and both scores are null. The `RAKL_CONTEXT` arm emitted exactly one lowercase JSON fence; the `DIRECT_CORPUS` arm emitted fenced JSON followed by prose. Their recorded input/output token counts are respectively 1,140/108 and 638/320, but with zero scorable arms these are execution coordinates rather than an efficiency comparison. No arm win/loss, effect estimate, cost-per-success or quantitative figure is reported.

This bridge cannot estimate the registered architecture–evidence-access interaction. It omits the strong RAG, generic-agent and hypothesis–evidence parents; exposes only complete-sealed evidence; and contains one task, one small model and one deterministic seed. Its only purpose was to test the real inference and receipt chain before expanding the task panel. The exact negative outcome is preserved in `research/paper2_microtrial_v4/PAPER2_V4_NATIVE_JOB_3475193_INGEST_RECEIPT_20260811.json`. An adaptive V4.1 parser-engineering replay, frozen after V4 result inspection but before any V4.1 output, fixes a non-heuristic serialization rule: bare JSON or exactly one lowercase `json` fence is admissible, while trailing prose, multiple/nonexact fences and substring extraction remain invalid. It is fresh only to V4.1 outputs, is not an untouched discriminator, does not reinterpret V4, and cannot convert the V4 nulls into scores.

The V4.1 batch is explicitly an adaptive non-confirmatory replay fresh only to V4.1 outputs: V4 outputs were already known. Its separately versioned LUNARC packet requires an exact clean merged checkout, allocated-node revalidation of the normalization and negative-history bindings, semantic preflight before inference, separate `v4_1` run/receipt/log roots and exact model-snapshot attestations before and after execution. Native job 3475212 satisfied this receipt chain at merged checkout `4a8d5ff19e3e6b26b95cb7408bbf55475208989c`: it completed in 54 scheduler seconds with return code zero, and the eight-file snapshot identity was unchanged across inference. The governed harvest admits one task–seed unit with two arm records and no receipt failure.

This is a partial parser-engineering result, not comparative evidence. `RAKL_CONTEXT` is the sole parse-valid/scorable arm; it answers 3 of 5 conceptual fields correctly but fails the exact conceptual gate. `DIRECT_CORPUS` again appends prose after a JSON fence, so its score remains null under the frozen exact normalizer. Consequently V4.1 has one scorable arm and zero exact conceptual passes. A paired effect, arm win/loss and uncertainty interval are not estimable. Under the frozen zero-success convention, each arm's observed token count per valid scientific success is infinite because each used positive tokens and neither produced a valid scientific success. The recorded 1,140/108 versus 638/320 input/output-token counts and 15,789 versus 32,447 ms arm wall times are retained only as execution coordinates. Both local provider API charges are zero, but electricity, hardware, download and labour coordinates are unpriced; zero API charge is not a full cost estimate or a fully costed cost-per-success comparison.

No quantitative figure is produced from a single scorable arm. The exact admitted bytes and scheduler/checkout/snapshot/task–seed/result lineage are bound by `research/paper2_microtrial_v4_1/PAPER2_V4_1_NATIVE_JOB_3475212_INGEST_RECEIPT_20260811.json`. Tip-subject rebinds on the same non-confirmatory V4.1 protocol (native jobs 3476520, 3476521 and 3476524) reproduce the same residual: `HARVEST_V4_1_TASK_SEED_PASS_NONCONFIRMATORY`, zero exact conceptual passes, and no estimable arm comparison; see `research/PAPER2_V4_1_NATIVE_JOBS_3476520_3476521_3476524_INGEST_STATUS_20260811.md`. Those harvests are receipt-chain authority only. They are not experience-benchmark method authority, arm win/loss evidence, or promotional metrics, and they do not reinterpret V4.

A later adaptive V4.2 prompt-interface successor, frozen after the V4.1 diagnosis but before any V4.2 output, keeps the V4.1 exact normalizer and the unchanged exact conceptual gate while adding explicit field polarity and stop-after-JSON clipping. Native job 3476540 completes with both arms `parse_valid`/scorable: the DIRECT fence+prose serialization residual is repaired. Both arms still answer only 3 of 5 conceptual fields and fail `exact_conceptual_pass`. Under the frozen zero-success convention, token count per valid scientific success remains infinite for both arms; no paired effect, arm win/loss, promotional metric or experience-§B authority is estimable. The honest reading is a Qwen2.5-0.5B-Instruct capability / scientific-output limit under this sealed gate, not a scorer, parser or tip bug. Paper promotional numbers therefore remain blocked. The exact admitted lineage is `research/paper2_microtrial_v4_2/PAPER2_V4_2_NATIVE_JOB_3476540_INGEST_RECEIPT_20260811.json` (with the non-confirmatory note `research/paper2_microtrial_v4_2/PAPER2_V4_2_JOB_3476540_CAPABILITY_LIMIT_RESULT_20260811.md`). V4, V4.1 and V4.2 negatives are retained as typed linked experiences. A larger Instruct model is out of scope for the V4.2 packet, which binds only the staged 0.5B snapshot; softening `exact_conceptual_pass` is forbidden. The full matched empirical claim remains open.

14.3 Current matched continual-experience result

The registered RESET-versus-LEARNING experiment has now progressed beyond protocol freeze to one scientifically interpretable native result, while preserving the failed interface iterations that preceded it. The first v1 execution (job 3476542) completed the required $S_0 \to S_n$ development chronology and fresh-transfer reset rules, but its prompts omitted the evaluator's allowed verdict vocabulary; model outputs such as `REJECT` and `FAIL` therefore made all twelve records schema-invalid. V1.1 (job 3476546) repaired the verdict vocabulary but still produced CSV-like or otherwise non-JSON outputs, disproportionately in RESET, so any arm delta remained confounded by the response interface. Both are retained as negative process history rather than reinterpreted as learning evidence.

A versioned v1.2 packet was frozen before its outputs with the same tasks, model, evaluator and matched resource ceiling, but an explicit JSON-object response skeleton. Native job 3476548 completed successfully and produced all twelve schema-valid records with the registered RESET and LEARNING state chronology intact: RESET tasks begin from S_0; LEARNING development forms one $S_0 \to S_n$ chain; and every fresh-transfer LEARNING task begins independently from the same frozen S_n.

Under that packet, the success rate is 0 for both arms on development and fresh transfer. The development score delta (LEARNING minus RESET) is 0. A small partial-score difference on fresh transfer does not satisfy the registered success criterion and grants no global capability or method-promotion claim. The bound verdict is therefore `NEGATIVE_NO_TRANSFER_SUCCESS_LIFT_UNDER_0_5B`. This is an interpretable negative result for Qwen2.5-0.5B-Instruct under the frozen six-task packet and resource ceiling, not evidence that persistent experience can never help a stronger model, a different task family, or a differently powered RAKL configuration.

The exact paper-facing result note is `research/paper2_experience_benchmark_v1_2/PAPER_FACING_NOTE_ISSUE138_SECTION_B.md`; the landed packet also contains the native job receipts, validation output, analysis tables and reproducible benchmark/resource figures. Earlier V4–V4.2 pendulum microtrials remain separate non-confirmatory prompt-interface and capability-limit evidence and are not reused as continual-experience observations.

This result closes the basic two-arm empirical residual: the framework can now report what happened under a matched RESET/LEARNING execution rather than only preregistering the comparison. It does *not* complete the stronger novelty campaign. Model-evaluated authority-leakage baselines, the learning-by-authority-governance factorial, closest-parent/function-matched ablations, larger-model robustness and independent external review remain separate open evidence coordinates. Paper II therefore treats the present negative as a constraint on the claim and a justification for those stronger discriminators, not as a reason to redefine success post hoc.

15 Manuscript saturation as a framework specialization

Relation to v3 vector saturation. The v3 substrate tracks bounded flatness separately across knowledge, operator, experience-pattern, obstruction, relation, path and meta-method axes. Manuscript saturation is a publication projection of primarily the knowledge/relation/obstruction coordinates plus explicit proof and review obligations. A flat manuscript therefore does not imply that operator learning, experience consolidation or Self-RAKL evolution is flat, and a new method episode does not automatically reopen the manuscript unless it changes a publication-relevant semantic object. This separation prevents a living learning system from making every task episode a mandatory paper revision while still requiring any materially changed claim or evidence boundary to reopen the publication projection.

The paper itself is an object produced by the research method, so the stopping rule for manuscript revision must be explicit. We define a *manuscript semantic object* as a retained addition that changes at least one of the following: a scientific or method claim distinction; a citation lineage or nearest-prior-work relation; a novelty boundary; a proof or implementation obligation; an explanatory bridge needed to interpret an equation or component; a falsifier; or a reviewer-relevant evidence boundary. Sentence-level paraphrases and duplicate citations do not count as semantic growth.

For manuscript pass r, let

$$\mathcal{M}_r = \mathrm{Canon}(\text{retained manuscript semantic objects through pass } r), \qquad \Delta_r^{\text{paper}} = \mathcal{M}_r \setminus \mathcal{M}_{r-1}.$$

A pass is semantically flat when $\Delta_r^{\text{paper}} = \varnothing$ after deduplication and it creates no new unresolved proof obligation, novelty correction or reviewer-blocking contradiction. Flatness is recorded separately for the registered lenses: epistemology/formal methods, mathematical geometry, scientific-agent literature, and methods/reproducibility.

A *locally saturated publication projection* additionally requires: all mandatory search/review routes are covered or explicitly blocked; the freshness scan reaches its registered cutoff; nearest-work and citation-equivalence audits are complete for the main novelty claims; structural proof obligations are resolved or explicitly open; and every remaining manuscript issue is classified as empirical/deferred, out of scope, or blocked by unavailable evidence. A material open issue prevents the certificate. Page count never supplies saturation credit.

This protocol is intentionally a specialization of the existing saturation/stopping and review surfaces, not a new 25th high-impact research-method surface. The underlying mechanism is the same semantic-growth accounting already used for research fibers. What changes is the object being measured and the terminal classification of open items. Same-context manuscript review can support only *local manuscript flatness*. It cannot be relabelled as independent peer review or independent scientific saturation.

The reopen rule is strong. Any later exogenous concept, citation, counterexample, implementation result or reviewer comment that materially changes the novelty boundary, formal architecture or evidence interpretation adds a new manuscript semantic object and reopens the projection. This makes the paper a versioned interface to a living atlas rather than a claim that the literature has been permanently exhausted.

The protocol also disciplines length. In response to a request for "more depth," the correct action is not to multiply prose uniformly. A section grows when the reader currently lacks an intellectual ancestor, formal distinction, proof obligation, empirical boundary or explanatory bridge needed to evaluate the claim. A paragraph that restates an equation without adding such information is removed. Saturation therefore opposes both premature compression and decorative expansion.

15.1 Section purpose is itself a proof obligation

A saturation protocol needs a deletion rule as well as an addition rule. For every top-level section, the manuscript records a *reader obligation*: the question that section must answer which is not already answered elsewhere. If no such obligation can be named, the section should be merged or removed even when its prose is correct. This turns the editorial question "why is this page necessary?" into a structural audit rather than a stylistic preference.

The present paper uses the following section-purpose map. The Introduction establishes the research problem and the claim boundary. Intellectual lineage establishes what is inherited and therefore what novelty remains. Formal method defines the scientific state and transition semantics required to evaluate the claims. Scientific memory explains how raw evidence becomes that state rather than an ordinary RAG store. The atomic lifecycle translates the formalism into an auditable execution contract. The Obsidian–J-space–RAKL comparison prevents category mistakes among navigation, representation and epistemic geometries. Workspace and OWMD explain bounded cognition and ontology escape. The known-answer trace tests deterministic mechanics. Bounded scientific cognition states the resource boundary. Method learning and Self-RAKL state how the method may change without self-authorization. Reference-implementation correspondence binds formal claims to code. The preregistered evaluation programme defines what empirical evidence would justify stronger claims. Manuscript saturation defines the stopping rule for this publication projection. Discussion exposes rival interpretations and failure modes. Reproducibility identifies the artifact and what can be replayed. Structural proof obligations make the mathematical words falsifiable.

This map supplies a concrete redundancy test. If a paragraph in Discussion only restates a formal definition, it should be removed unless it interprets a consequence or boundary. If Related Work merely names systems already used to narrow a specific contribution, it should be reorganized around the novelty claim rather than expanded as a catalogue. If the known-answer demo repeats the lifecycle description without showing a selective failure or invariant, it adds no evidential role. Conversely, a short section may need expansion when its reader obligation cannot yet be discharged. The three-geometry section grew for this reason: the earlier analogy to Obsidian did not establish how a note graph, a cone-like representation space and a typed epistemic state differ mathematically.

The same rule applies below the section level. A paragraph should normally contribute one of a small number of functions: define an object, locate intellectual ancestry, state evidence, derive a consequence, distinguish nearby concepts, expose a limitation, specify a falsifier, or connect a formal object to implementation/evaluation. A paragraph whose only function is rhetorical emphasis is a candidate for deletion. Equations and tables receive the same treatment. An equation is justified when its symbols or invariants matter to later reasoning; a table is justified when simultaneous comparison prevents ambiguity that prose would hide. Mathematical decoration and citation decoration are both failures of purpose.

This purpose audit is deliberately coupled to depth. "Concise" does not mean omitting the argument required to assess a claim, and "comprehensive" does not mean saying the same thing three ways. The target is *epistemic density*: retained words should increase the reader's ability to locate the claim, understand its ancestry and assumptions, reproduce its derivation, test its boundary, or identify the evidence needed to change it. Length may therefore increase sharply during an early saturation pass and then decrease during canonicalization as duplicate explanations are merged.

The page budget creates a projection problem rather than a truth problem. If a journal format cannot contain all canonical detail, the solution is to move reproducible derivations, extended lineage matrices or machine-readable receipts to appendices and repository artifacts while preserving enough in the main paper to make every central transition intelligible. The omitted material remains addressable through stable pointers. Silent deletion of a blocking assumption or counterexample to meet a page limit is not acceptable compression; it changes the epistemic content of the publication projection.

A final editorial pass therefore asks two questions in both directions. For every retained page: what reader obligation would become materially harder to discharge if this content disappeared? For every central claim: is there a page that actually discharges the corresponding ancestry, formal, evidence and boundary obligations? A "no" to the first question triggers deletion or merging. A "no" to the second triggers a new manuscript fiber. The paper stops changing only when these operations, together with the external literature and formal audits, become semantically flat. The criterion is intentionally symmetric: it prevents ornamental pages while reopening underdeveloped claims whose evidence, ancestry or proof obligations remain hidden.

15.2 Operational saturation ledger

A manuscript-saturation run maintains a ledger rather than a subjective sense of completion. Each review pass records its lens, route, context identifier, sources inspected, semantic objects proposed, semantic objects retained after canonicalization, and open items created or closed. A retained object receives a stable identifier so later passes can distinguish new content from repeated rediscovery.

The ledger separates seven semantic kinds: claim distinction, citation cluster, novelty-boundary change, proof obligation, explanatory bridge, falsifier and review repair. This taxonomy is intentionally coarse. Its purpose is to answer whether the paper's epistemic content grew, not to grade prose. A new citation that supports an already well-established sentence without changing lineage may be useful copyediting but adds no semantic object. A single prior paper that shows a claimed contribution is established can create a major novelty-boundary object even though it adds only one reference.

Flatness is computed after canonicalization. If three review lenses independently rediscover the same missing distinction, the first retained instance is growth and the others are corroboration of its importance. Conversely, splitting one idea into many headings does not manufacture novelty. This makes semantic saturation resistant to cosmetic expansion.

Open items have terminal classifications. 'MATERIAL_OPEN' blocks local saturation because it could change the paper's central claims or architecture. 'EMPIRICAL_DEFERRED' identifies a claim that can only be resolved by the preregistered experiment or other future evidence. 'OUT_OF_SCOPE' records a neighboring question whose exclusion is deliberate and justified. 'BLOCKED_MISSING_EVIDENCE' records a relevant question that cannot currently be checked. These states are visible in the saturation receipt so the absence of prose is not misread as an implicit solution.

A route or lens can be flat while the whole manuscript remains active. For example, a mathematics pass may add no new object while a literature freshness pass discovers a new scientific-agent benchmark that changes the evaluation section. That new object resets the global flat tail and can reopen earlier sections because its implications propagate. The stopping rule therefore operates on the manuscript dependency graph, not on a fixed number of passes per section.

Saturation also has a cost boundary. The process is not required to read every paper ever written. Instead it must show that the registered routes reached local semantic flatness, that nearest competing work was explicitly sought, and that unresolved omissions are classified. A future external reviewer can then challenge the certificate constructively by naming the missing semantic object. If the challenge is material, the proper response is to reopen the paper, not to defend the word "saturated."

In the present release, the four review lenses were chosen because they correspond to the most consequential failure families: epistemology/formal methods can expose authority or identification errors; mathematical geometry can expose category mistakes in lattice, graph, cone and sheaf language; scientific-agent literature can expose novelty and evaluation gaps; and methods/reproducibility can expose untraceable claims or artifact drift. Other journals or domains can register additional lenses without changing the underlying saturation mechanism.

15.3 What a manuscript-saturation certificate actually certifies

A saturation certificate must identify its universe. For a manuscript projection P, define a registered audit scope

$$\Sigma_P = (Q_P, L_P, R_P, D_P, t^\star, B_P),$$

where Q_P is the set of central paper claims and proof obligations, L_P the review lenses, R_P the search-route families, D_P the declared source universe or source classes, $t^\star$ the freshness cutoff, and B_P the resource and page/context budget. The certificate says that semantic growth became flat under Σ_P after the registered audits. It says nothing about sources, disciplines or future evidence outside that tuple.

The route set matters because repeated searching with one vocabulary can create false flatness. A claim may appear saturated under exact-term search while remaining incomplete under historical terminology, mathematical equivalence, adjacent-domain function search or citation-neighborhood expansion. The paper therefore inherits the OWMD lesson: at least one route for each novelty-critical mechanism should be capable of succeeding without repeating the framework's own labels. The Obsidian miss is the motivating counterexample. The concept was easy to recognize after it was supplied, yet the earlier route family had not surfaced it.

The claim set matters for a different reason. A manuscript can be globally long and still be locally thin at its most important inferential bridge. Saturation is not uniform word density. The audit asks whether every central claim has enough ancestry, formal semantics, evidence boundary and falsifier for a reviewer to determine what would make it wrong. A background paragraph can remain short because its only job is orientation. A geometry section may need substantial depth because a category mistake there would invalidate several downstream claims. The page budget therefore constrains presentation, not the amount of canonical research state that must exist behind the paper.

Proof obligations are treated as semantic objects because a newly discovered obligation changes what the paper must establish even before the proof is supplied. For example, using the word "lattice" creates meet/join or closure-system obligations; using "sheaf" creates locality, restriction and gluing obligations; claiming independent corroboration creates evidence-lineage obligations; claiming a matched experiment creates treatment-definition and resource-comparability obligations. A pass that discovers such an obligation is non-flat even if it adds no new empirical result. The manuscript remains active until the obligation is discharged, explicitly weakened, or classified open.

Citation saturation is similarly semantic rather than numeric. One hundred references that all repeat the same conceptual neighborhood may add less coverage than one remote paper that changes the novelty boundary. Each source is therefore mapped to a role: foundational ancestor, nearest competing method, mathematical precedent, measurement/evaluation precedent, counterexample, contemporary benchmark, implementation analogue or scope boundary. The useful question is not "does every paragraph contain citations?" but "could a knowledgeable reviewer name a materially closer ancestor or counterexample that the current claim map does not represent?" When such a source appears, the novelty coordinate reopens.

The certificate also separates *flatness* from *closure*. A pass can add no new semantic object because an important database is unavailable. That is flat observation, not closure. Likewise, a route can be exhausted because its token budget was reached while new concepts were still arriving. The correct states are blocked or active-non-flat. Closure requires that material open items are absent and that deferred empirical questions are explicitly labeled. This prevents resource exhaustion from masquerading as knowledge saturation.

Same-context recursion introduces a final dependence problem. The same researcher or model can run multiple lenses and still share hidden assumptions, search habits and evidence ancestry. Repeated flat passes are useful engineering evidence that the current context has stopped generating new repairs, but they cannot establish independent literature completeness or peer-review consensus. The receipt therefore has two separate fields: same-context local manuscript saturation and independent review status. In this release the former can become true while the latter remains false.

A practical certificate should be falsifiable by a small object. An external reviewer does not need to refute the entire paper to reopen it; supplying one materially new prior method, one counterexample to a formal claim, one missing proof obligation, one unrepresented failure mode or one artifact mismatch is enough. The challenge is canonicalized into the ledger, the affected dependency neighborhood reopens, and later flatness must be re-earned. In this sense saturation is a versioned state, not a medal attached permanently to the manuscript.

This interpretation makes the paper itself an executable example of epistemic mechanics. The canonical project atlas can be richer than the publication, while the publication declares which slice it projects and which open coordinates it intentionally leaves out. A reader can therefore distinguish three statements that conventional writing often conflates: "the authors stopped editing", "the registered review process stopped producing new material objects", and "the scientific field contains no relevant unknowns." Only the second is the manuscript-saturation claim made here.

16 Discussion and falsifiers

RAKL is intended as a scientific authority and research-control layer around a replaceable LLM, not a claim that one agent architecture has solved scientific reasoning. The architecture externalizes evidence-bearing knowledge, explanatory models, procedural skills, a research agenda, failure history, metacognitive control and resource constraints. Human-inspired terms are operationalized rather than anthropomorphized: intellectual humility becomes evidence-triggered revision, curiosity becomes information gain or learning-progress exploration, reflection becomes diagnosis plus a costed control action, and experience becomes a reusable skill only after transfer.

The self-bootstrap requirement is not circular because self-application is separated from self-authorization. RAKL may discover a weakness in itself and propose a repair, but acceptance comes from a frozen harness and fresh tasks outside the challenger's control. Formal closure likewise does not mean the method is finished forever. It means the current registered high-impact inventory has no known unowned specification surface. The Polymarket/spot project is deliberately the next integration stress test. If that project exposes a high-impact operation absent from the current 24-surface system, the correct outcome is to reopen scoped formal closure.

The paper registers direct falsifiers. If direct prompting, RAG or a simpler agent workflow matches RAKL on registered process and final-task outcomes at lower cost without more validity failures, the framework complexity is not justified. If removing a mechanism does not selectively worsen its preregistered failure mode, that mechanism's contribution is unsupported. If method changes improve only development tasks, the strong self-evolution claim fails. If a simpler context policy matches mandatory evidence retention and task quality under matched budgets, the complex compiler has not earned its cost. If the registered quant information set contains no material 5/15-minute predictive information after causal controls, RAKL must report that ceiling or the missing observable rather than search until significance appears. If semantically equivalent prior work is found, the novelty claim narrows regardless of terminology.

Two additional falsifiers follow from the hardening pass. First, longitudinal atlas-volume or density claims are invalid if their measurement-basis fingerprint changes; ontology refinement cannot count as knowledge growth by itself. Second, external-framework or novelty saturation is falsified for a scope when a later candidate materially overlaps registered functions yet was missed by a route ensemble that had declared saturation. In that case the missed candidate remains failure memory, the affected discovery scope reopens, and a repair earns method credit only prospectively on fresh hidden concepts.

16.1 What would count as a theoretical failure of the architecture?

The framework can fail before any empirical benchmark is run. A theoretical failure occurs if two objects that the architecture treats as distinct are shown to require a licensed coercion it forbids, or if an object the architecture treats as one type is shown to mix incompatible scientific roles. For example, if a general and defensible map from predictive authority to mechanism authority were established under the framework's claimed scope, the non-escalation rule would need revision. If a scalar authority score could preserve every blocking condition and every relevant partial order without loss, the multi-axis representation would be unnecessary. If a simpler provenance graph plus ordinary belief revision reproduced the same update semantics, the integrated architecture's novelty would narrow.

A second failure class is *formal decoration*: using mathematical vocabulary without operational consequences. The explicit compatibility-complex boundary is designed to prevent this. Calling an object a lattice, sheaf, manifold or geometry should create proof obligations that can fail. If no operation, invariant or falsifier depends on the structure, the terminology should be removed or downgraded to analogy.

A third failure class is *governance without value*. It is possible for an evidence-governed workflow to be safer but so expensive or conservative that it is scientifically impractical. The matched evaluation therefore includes cost, wall time and opportunity costs, and permits simpler systems to win. Scientific integrity is a blocking requirement for strong claims, but among integrity-preserving systems efficiency remains a legitimate objective.

16.2 Plural scientific state is not indecision

A recurring concern is that preserving identified sets, contextual charts and unresolved fibers may prevent the system from ever making decisions. The answer is to separate scientific identification from decision robustness. A policy can be chosen under model uncertainty by optimizing worst-case, Bayesian or other registered decision criteria without pretending that the underlying mechanism has been identified. Manski's work on decision-making under partial identification makes this point explicit [79]. RAKL stores the decision rule and its loss/context certificate separately from the mechanism certificate.

This separation can make the final report more decisive, not less. Instead of one hedged paragraph that mixes uncertainty about facts, mechanisms and actions, the atlas can state: which observations are established; which mechanism classes remain possible; which additional observation would discriminate them; and which current decision is robust across the survivor set. Different readers can then reuse the same state for different quantities of interest without inheriting a false singular theory.

16.3 A catalog of framework-level failure modes

The integrated architecture creates new failure modes in addition to the ones it is designed to prevent. *Schema ossification* occurs when the typed state becomes so rigid that genuinely new scientific objects are forced into old categories. OWMD and method-basis fibers are intended as escape routes, but they do not guarantee success. *Governance overload* occurs when verification requirements consume more resources than the scientific value of the claim justifies. The response should be tiered authority and target-specific assurance, not removal of all gates.

False independence occurs when provenance graphs miss a shared evidence root and the system overcounts corroboration. *False identity* occurs when semantic canonicalization merges distinct claims. *False novelty* occurs when remote prior work is not retrieved. *False obstruction* occurs when a bad mapping makes compatible charts appear incoherent. *Context omission* occurs when an important coordinate is not represented at all, making later alignment impossible. Each failure has a different remediation route.

Metric gaming can occur inside the framework. Once unsupported-authority rate, semantic novelty or evidence-lineage diversity become optimization targets, an agent can improve the metric superficially: by abstaining excessively, splitting claims into many "novel" atoms, or manufacturing nominally separate evidence paths. The protected invariants and multi-coordinate evaluation reduce this risk but do not eliminate Goodhart effects. Metrics are diagnostic projections of state, not substitutes for scientific review.

Over-conservatism is equally important. A fail-closed system can return `CANNOT_CHECK` too often and avoid useful exploratory leaps. The architecture therefore grants proposal authority to analogies, hypotheses and teacher models without granting them scientific authority. Exploration is permitted to be bold so long as promotion remains strict. This separation is intended to preserve creativity without confusing creativity with confirmation.

Formalism debt arises when named mathematical structures accumulate faster than their proof obligations. The compatibility-complex terminology repair is an example of paying such debt. Future additions are expected to include explicit operations, invariants and falsifiers or remain analogies. A dense paper with many equations is not necessarily a more formal method.

Artifact drift occurs when code, receipt, figure and paper describe different versions. Exact-subject builds address this mechanically. *Review monoculture* occurs when all critics share the same context or conceptual assumptions. Independent external review remains an open coordinate specifically because same-context recursive review cannot close it.

This catalog is not complete. Its role is to show how the framework can fail in ways that are themselves represented as fibers. A useful self-evolving methodology must make its own attack surface legible rather than presenting governance as a solved problem.

16.4 Why the saturation claim remains bounded

The expanded literature review itself demonstrates why universal saturation would be indefensible. Recent work on scientific-agent evidence boundaries, mechanism-centric world models, sheaf-theoretic theory-shift detection and confidence-based stopping materially sharpens the framework's positioning [73, 76, 86, 88]. Some of these sources appeared only shortly before this release. A permanent completeness claim would be falsified by the ordinary passage of time.

The correct claim is therefore procedural: the present version can document which route families and review lenses were searched, what semantic objects they added, what remained flat, and which empirical coordinates remain open. This turns "we searched enough" from an intuition into an auditable, reopenable statement. It is a weaker claim than universal completeness and a stronger claim than stopping because the author ran out of pages.

16.5 Language is part of the formal boundary

The framework uses technical metaphors only when their inferential consequences are stated. "Atlas" is retained because local charts, overlap alignment and failure to obtain one global representation are genuine organizing ideas. "Lattice" is retained in the project name for historical continuity but is qualified by the compatibility-complex boundary whenever the generic atom structure is discussed. "Workspace" denotes bounded computational access and not consciousness. "Self-evolution" is reserved for method changes that survive protected assurance rather than ordinary iterative editing.

This vocabulary discipline is scientifically important because anthropomorphic or mathematical language can mint authority by implication. Saying that an agent "understands" a mechanism can be read as a stronger claim than saying it produced a model that passed registered tests. Calling a visualization a "knowledge graph" can imply typed semantics absent from ordinary hyperlinks. Calling repeated self-critique "independent review" can imply evidence separation that does not exist. The paper therefore prefers operational predicates over evocative labels when the latter could change the reader's interpretation of evidence.

The same rule applies to certainty verbs. "Demonstrates" is used for results whose design directly supports the stated relation; "is consistent with" is used when alternatives remain; "identifies" is reserved for an explicit identification condition; "suggests" remains proposal-side unless supported by the relevant certificate. These are not cosmetic hedges. They mirror the typed authority model in natural language.

A mature implementation could lint publication text against authority certificates, flagging mechanism verbs unsupported by mechanism-level evidence or universal quantifiers attached to scoped tests. The current release does not claim a complete scientific-language type checker. It adopts the more modest discipline manually and records automated language-policy enforcement as a possible future method fiber.

Reproducibility and AI-use statement

The public repository is `https://github.com/SzeChunYiu/RAKL`. It maintains frozen benchmarks, content-addressed execution receipts, exact-subject CI, negative-history records, release manifests and editable paper/figure sources. The V2.1 engineering figures are rendered with deterministic Matplotlib code from `research/MINI_RESEARCH_DEMO_043_RECEIPT.json`, `research/MINI_RESEARCH_METROLOGY_044_RECEIPT.json` and `research/MINI_ARCHIVE_STORAGE_044_RECEIPT.json`; CI checks the plotted source coordinates and exports PDF, editable SVG, PNG preview and machine-readable figure source data. Structural arrows are intentionally unlabelled in the paper schematics, and quantitative arrow callouts are forbidden by repository tests. The first matched-study attempt is preserved separately as an executable freeze packet, a fail-closed preflight receipt with zero evaluated task records and a non-scientific provider-connectivity receipt. These are readiness artifacts, not model-performance data. Language models are research, coding and drafting tools rather than authors; their proposals do not possess canonical scientific authority. Same-context role simulations and the present internal Nature-style review rounds are not counted as independent peer review. The later journal-results manuscript will bind exact data/transformation identities, model-run receipts and all quantitative figure source data.

.1 Reproducibility as a chain of identities

Reproducibility is not one binary property. Computational reproducibility asks whether the same code, environment and data can regenerate the result. Analytical reproducibility asks whether the transformation from data to claim is specified enough to inspect. Experimental reproducibility asks whether a new realization of the scientific process yields compatible evidence. Conceptual replication asks whether a result survives a different operationalization of the same underlying question. These levels can disagree.

The repository infrastructure mainly addresses the first two. Exact subject SHAs, content hashes, deterministic receipts, dependency declarations and publication manifests allow a reader to identify the artifact that generated a figure or software claim. Established reproducible-research guidance motivates this bookkeeping [51, 52], and broader reproducibility reform emphasizes transparency across design, analysis and reporting [91]. The framework does not claim that such bookkeeping produces independent replication.

This hierarchy explains why artifact identity is printed inside the manuscript. If a software contract changes after a paper is rendered, the old PDF still describes the old subject. If the release process instead prints "latest" or an unpinned package version, future readers cannot know which behavior was tested. Content addressing converts that ambiguity into an immutable reference.

Data and model availability can still limit reproducibility. Proprietary APIs can change, source websites can disappear, and external models can be revised behind stable product names. When exact replay cannot be guaranteed, the receipt records the strongest available identity, raw outputs where permitted, timestamps, settings and hashes of locally controlled inputs. The residual limitation is reported explicitly rather than masked by a reproducibility label.

AI-in-the-loop scholarship is also moving toward executable provenance. F(AI)2R, for example, treats the paper-production process itself as an auditable provenance graph and uses continuous integration to check graph conformance [99]. The important lesson for RAKL is not that CI can certify scientific truth. It is that manuscript text, code, evidence and verification acts can be made addressable enough that later reviewers can ask which artifact authorized a sentence. The present release follows the same direction by binding generated engineering claims to exact source subjects and by separating same-context manuscript review from independent scientific replication.

Independent replication then sits above the artifact layer. A fresh group using the published method and independently acquired evidence can test whether the scientific effect transports. That event would upgrade a distinct authority coordinate. The current paper has not earned that coordinate and says so.

.2 Publication artifact identity and chaptered source

The public paper package is generated as a chaptered LaTeX project. Each major top-level section is stored in its own source file and assembled by a small `main.tex`; quantitative figure sources remain receipt-bound; and a source manifest records hashes for the assembled paper and section files. The compiled PDF is treated as a derivative release artifact rather than as the primary editable record.

The build identity deliberately distinguishes the *scientific software subject* from the later commit that stores rendered artifacts. The manuscript prints the exact implementation SHA evaluated by the build. CI runs the software tests, regenerates receipt-bound figures, stages the paper from that exact subject, compiles with blocking LaTeX warnings enabled, inspects PDF metadata and embedded fonts, renders every page, and only then may store the PDF alongside the TeX package. This avoids a recursive identity problem in which committing the PDF changes the source SHA that the PDF itself claims to describe.

Visual inspection remains necessary even when LaTeX exits successfully. Broken glyphs, clipped panels, unreadable figure labels and page collisions are artifact failures that a source-level unit test may not detect. The release process therefore renders all pages to images and treats the rendered pages, not the source alone, as the authoritative layout check.

A Structural proof obligations

Core proof/test obligations include generation non-authority, mechanism non-upgrade, typed relation non-escalation, non-forced gluing, negative-history monotonicity, conservative evidence-lineage saturation, missing-evidence honesty, residual reopening, meta-evaluator separation, measurement non-escalation, explicit uncertainty assumptions and formal-closure honesty. These are structural constraints on the method; they do not prove empirical scientific performance.

A.1 Additional obligations exposed by the saturation pass

The deeper comparison adds several obligations that were implicit in earlier versions.

Geometry obligation. Any future claim that a RAKL object is an order-theoretic lattice must name the scoped order and demonstrate the required meet/join or closure-system properties. Any sheaf claim must name the local objects, restriction maps and gluing condition. Navigation-graph adjacency and J-space vector geometry do not satisfy those obligations by analogy.

Cognition/evidence separation obligation. A workspace intervention or interpretability observation can establish computational influence only. Promoting it to scientific evidence requires an explicit bridge to the external scientific object and the normal verification path.

Manuscript-saturation obligation. A paper-level saturation certificate must report its source universe, freshness cutoff, route coverage, review lenses, last material semantic additions and remaining open classifications. Same-context flatness must never be described as independent peer review.

Novelty-reopen obligation. Discovery of prior work that semantically subsumes an integrated RAKL claim must narrow the novelty claim even if the terminology differs. The prior search transcript remains negative method history and the affected saturation certificate is reopened.

Evaluation obligation. Scientific-agent superiority cannot be inferred from software-test count, toy-world correctness, context compression or same-context self-repair. It requires the preregistered matched and fresh evaluations under the published resource and authority criteria.

A.2 Non-equivalence obligations

Several negative claims in the paper are structural and should be read as non-equivalence obligations rather than empirical performance claims.

Graph is not authority. There is no generic homomorphism from note-link degree, citation count or retrieval centrality to scientific authority that preserves the blocking coordinates of the authority poset. A visualization may correlate with relevance while remaining non-authoritative.

Access is not evidence. There is no generic map from J-space occupancy or workspace salience to external scientific truth. Such a map requires an additional empirical bridge specific to the scientific object. The default relation is cognitive provenance only.

Compatibility is not a lattice. Pairwise or higher-order co-admissibility does not supply an order, meet or join. An order-theoretic lattice claim requires separately verified structure; FCA illustrates one legitimate route [85].

Global existence is not identification. The existence of a global section or coherent global model does not imply uniqueness, and uniqueness of a mathematical representation does not imply evidence sufficient for scientific authority. These predicates live on different certificates.

Traceability is not independence. Two results with complete provenance can still share one evidence root. Independent-corroboration credit therefore requires evidence-lineage analysis in addition to ordinary provenance.

These obligations are useful because they define simple reviewer attacks. A counterexample that violates one non-equivalence under the claimed scope is enough to reopen the corresponding part of the framework. They are also implementation tests: any code path that raises authority because an item was frequently retrieved, highly linked or present in a workspace violates the architecture even if downstream answers improve.

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